



SCHOOL
FOR ADVANCED
STUDIES
LUCCA



EEG in Practice

*Methods for EEG Data
Acquisition and preprocessing*

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Assistant Professor

MOMILAB, School for Advanced Studies, Lucca

Lucca, 10° December 2025

Molecular Mind Laboratory

MoMiLab is a research group of the IMT School for Advanced Studies Lucca (Italy)

Topics and Research Areas

- Sensory Experience Dependence and Plasticity - SEED
- Social and Affective Neuroscience - SANE
- Sleep, Plasticity and Conscious Experience - SPACE
- Affective Physiology and Interoception - API
- Forensic Neuroscience and Psychiatry - FoNeP
- Sensorimotor Experiences and Mental Representations - SEMper
- Models, Inference, and Decisions - MInD
- Methods for Advanced Biosignal Analyses - MABA
- Movement and Action Representation TMS-EEG Lab - MARTLab

Who We Are

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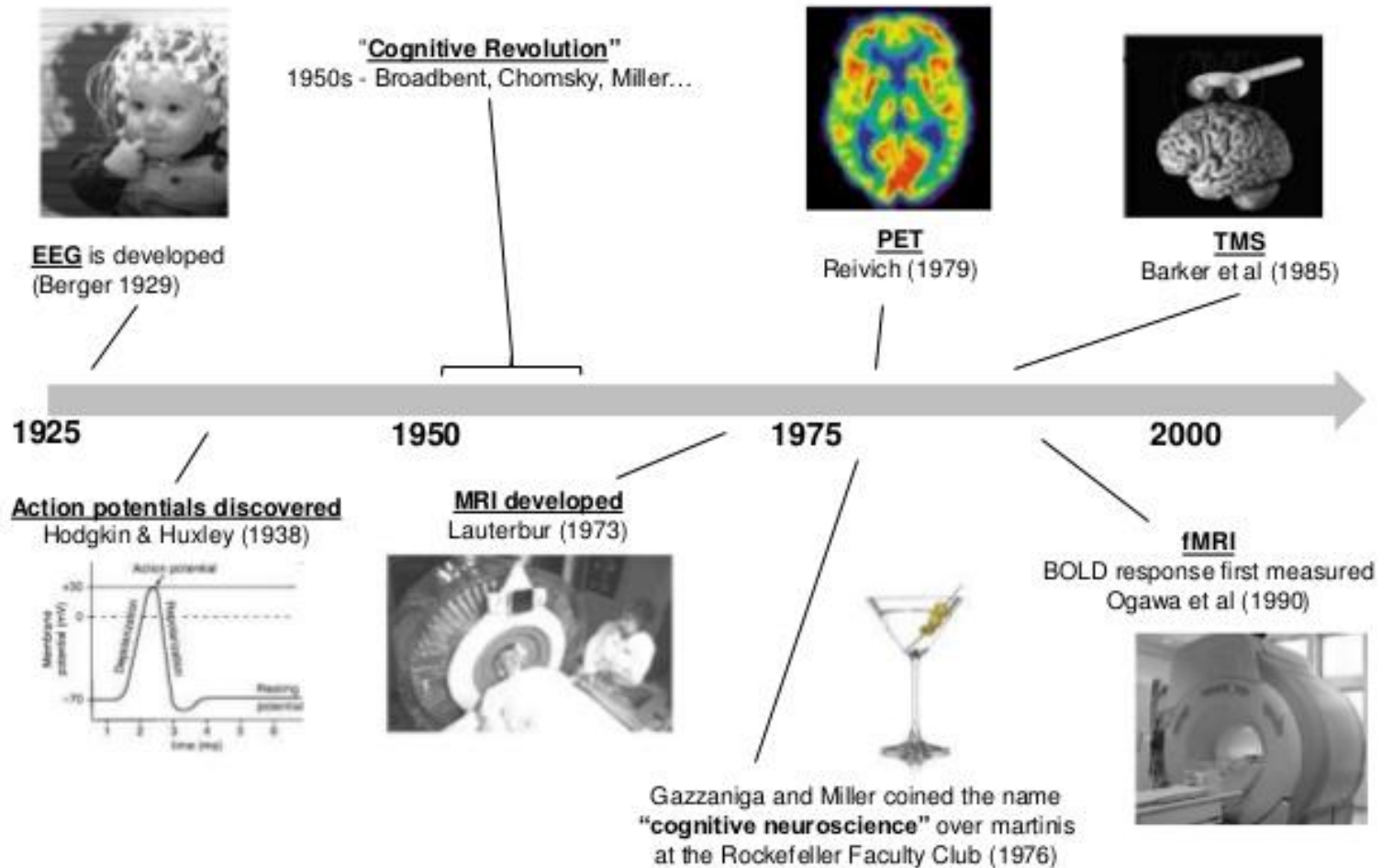
PhD Student
Psychologist

Agenda

- 1) Introduction to Electroencephalography (EEG):
Neurophysiological basis and signal acquisition systems
- 2) Practical demonstration: setup of a recording session using an high-density EEG cap
- 3) Methods for EEG signal Preprocessing

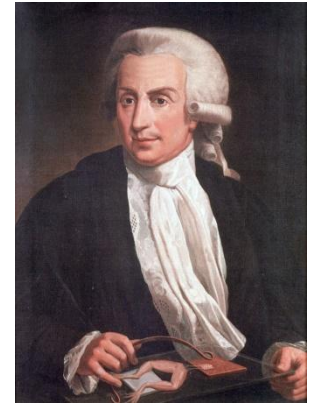
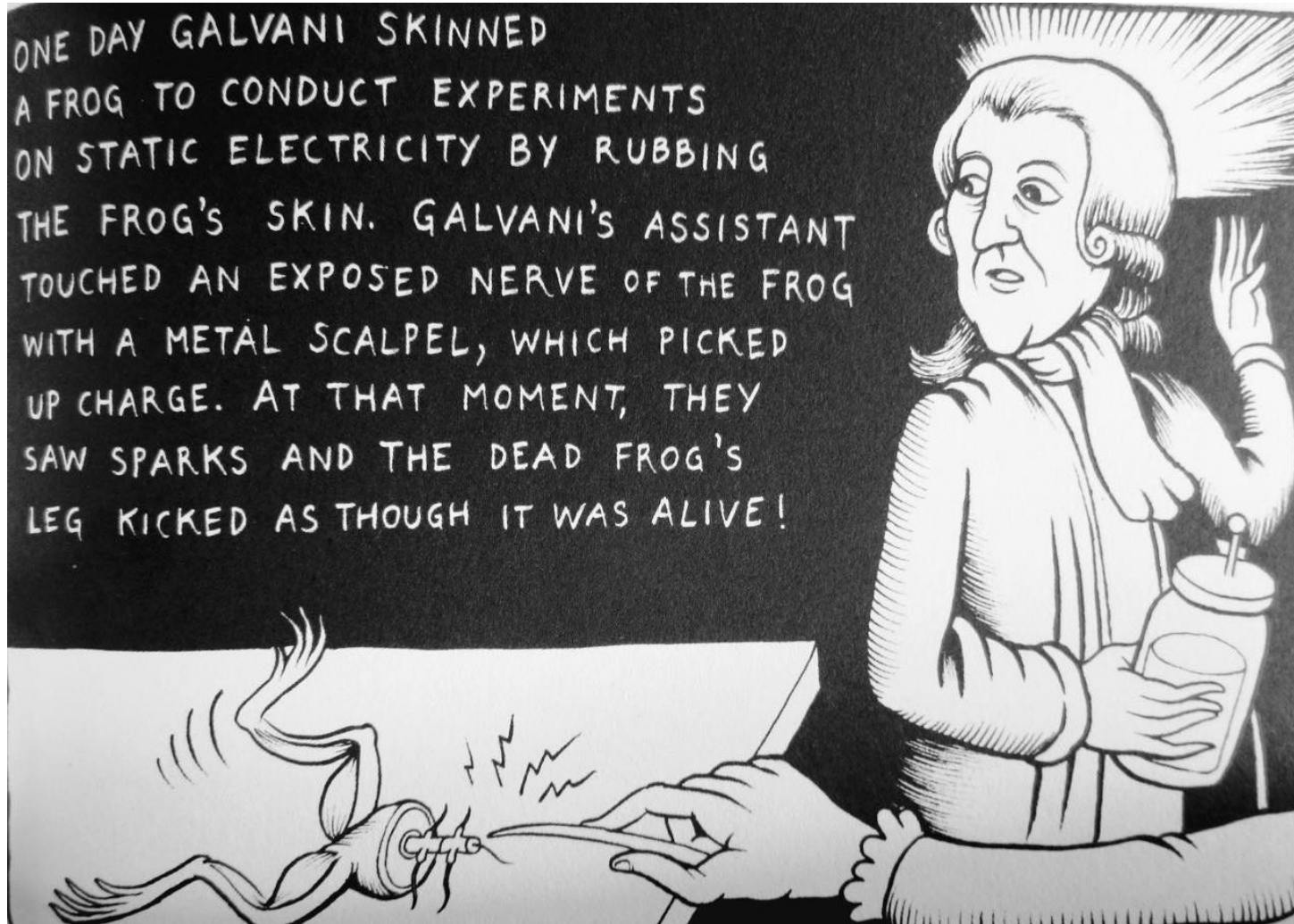
Historical Landmarks:

Cognitive Neuroscience Timeline



Adapted from: The Student's Guide to Cognitive Neuroscience by Jamie Ward, 2006

Historical Landmarks: Pioneers in Electrophysiology



Luigi Galvani
(1737-1798)

Historical Landmarks: Pioneers in Electrophysiology



Carlo Matteucci
(1811-1868)



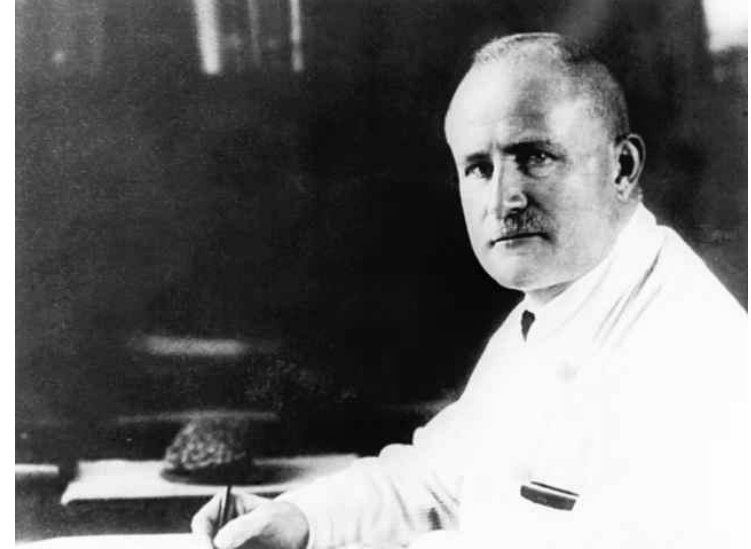
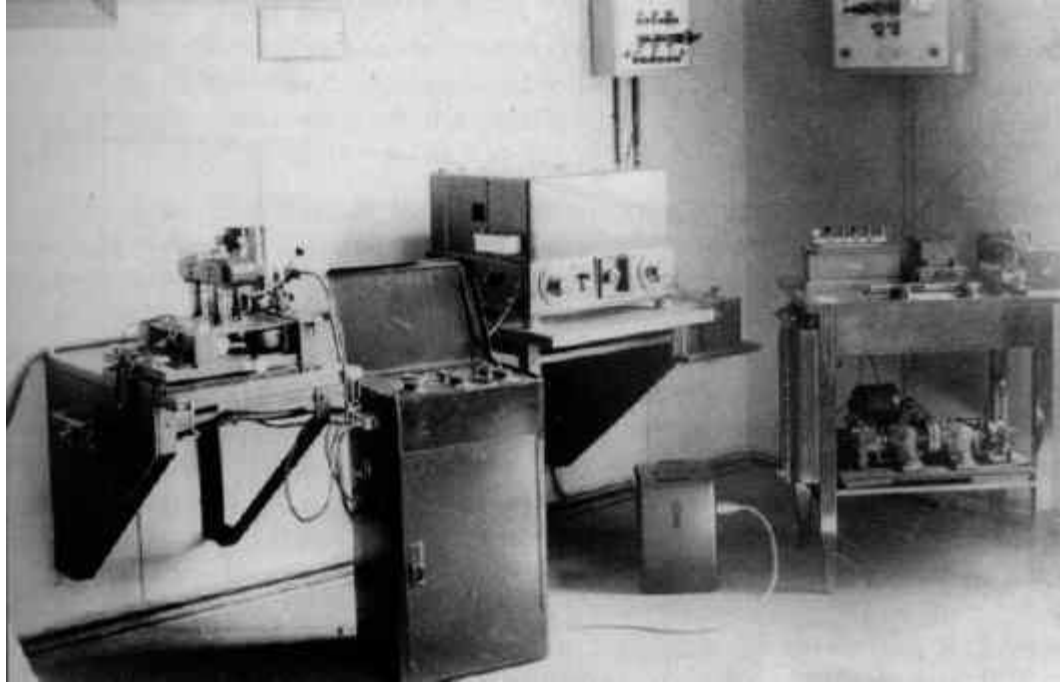
Emil du Bois-Reymond
(1818-1896)



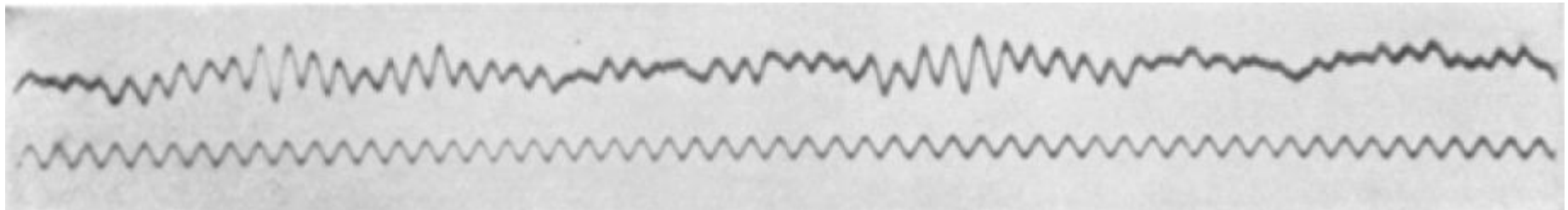
Richard Caton
(1842-1926)

- *First quantitative evidence of bioelectric currents*
- *Discovery of the action potential*
- *First recording of electrical activity from the mammalian cerebral cortex*

Historical Landmarks: the first EEG!



German physiologist and psychiatrist Hans Berger (1873–1941) recorded the first human EEG in 1924.



Historical Landmarks

Year	Name	Description
1875	R. Caton	First tracing in animals of fluctuating potentials that constitute the EEG
1924	H. Berger	First human EEG measurement
1929	H. Berger	First human EEG publication in <i>Archiv für Psychiatrie und Nervenheilkunde</i>
1932	J.T Toennies	First ink-writing biological amplifier
1932	G. Dietch	First application of Fourier analyses on human EEG
1934	F. Gibbs	First systematic application of the EEG to epilepsy
1935	A.L. Loomins	First systematic application of the EEG to sleep
1936	W.G. Walter	Discovery of slow (delta) activity in the presence of tumors
1942	K. Motokawa	First EEG brain map
1943	I. Bertrand and R.S. Lacape	First book on EEG modeling
1947	N/A	American EEG Society is founded
1947	G.D. Dawson	First demonstration of human evoked potential responses
1949	N/A	<i>Electroencephalography and Clinical Neurophysiology</i> , the first EEG journal, is launched
1952	A. Remond and F.A. Offner	First topographic analyses of occipital EEG
1952	M.A.B Brazier and J.U. Casby	Introduction of auto- and cross-correlation function
1955	A. Remond	Application of topographical EEG analyses
1958	H. Jasper	Introduction of 10-20 system for standardized electrode placement
1960	W.R. Adley	Introduction of Fast Fourier Transformation (start of computerized spectral analyses)
1961	T.M. Itil	Application of EEG analyses for classification of psychopharmacological agents
1963	N.P. Bechtereva	Localization of focal brain lesions by EEG
1965	J.W. Cooley and J.W. Tukey	Introduction of fast Fourier algorithm
1968	D.O. Walter	Introduction of coherence analyses for the human EEG
1970	B. Hjorth	Development of new quantitative methods, including source derivation
1971	D. Lehmann	First multichannel topography of human EEG fields
1973	Matousek and Petersen	Development of age-corrected EEG spectral parameter for detecting pathology (qEEG)
1977	E.R. John et al.	Introduction of "neurometrics" (standardized qEEG analyses with normative databases)
1978	R.A. Ragot and A. Remond	EEG field mapping
1979	F.H. Duffy	Introduction of brain electrical activity mapping (BEAM)

Number of Sensor in EEG System



Clinical EEG & PSG

1-21 Electrodes

Number of Sensor in EEG System



Clinical EEG & PSG
1-21 Electrodes



Standard Research EEG
32-64 Electrodes



Number of Sensor in EEG System



Clinical EEG & PSG
1-21 Electrodes

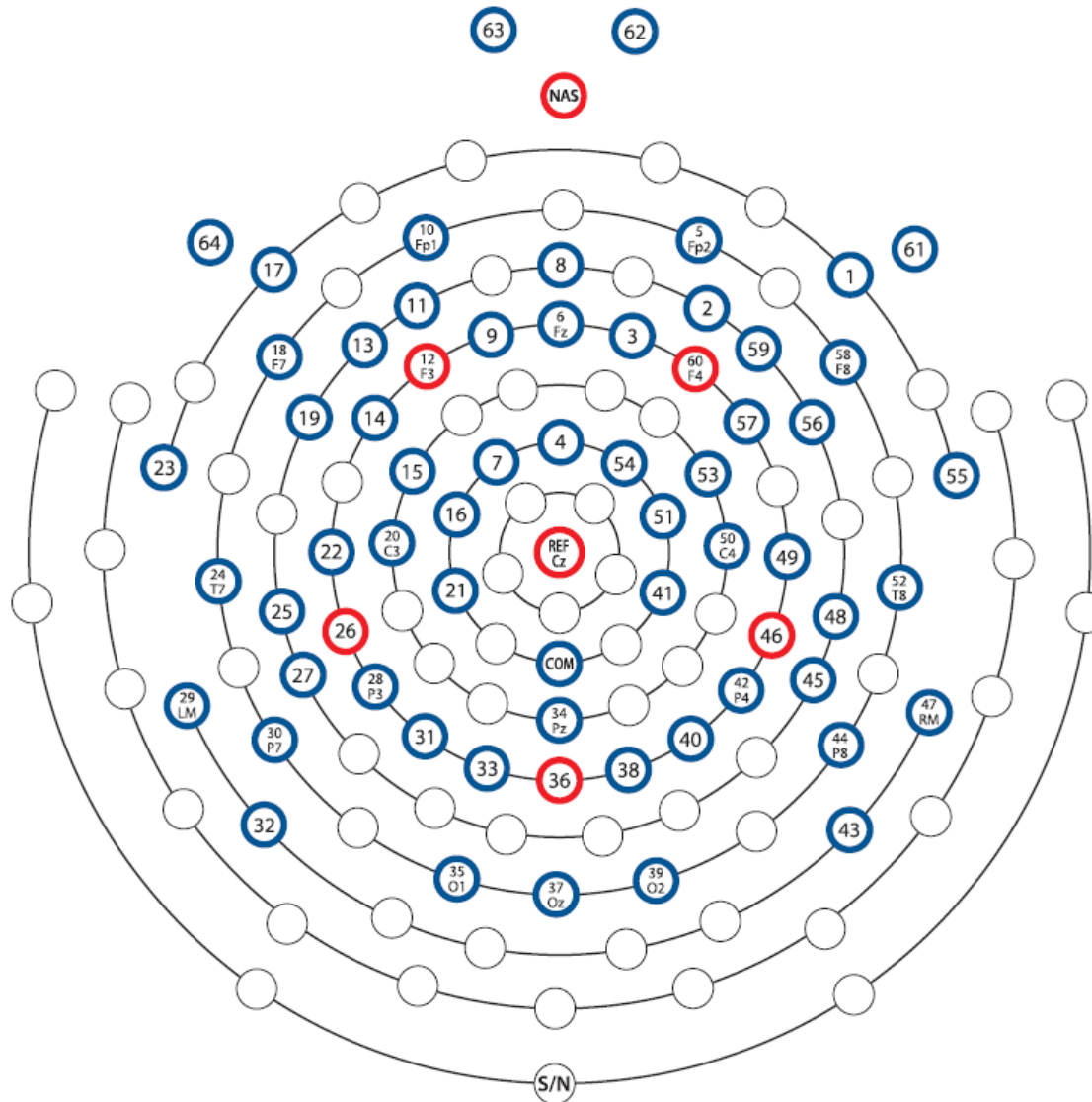


Standard Research EEG
32-64 Electrodes

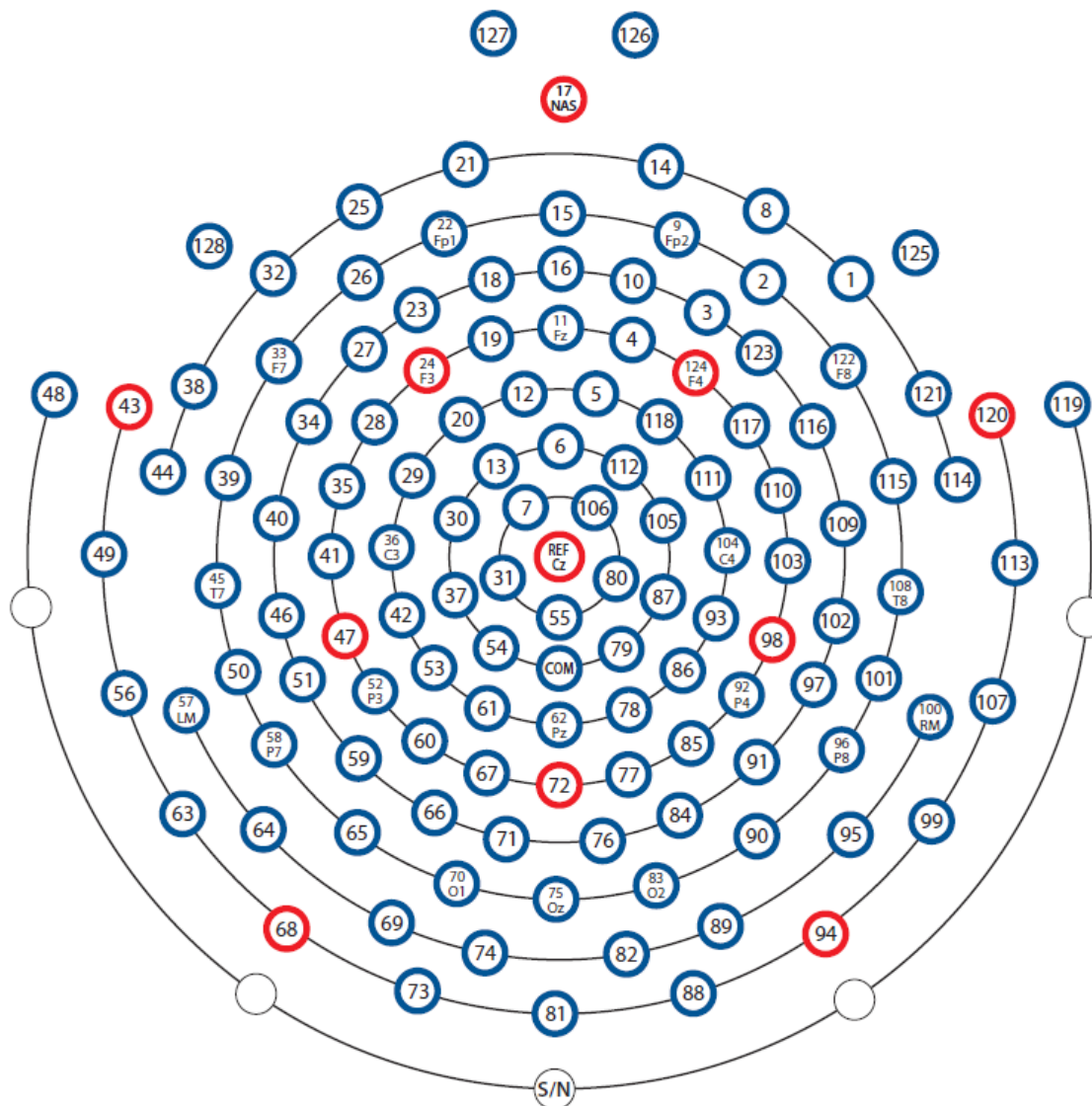


High-Density EEG
128-256 Electrodes

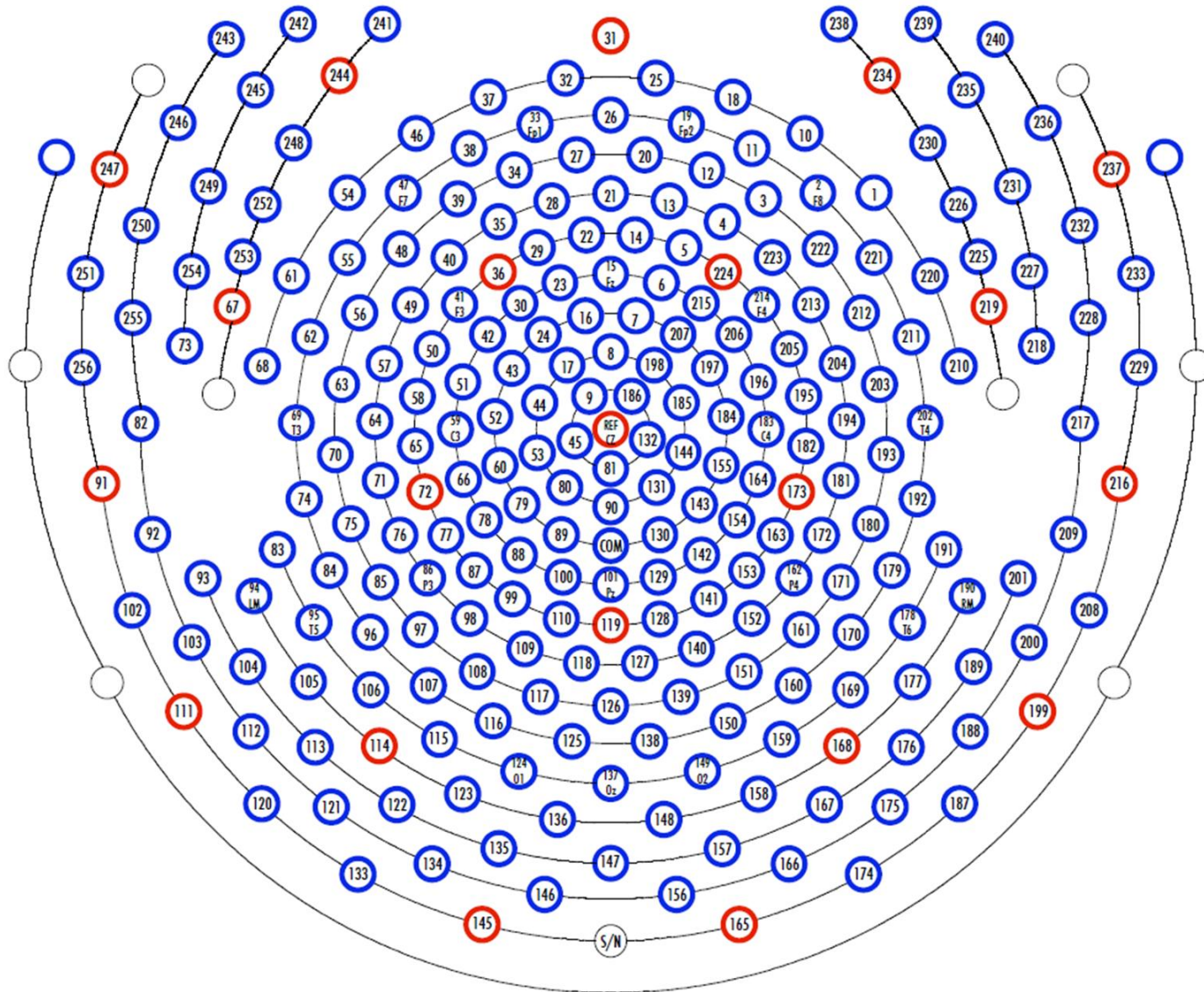
64 Electrodes + Reference Electrode



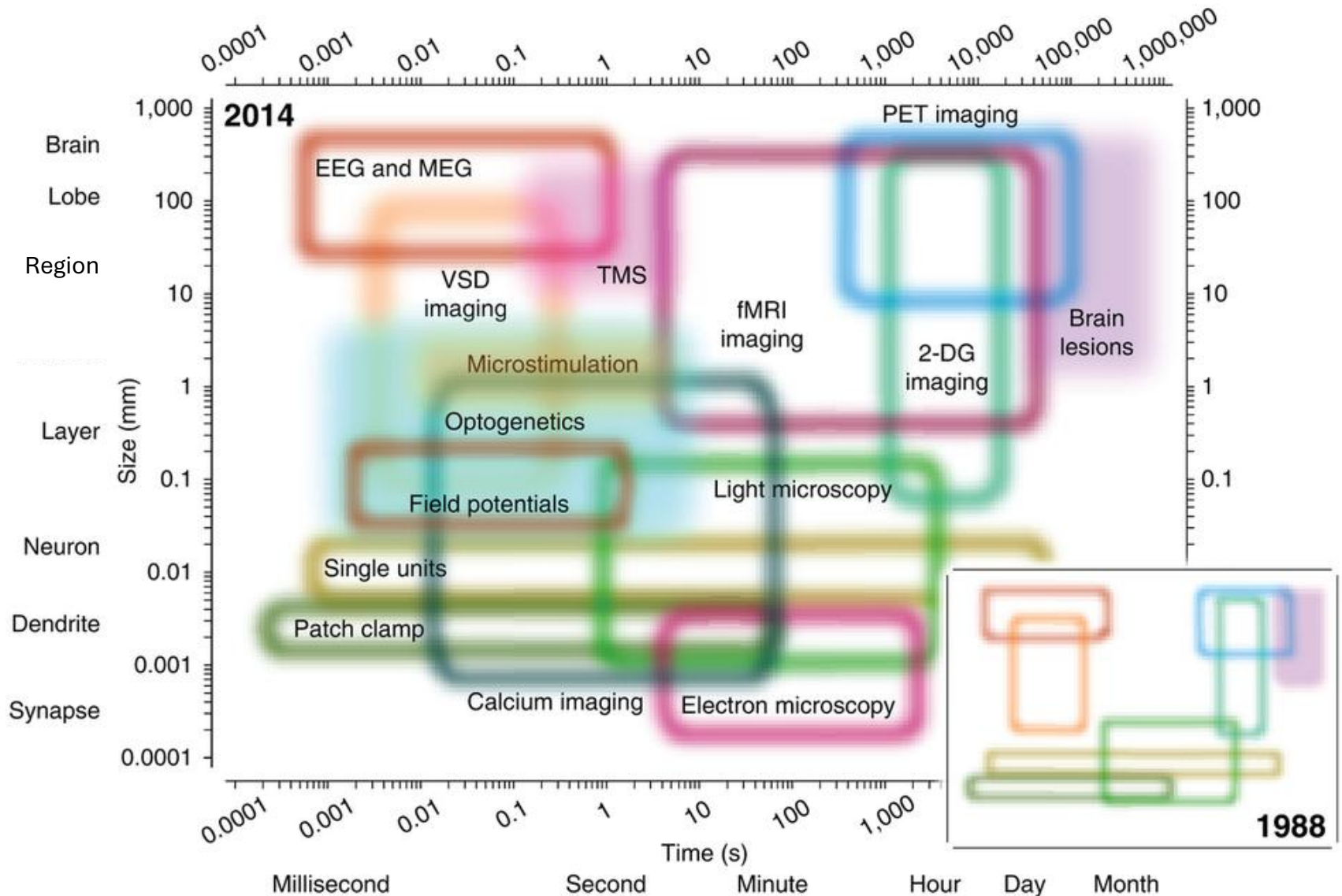
128 Electrodes + Reference Electrode



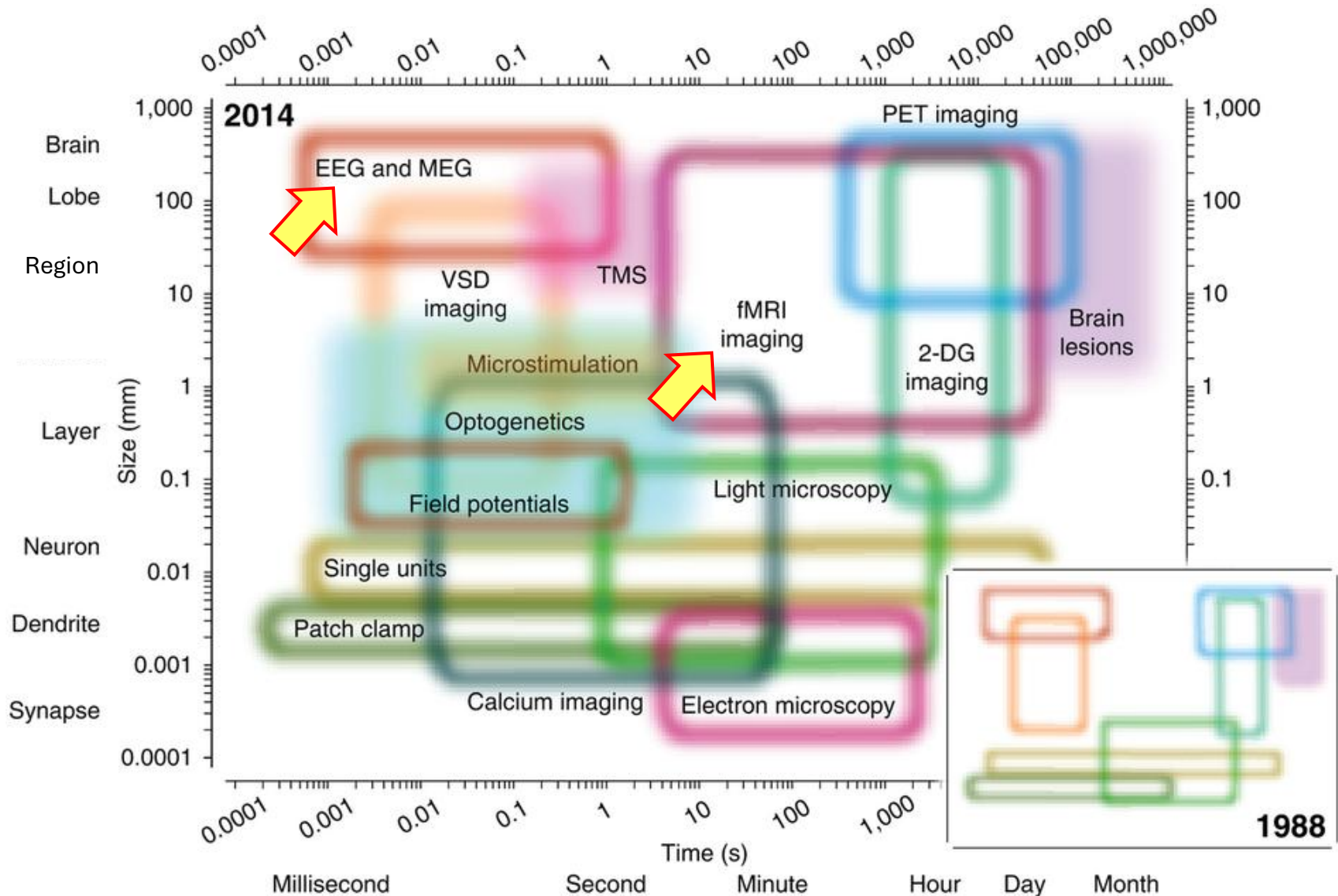
256 Electrodes + Reference Electrode



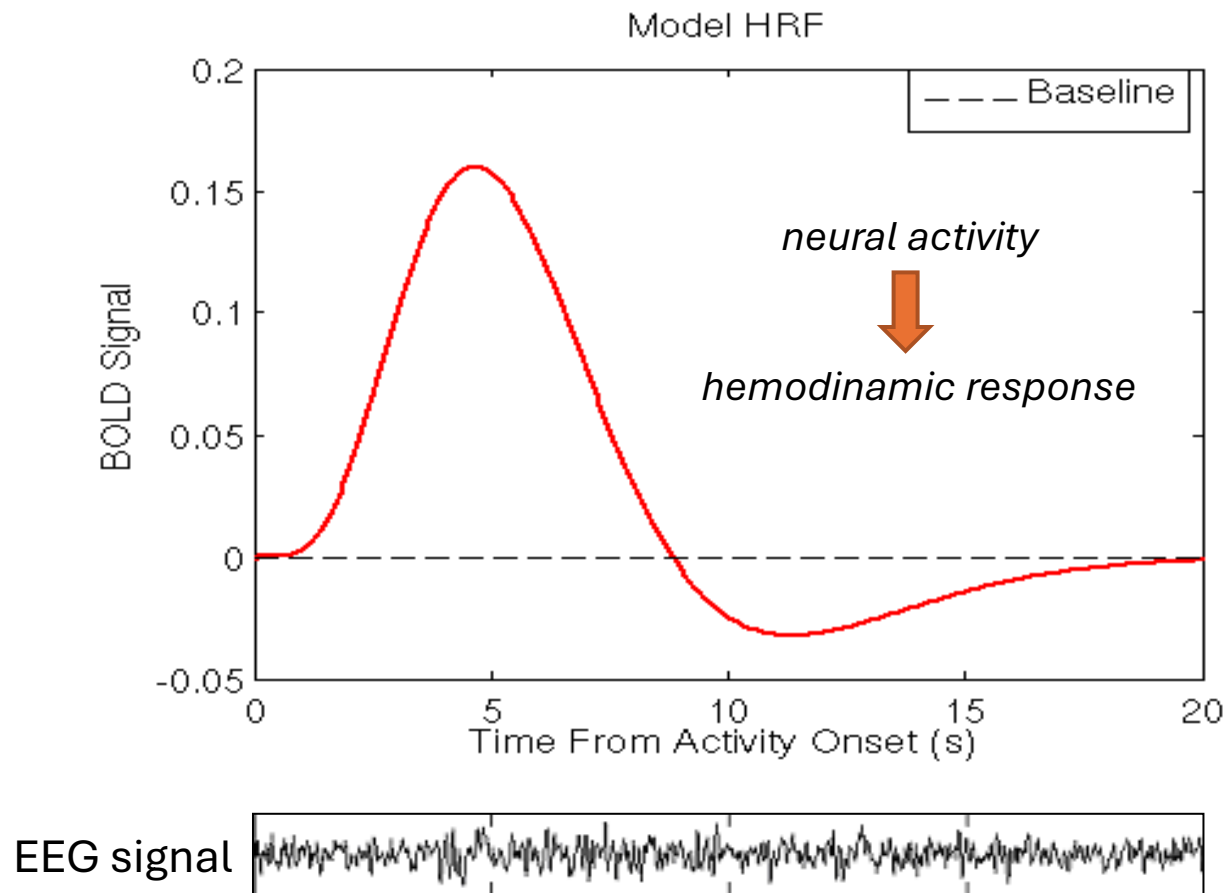
Temporal and Spatial Resolution



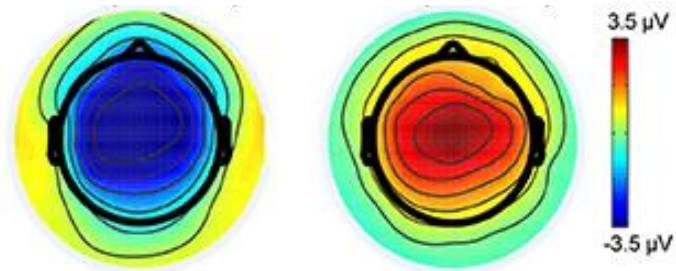
Temporal and Spatial Resolution



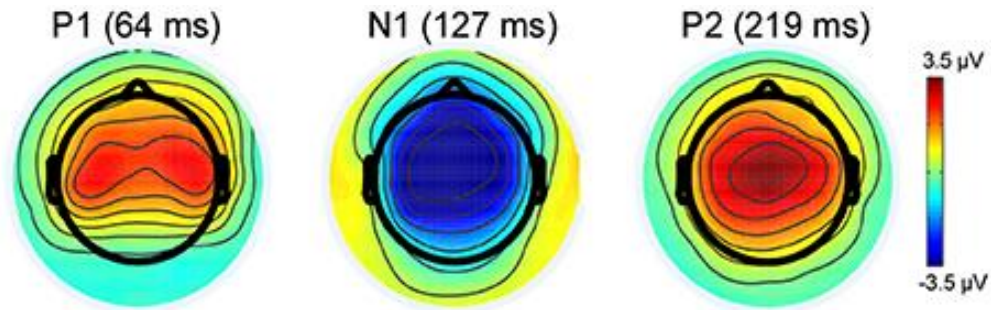
EEG Vs fMRI



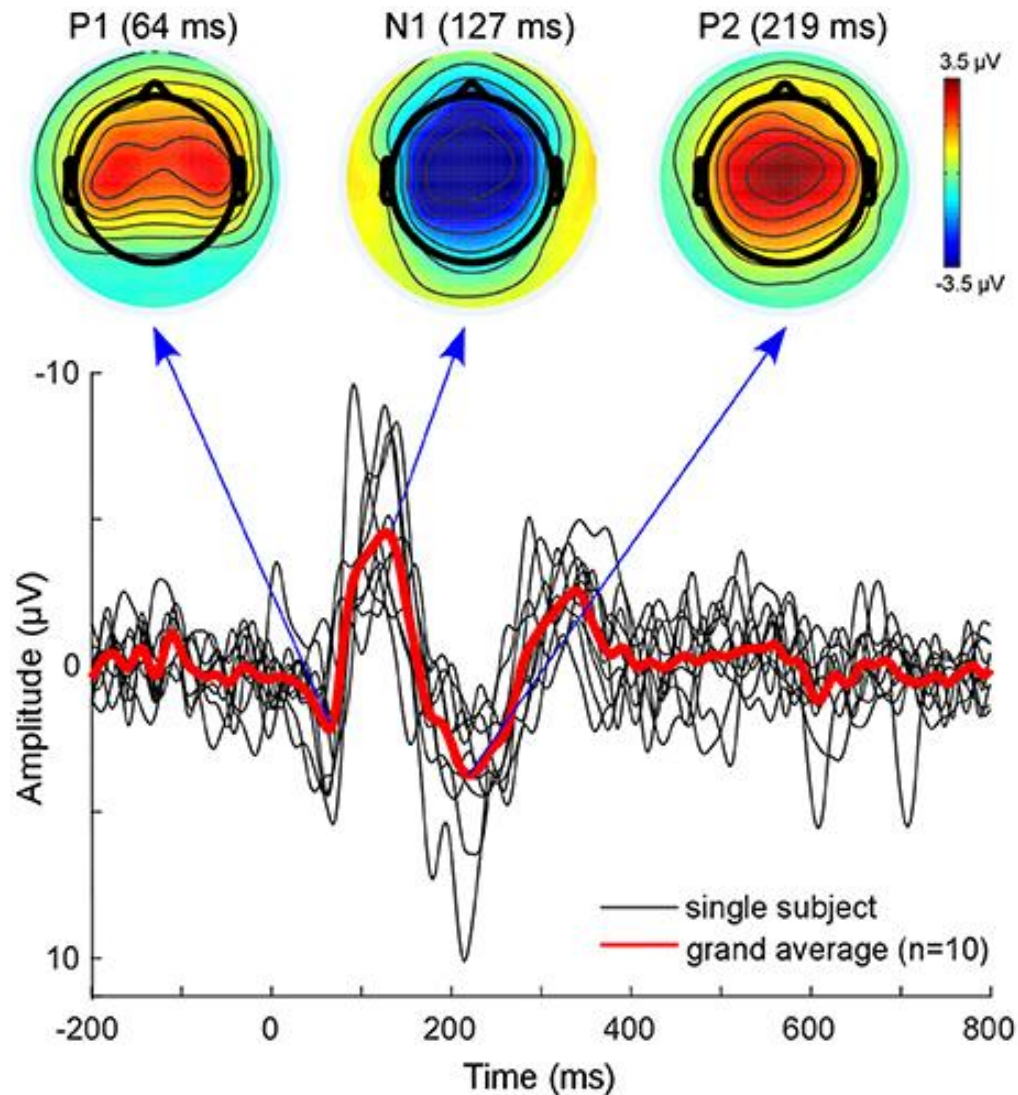
EEG spatial resolution



EEG spatial resolution

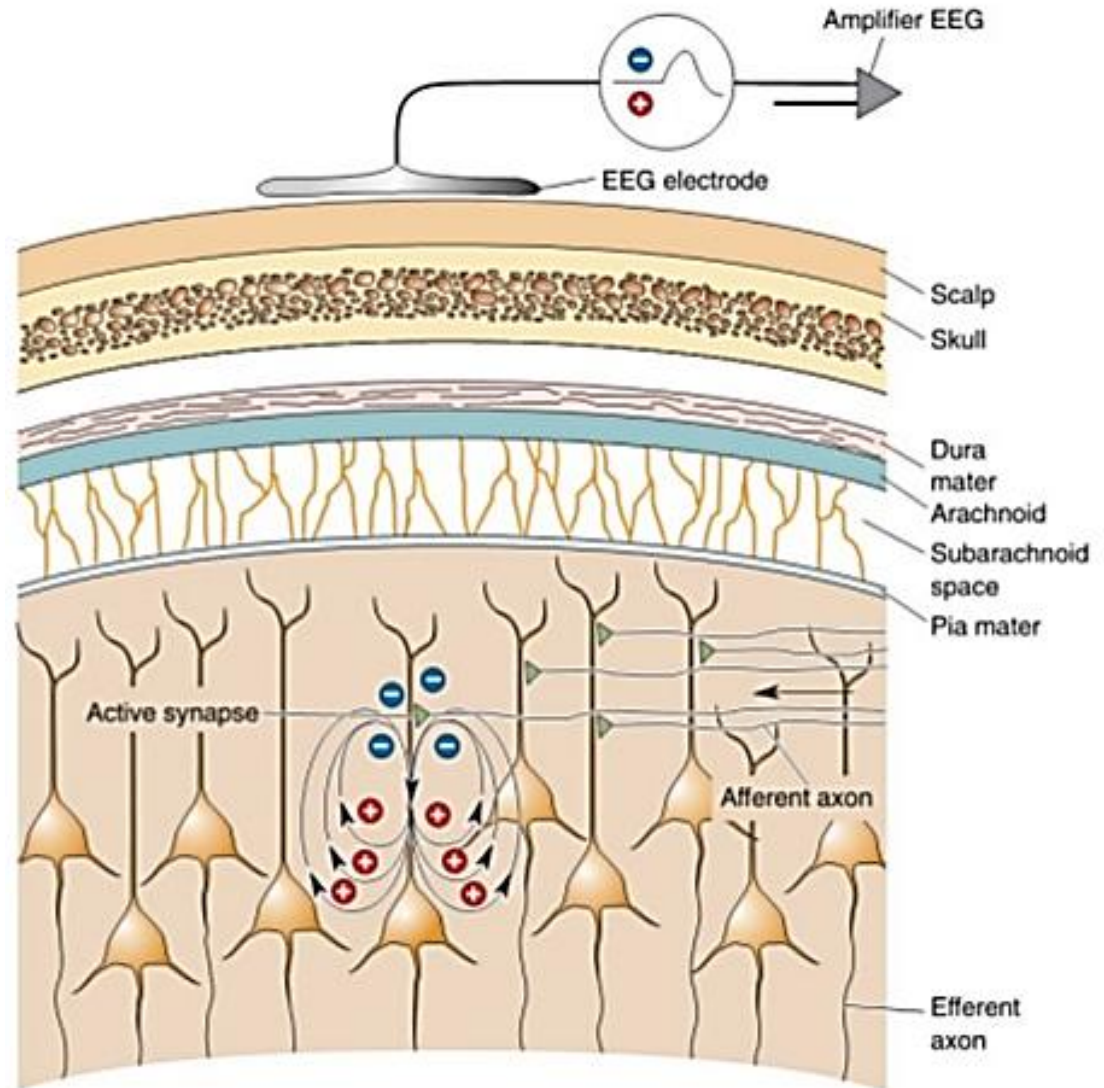


EEG spatial resolution



Physiological Basis of the EEG

- The EEG signal reflects the electric signal generated by the cooperative activity of brain cells (in particular pyramidal cells), or more precisely the **time course of extracellular FIELD POTENTIALS** generated by their synchronous action
- Only if thousand of cells contribute their small voltage is the signal large enough to reach the scalp surface.

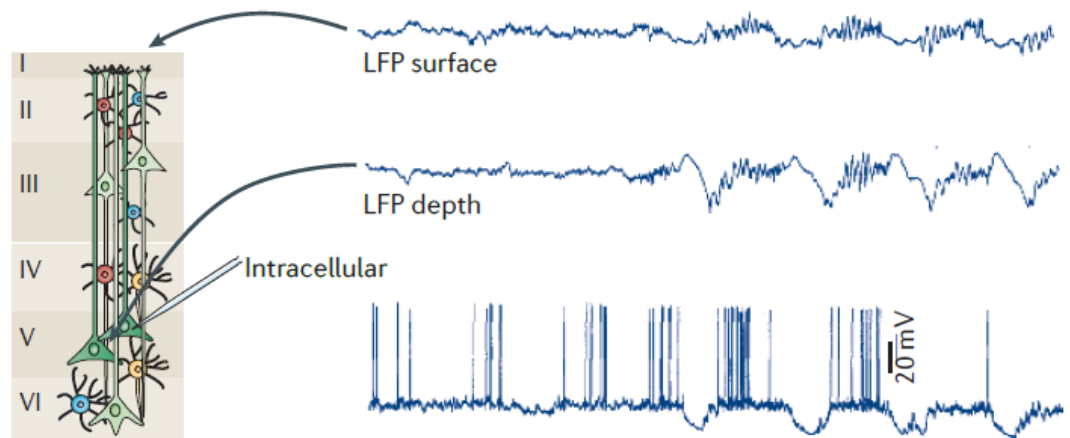
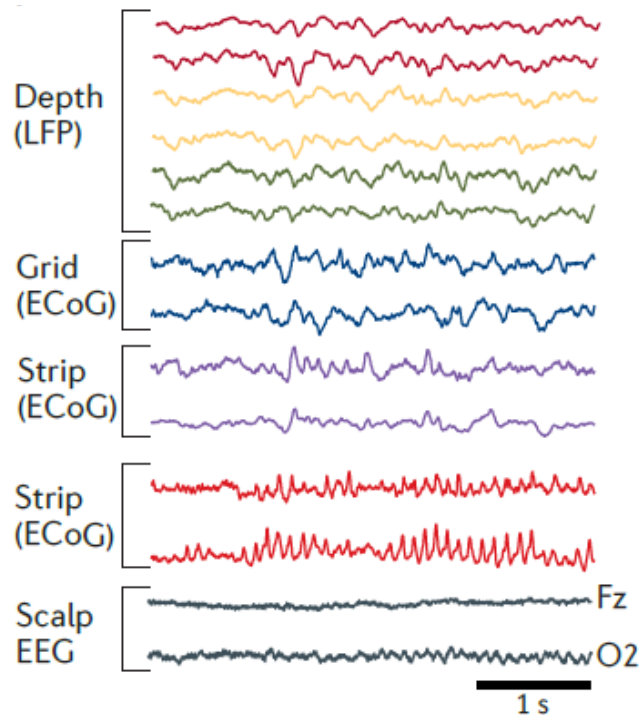


Contributors to the Extracellular Fields

- Any excitable membrane, whether it is a spine, soma, axon or axon terminal, contributes to the extracellular field and any transmembrane current leads to extracellular voltage deflections
- The extracellular field is the superposition of all ionic processes, from fast action potentials to the slowest fluctuation in the glia cell (post-synaptic potentials, glial currents and ionic concentration shift)
- The characteristic of the LFP (Local Field Potentials) waveform we measure at each point, such as amplitude and frequency content, depend on many factors:
 - the **proportional contribution of the multiple sources**
 - the **various properties of the brain tissues**
 - the **temporal coordination of the respective current sources (that is, their synchrony)** shapes the extracellular field

Characteristic of the Extracellular Fields

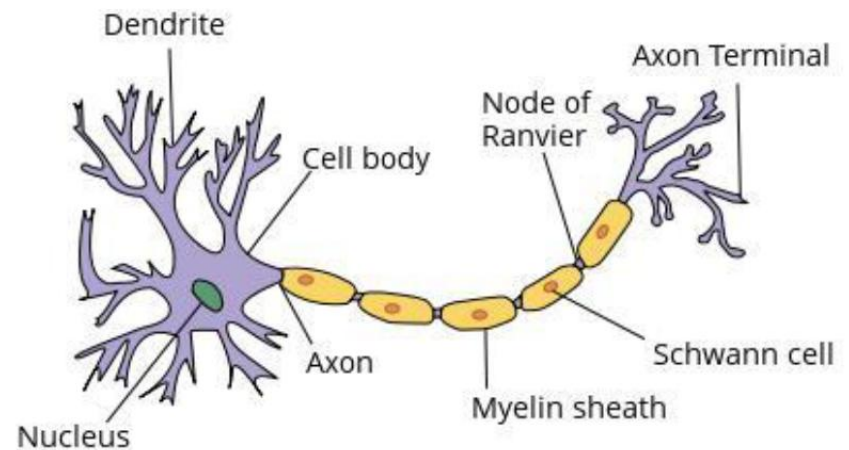
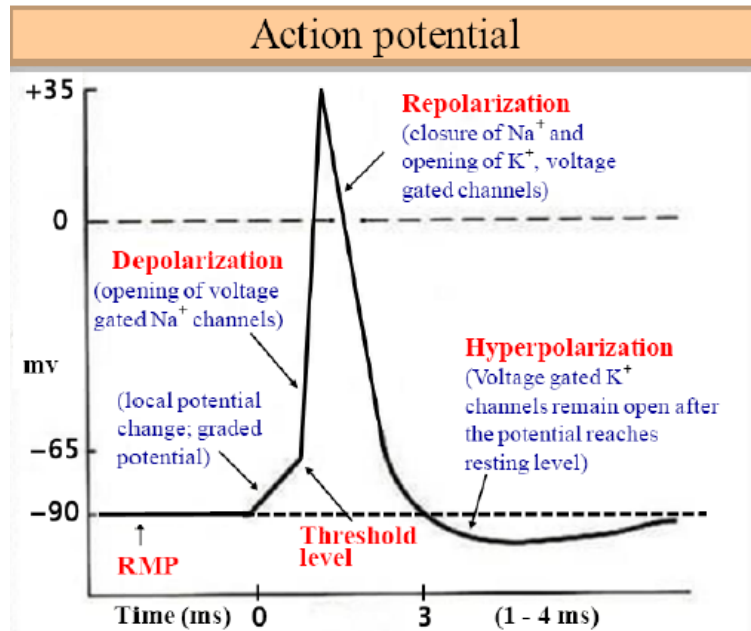
- The larger the distance of the recording electrode from the current source, the less informative the measured LFP becomes about the events occurring at the location of the source
- This is mainly owing to the fact that the V_e amplitude scales with the inverse of the distance r between the source and the recording site, and to the inclusion of other (interfering) signals (leading to ‘spatial averaging’).



RIGHT: Simultaneous recordings from three depth electrodes (amygdala and hippocampus), a 3x8 subdural grid electrode array placed over the temporal cortex, two strips placed under the inferior temporal surface, a strip placed over the left obifrontal surface, and scalp electroencephalography

LEFT: Simultaneous recorded LFP traces from superficial and deep layers of the motor cortex in an anaesthetized cat and intracellular trace from a layer 5 pyramidal neuron

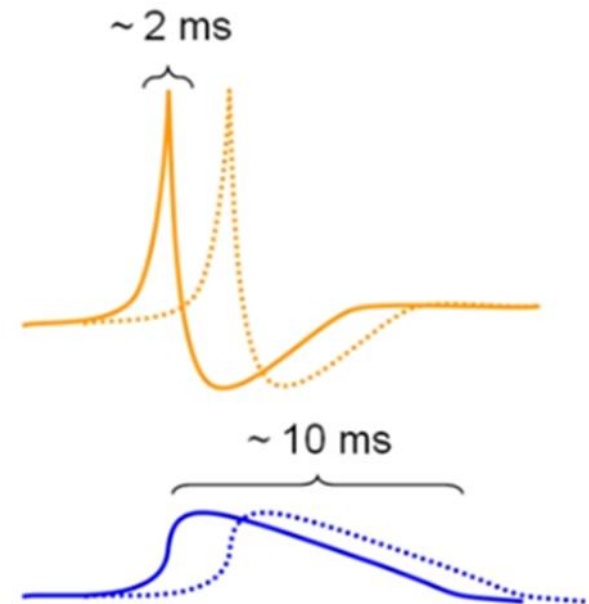
Sources of the Extracellular Fields



- The **action potential** consists of rapid change in membrane potential such that intracellular potential suddenly jumps from negative to positive and quickly (**1-2 ms**) returns to the resting intracellular negativity. In this way, an impulse is generated that has the remarkable property of propagating along axons without any loss of amplitude.
- Extracellular currents from many individual compartment must overlap in time to induce a measurable signal, and such an overlap is most easily achieved for relatively slow events.

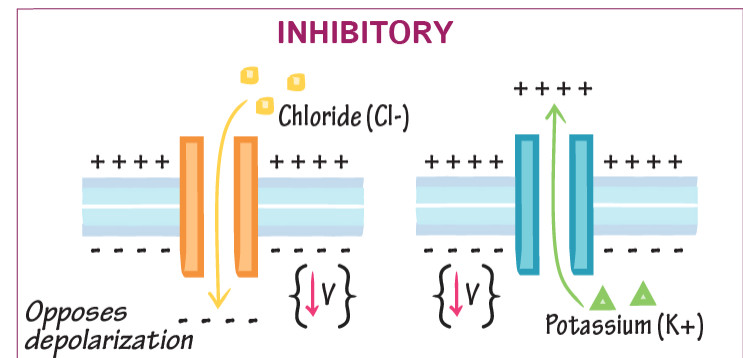
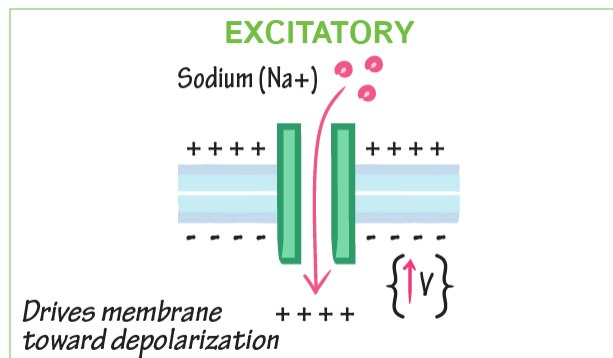
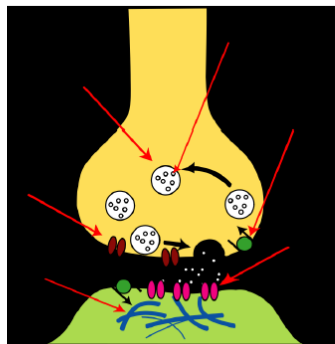
Sources of the Extracellular Fields

- Although Na⁺ spikes generate large amplitude voltage deflection near the soma, they are thought not to contribute substantially to the traditionally considered LFP band (<100 Hz) or to the scalp-recorded EEG, because the stronger fields they generate are of short duration and nearby neurons rarely fire synchronously in such short time window.
- Action potentials are biphasic - not ideal for summation due to cancellation
- Postsynaptic potentials are monophasic and slower – ideal for summation

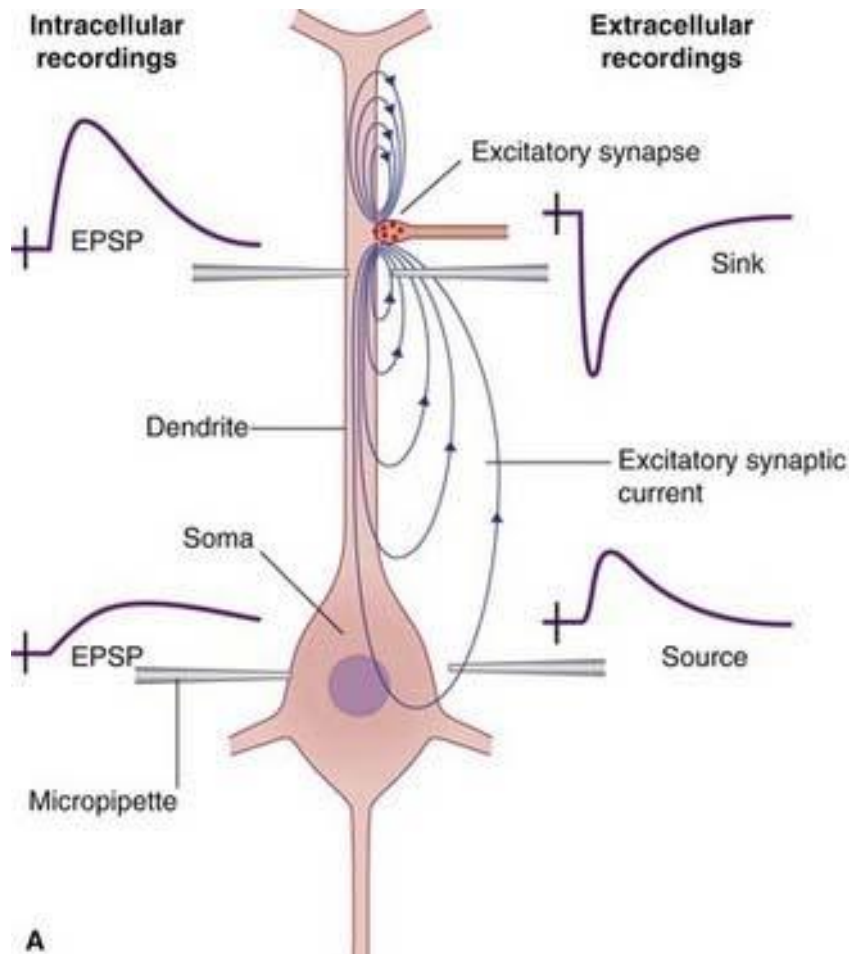


Sources of the Extracellular Fields

- **Graded potentials** are changes in the membrane potentials, arising in dendrites and soma, mainly induced by the synaptic activity whose magnitude is variable, is proportional to the magnitude of the stimulus and decreases with distance from the point of generation
- Regarding the slower **postsynaptic potentials** two main types can be distinguished:
 - If an action potential travels along a fiber which ends in an excitatory synapse an **EXCITATORY POST SYNAPTIC POTENTIALS (EPSP)** occurs in the following neuron. The excitatory synapse induces an **influx of positive Na^+ ions at apical dendrites causes DEPOLARIZATION OF THE POSTSYNAPTIC CELL.**
 - If an action potential travels along a fiber which ends in an inhibitory synapse an **INIBITORY POST SYNAPTIC POTENTIALS (IPSP)**. The inhibitory synapse induces an **influx of negative Cl^- ions causes HYPERPOLARIZATION OF THE POSTSYNAPTIC CELL.**



Sources of the Extracellular Fields

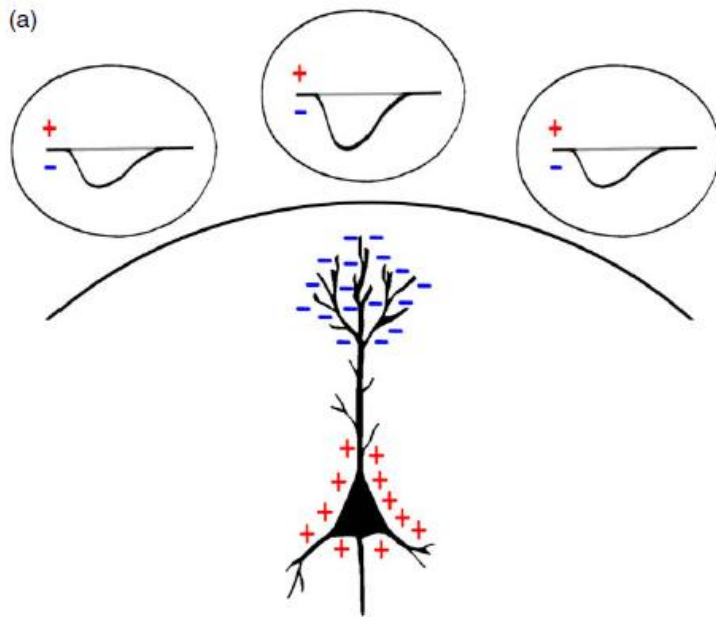


- The influx of cations from the extracellular into the intracellular space accompanying the **EPSP** gives rise to a local extracellular active **SINK** (= by convention a site on the neuronal membrane where positive charges enter the neuron).
- As a consequence a potential gradient develops all along the cellular membrane both in the intra- and extracellular space.
- This gradient gives rise to secondary compensating ionic current along the membrane that flow in opposite directions in the intracellular and extracellular spaces. (these current)
- The extracellular sink is compensate for the sake of electroneutrality by an extracellular **SOURCE** (= a site along the membrane where positive charges flow out of the neuron)/(distributed passive sources along the soma-dendritic membrane)
- This active extracellular SINK-SOURCE configuration form an electric DIPOLE in the extracellular medium around the neuron

Sources of the Extracellular Fields

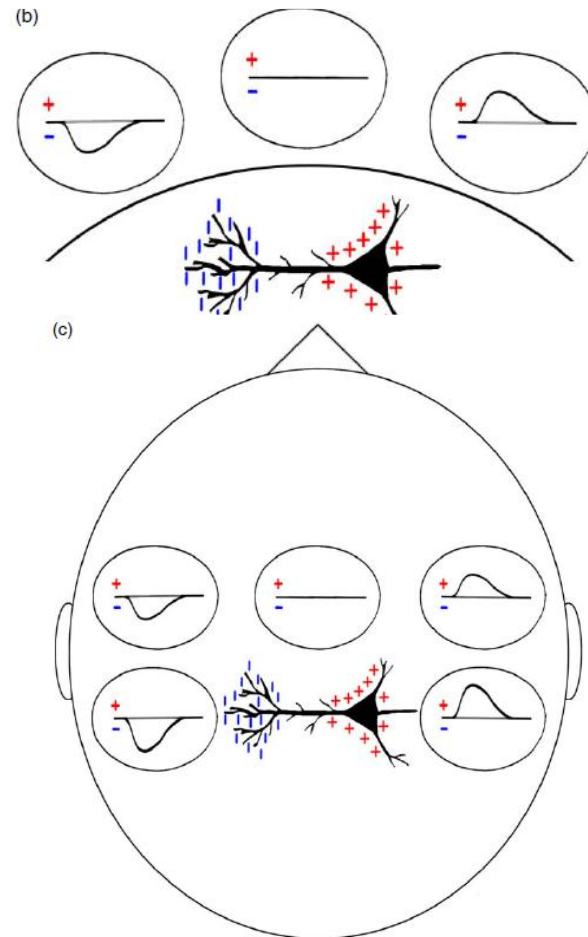
RADIAL DIPOLES:

Oriented perpendicular to the scalp surface



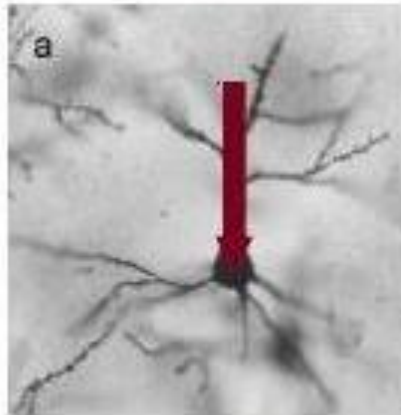
TANGENTIAL DIPOLES:

Oriented parallel to the scalp surface



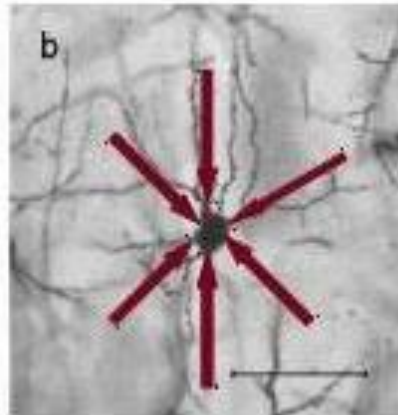
Importance of cell geometry

- The relative contribute to the extracellular field of each neuron depends on the shape of the cell.
- Pyramidal cells have long thick apical dendrites that can generate strong dipoles along the somato-dendritic axis. Such dipoles give rise to an **OPEN FIELD**, as there is considerable spatial separation of the active sink (or source) from the return currents, inducing substantial ionic flow in the extracellular medium.



OPEN FIELD:

Dipoles are parallel and aligned



CLOSED FIELD:

Dipoles are arranged in spherical geometry

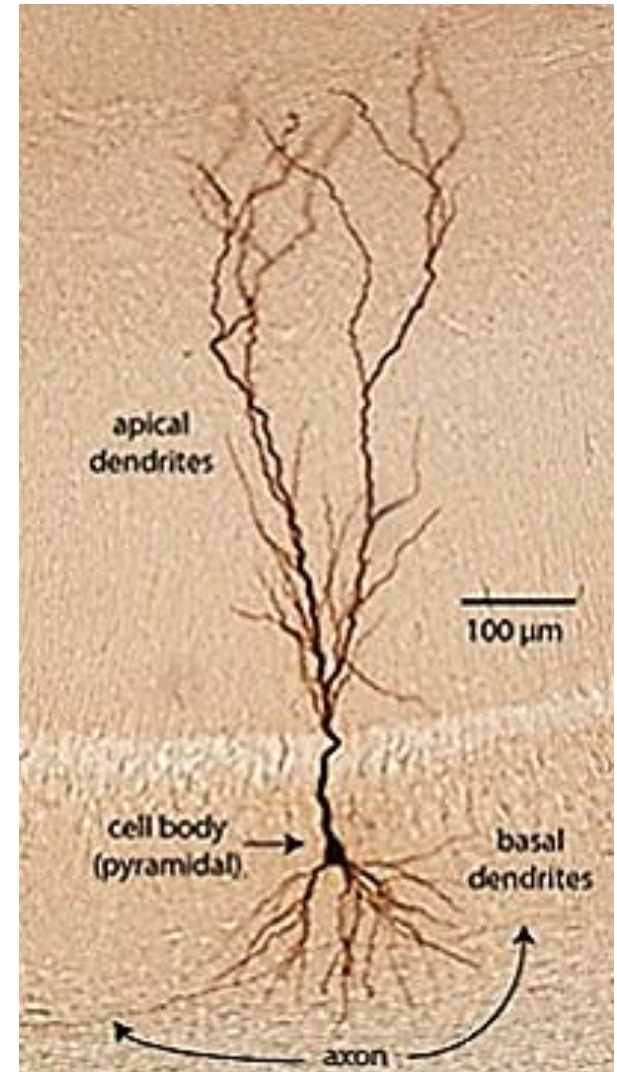
- Neurons that generate open fields make a sizeable contribution to the extracellular. By contrast spherically symmetric neurons that emanates dendrites of relative equal size in all direction can give rise to a **CLOSE FIELD** (when all dendrites are simultaneously activated)

Pyramidal Neurons

Pyramidal cells are found in **forebrain structures** such as the cerebral cortex, hippocampus, and amygdala, but not in the olfactory bulbs, striatum, midbrain, hindbrain, or spinal cord.

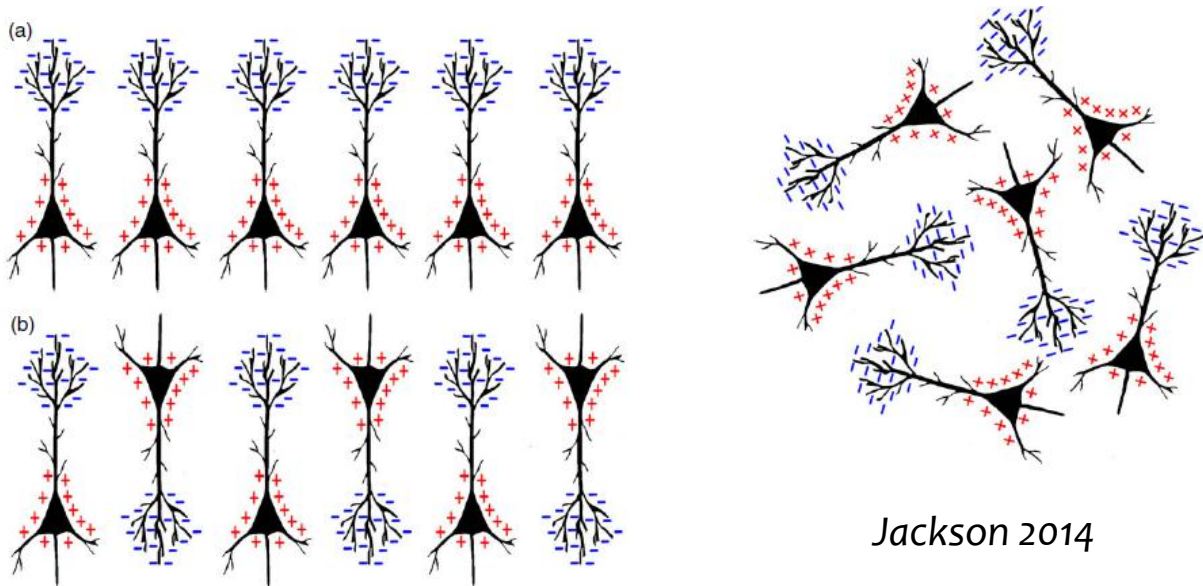
They are the most numerous **excitatory cell type** in mammalian cortical structures.

Pyramidal neurons have multiple dendrites (input) and a single axon (output), but both dendrites and axons branch extensively.



Neuronal populations architecture

- *A single neuron's dipole is too small to be measured as far away as the scalp*
- *Many individual dipoles from a large number of neurons in a area must sum together in order to produce a dipole measurable at the scalp*
- *In order to produce a larger signal (effectively sum the potentials) neurons must be both (a) arranged in a parallel fashion and (b) synchronously active*



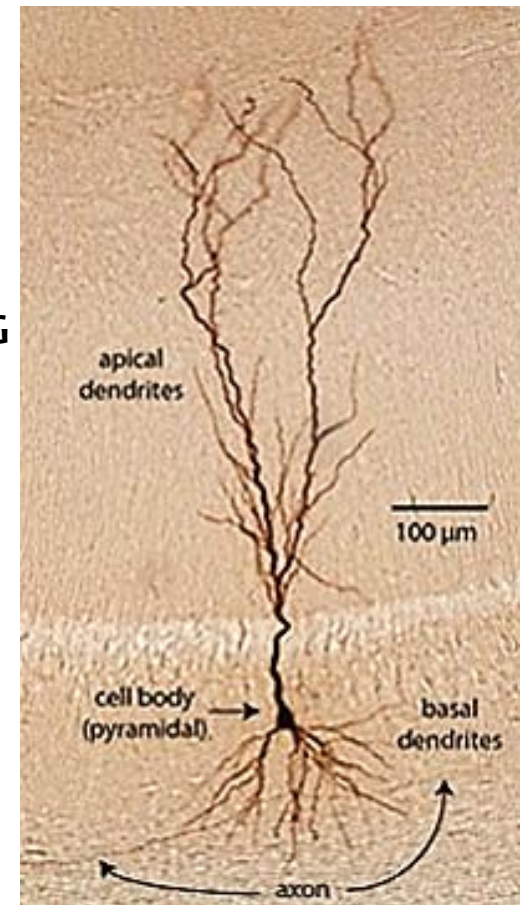
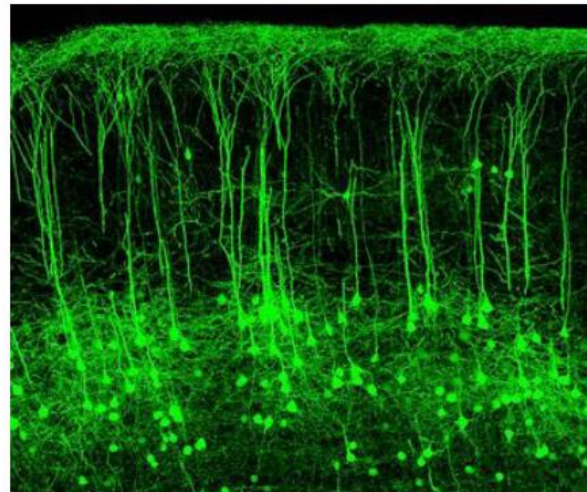
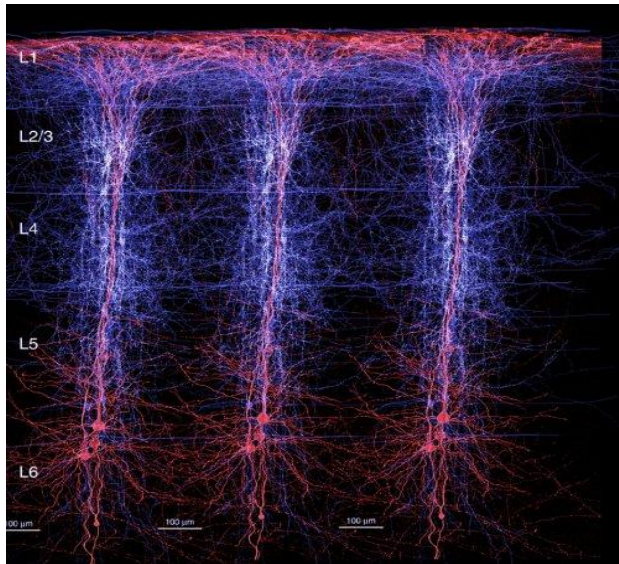
Jackson 2014

Neuronal populations architecture

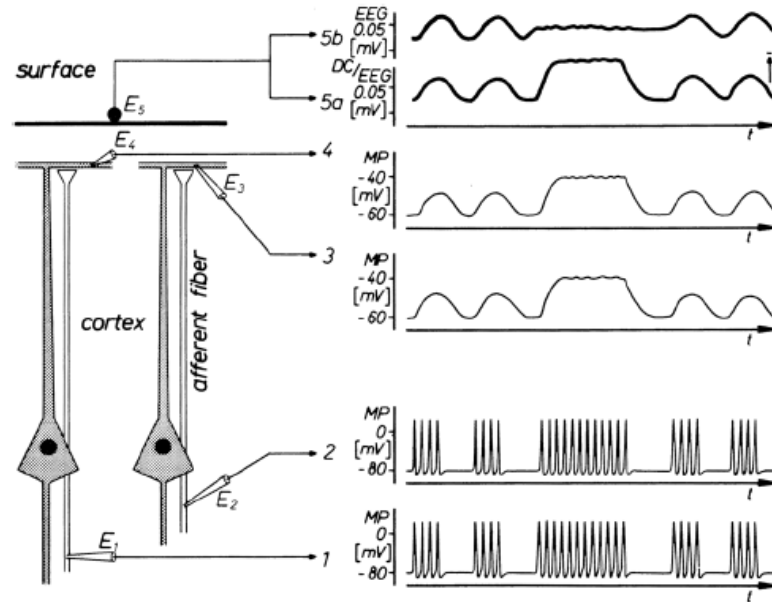
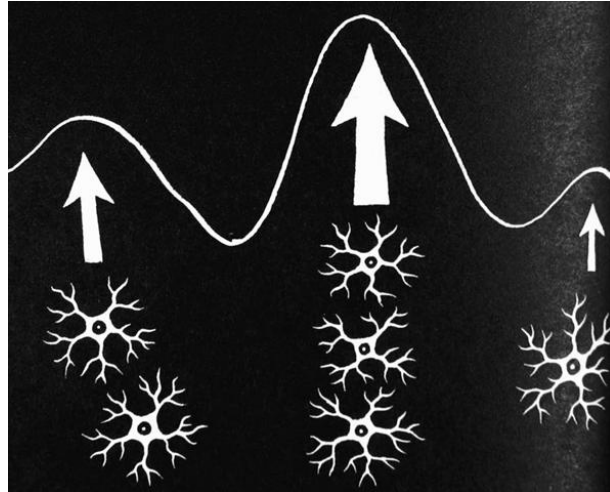
Layer II/III and V pyramidal cells are ideal as current generators because they are:

1. spatially aligned
2. perpendicular to the cortical surface
3. recurrently connected
4. receive synchronous inputs

A combination of thalamocortical and corticocortical connections are thought to drive the synchronous activity that gives rise to EEG



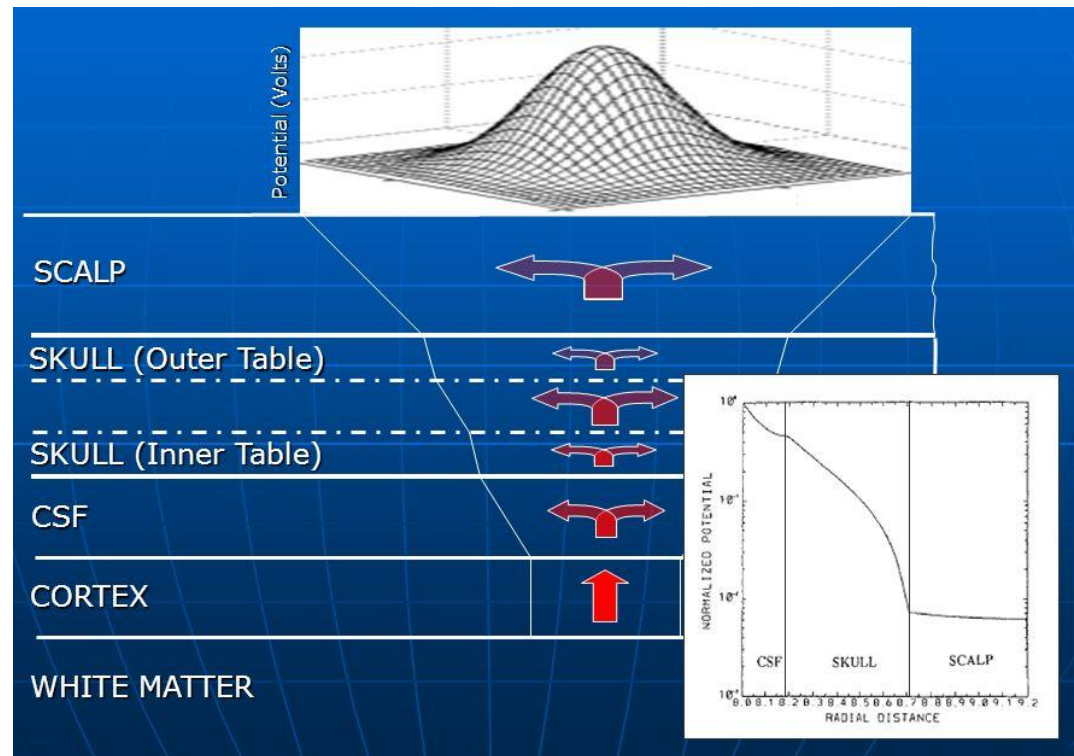
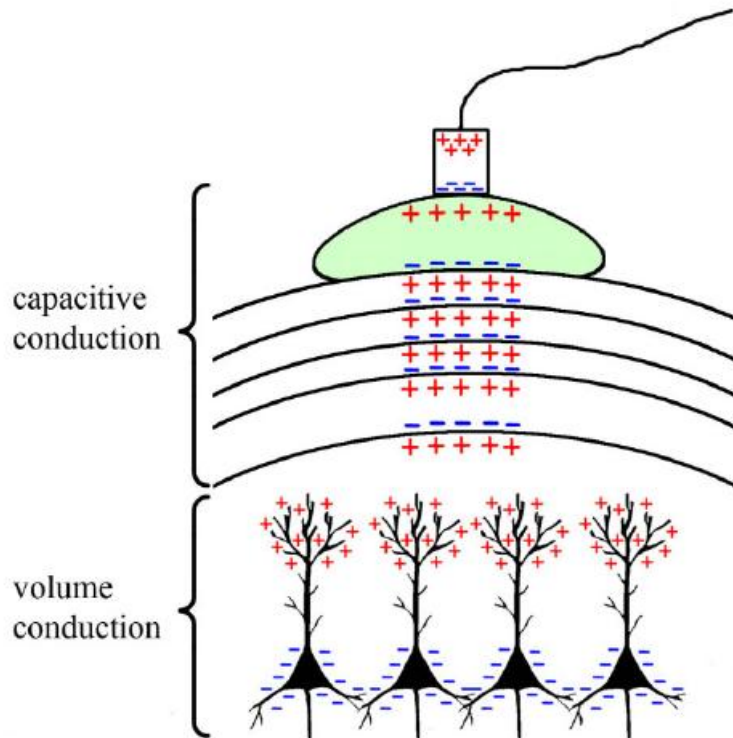
Wave Generation



- ✓ The amplitude of EEG oscillations correlates with the activity of cortical neurons
- ✓ Peaks correspond to moments of maximum synchronicity within oscillating neuronal populations (synchronous influx in afferent fibers)
- ✓ Periods of synchronization/de-synchronization correspond to changes in local cortical activities (periodic sequence of synchronous afferent burst)
- ✓ Waves coordinated between distant electrodes can indicate underlying brain networks and information flow

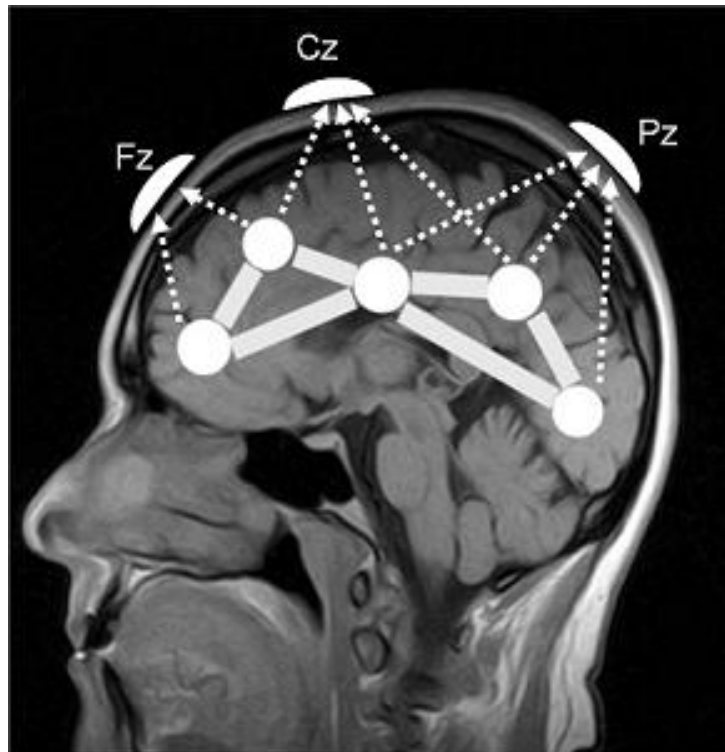
Volume Conduction

In bioelectromagnetism is defined as the transmission of electric and magnetic fields from an electric primary current source through biological tissue . In the low-frequency band (<1000Hz) considered the capacitive component of tissue impedance and the inductive effect can be neglected. The propagation can be thought ad quasi-instantaneous and frequency independent

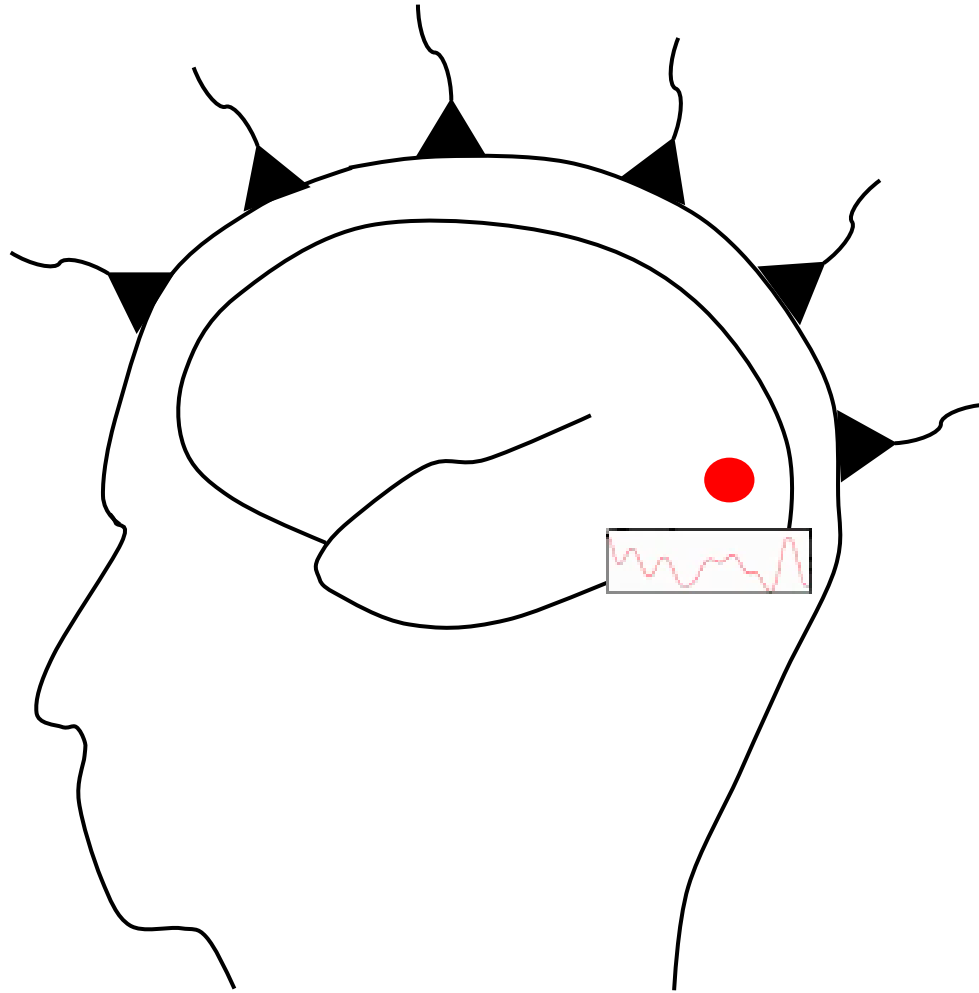


Volume Conduction

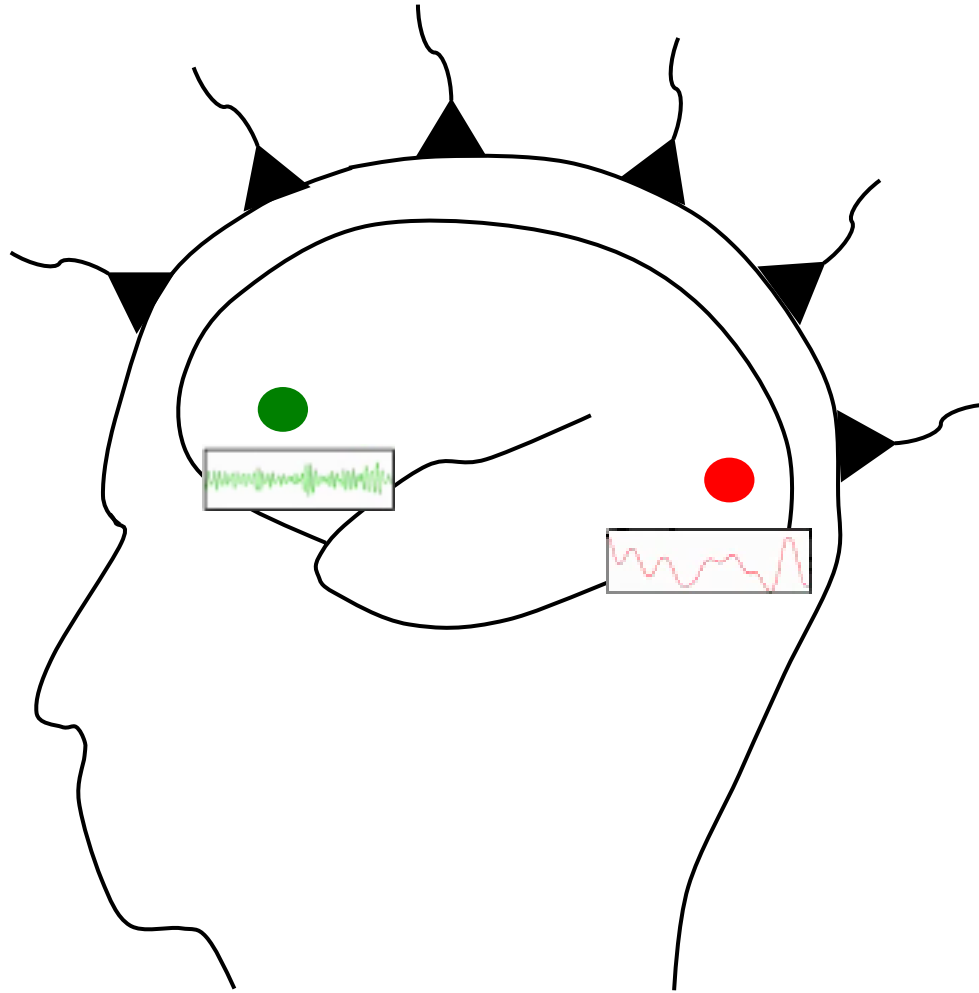
- Volume conduction is linear. Each sensor measures a (weighted) sum of each neuron's activity.
- Volume conduction in EEG may lead to excessive colinearity among EEG time series in sensor space (this is less of a problem for MEG).



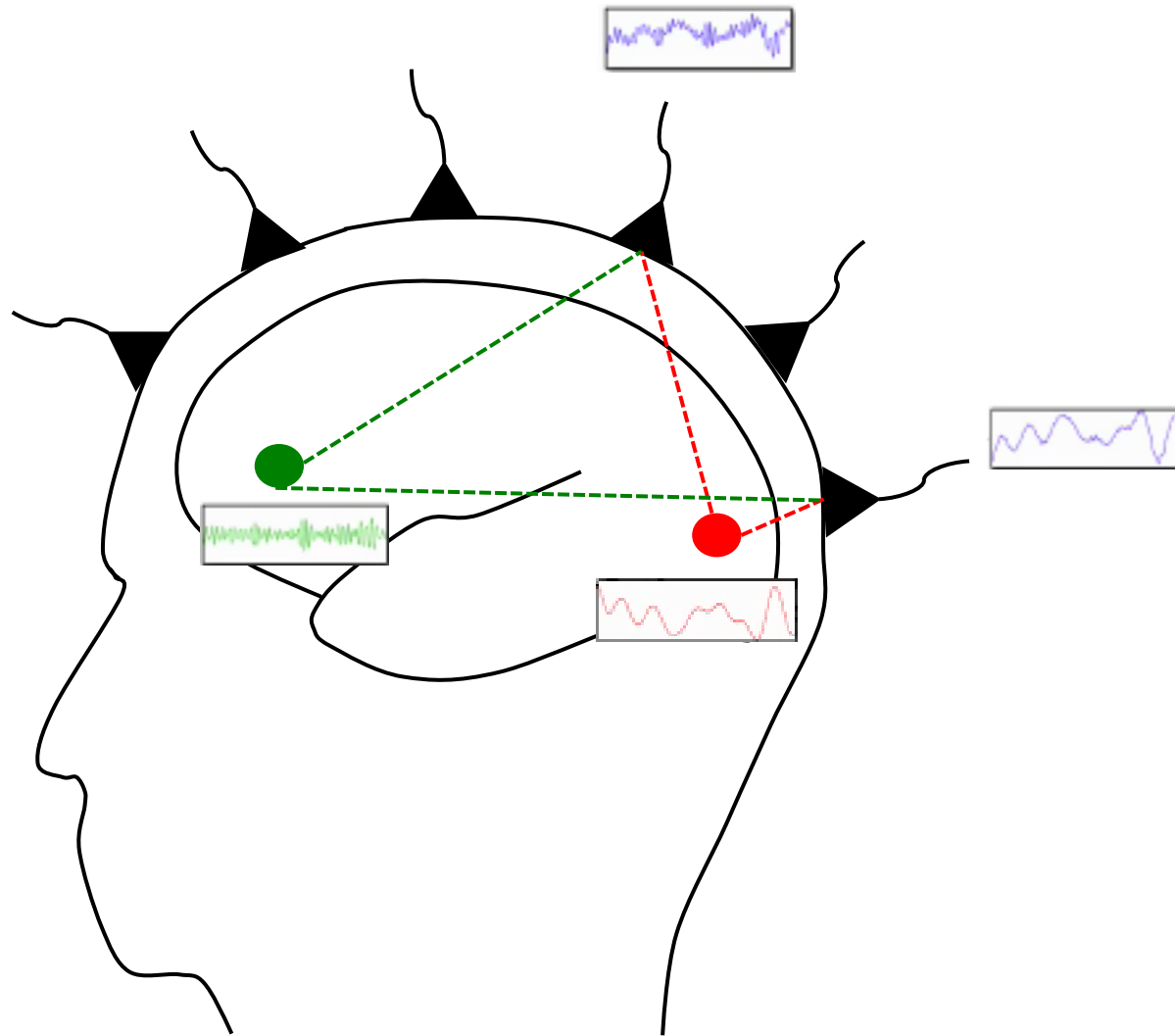
Superposition of Source Activity



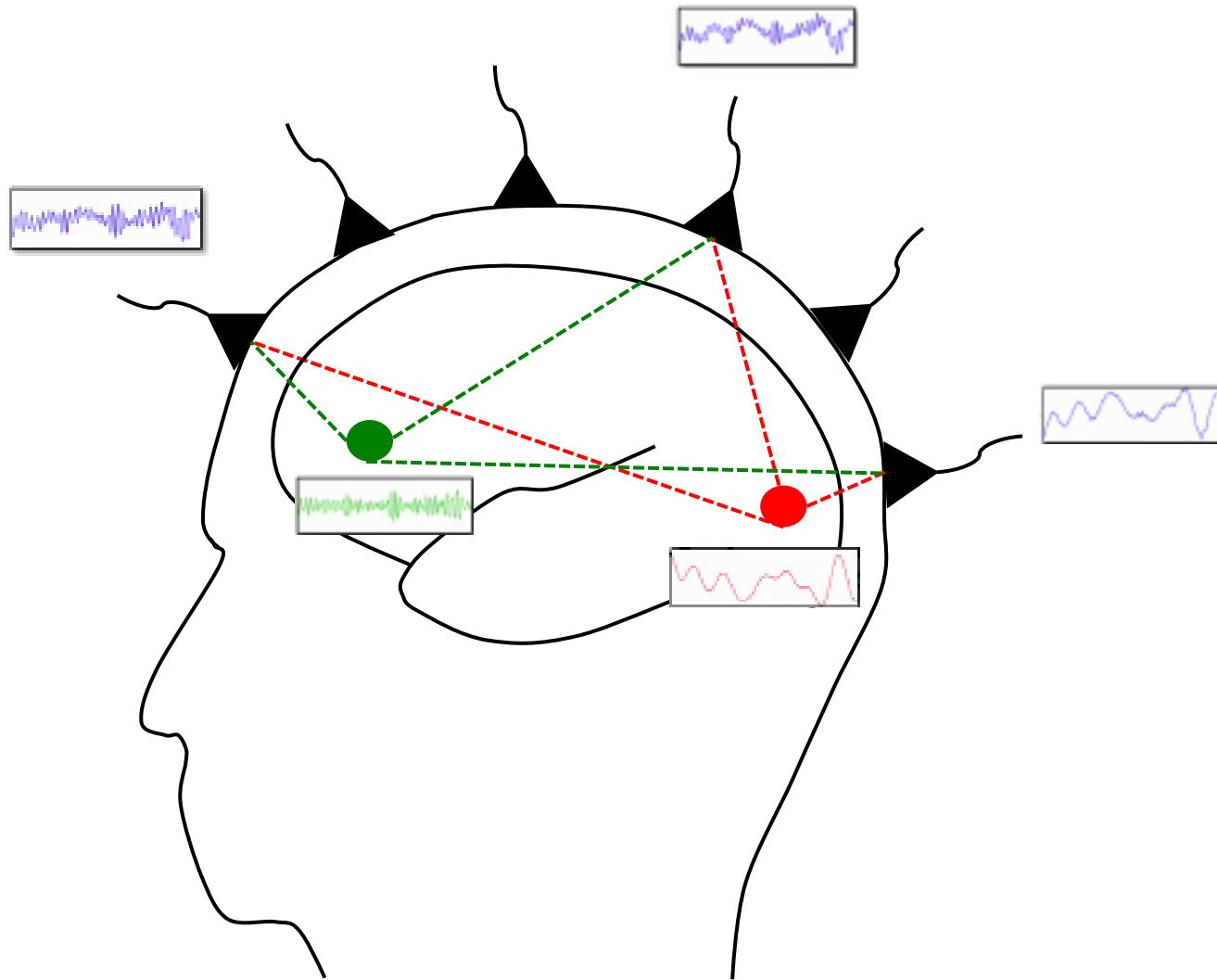
Superposition of Source Activity



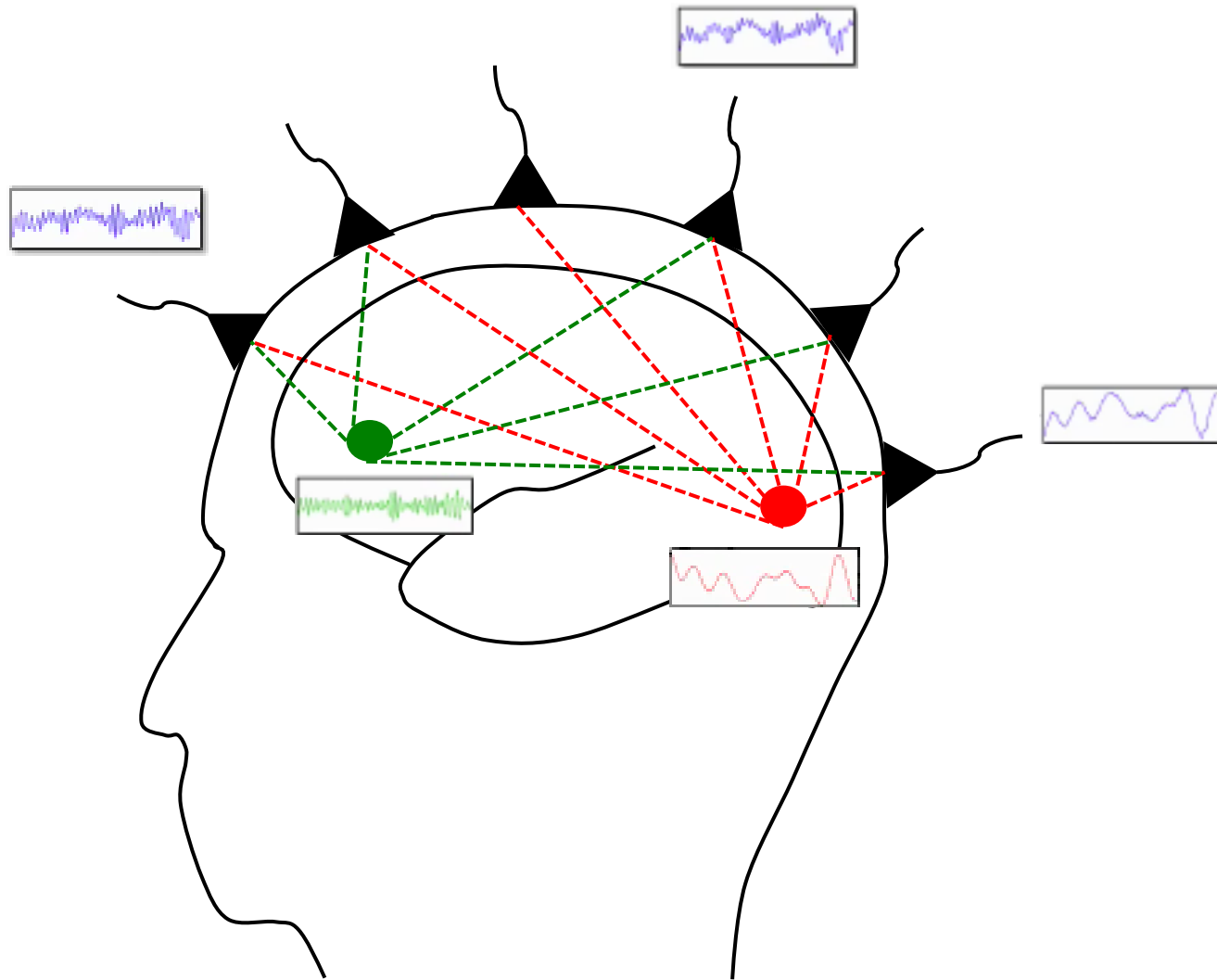
Superimposition of Source Activity



Superimposition of Source Activity



Superimposition of Source Activity



High-density EEG recording Systems

Hd-EEG systems include **128-256 electrodes**.

With **10-20 system** placement of conventional EEG, the electrodes are as far as **7 centimeters apart**. **High-density EEG** can reach an intersensor distance of **1-2 cm** and allow to cover areas of the head that are not covered in conventional EEG systems.

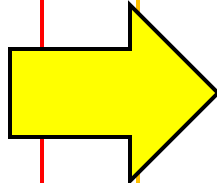
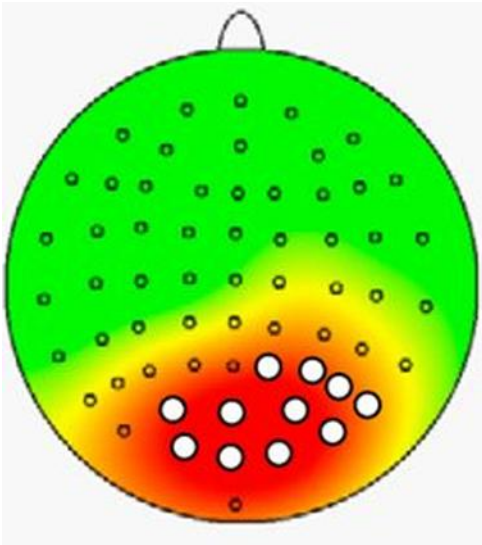
Improved spatial resolution, especially in combination with source reconstruction approaches.

Biophysical analyses have shown that information is lost unless an intersensor distance of 1 to 2 centimeters is achieved with EEG sampling. In fact, precision of source reconstruction methods is greatly increased if **more than 64 electrodes** are used.



Spatial resolution

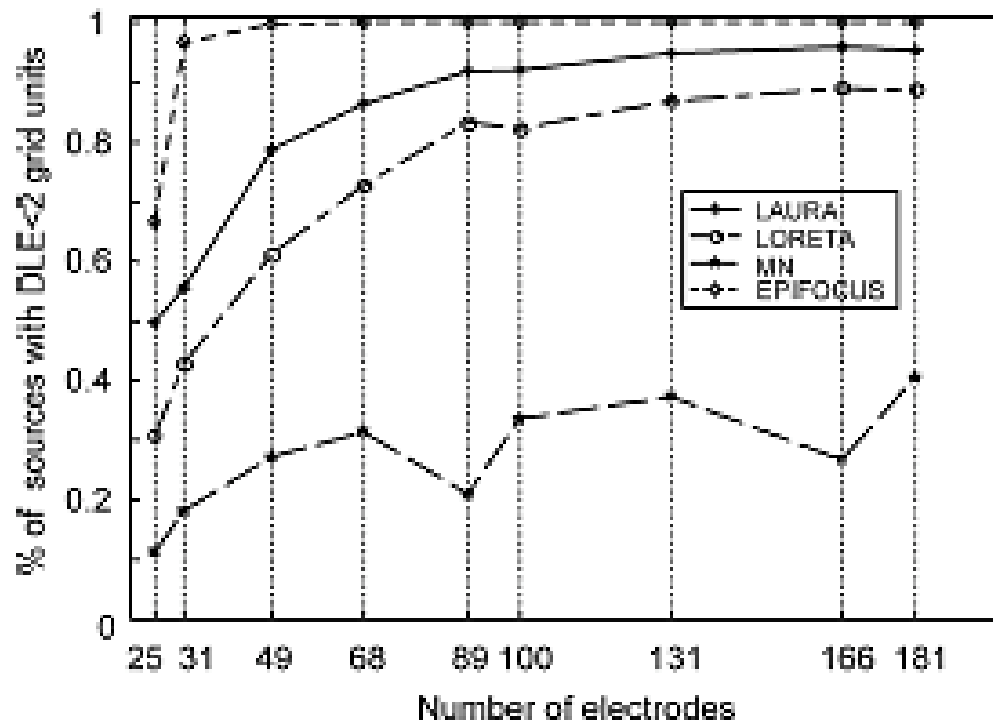
Scalp EEG



Source Localization



Spatial resolution



Accuracy of localization of a focus epilepticus (using source modelling)

Spatial resolution

- Theoretically
 - Bridging issues
 - Long preparation time

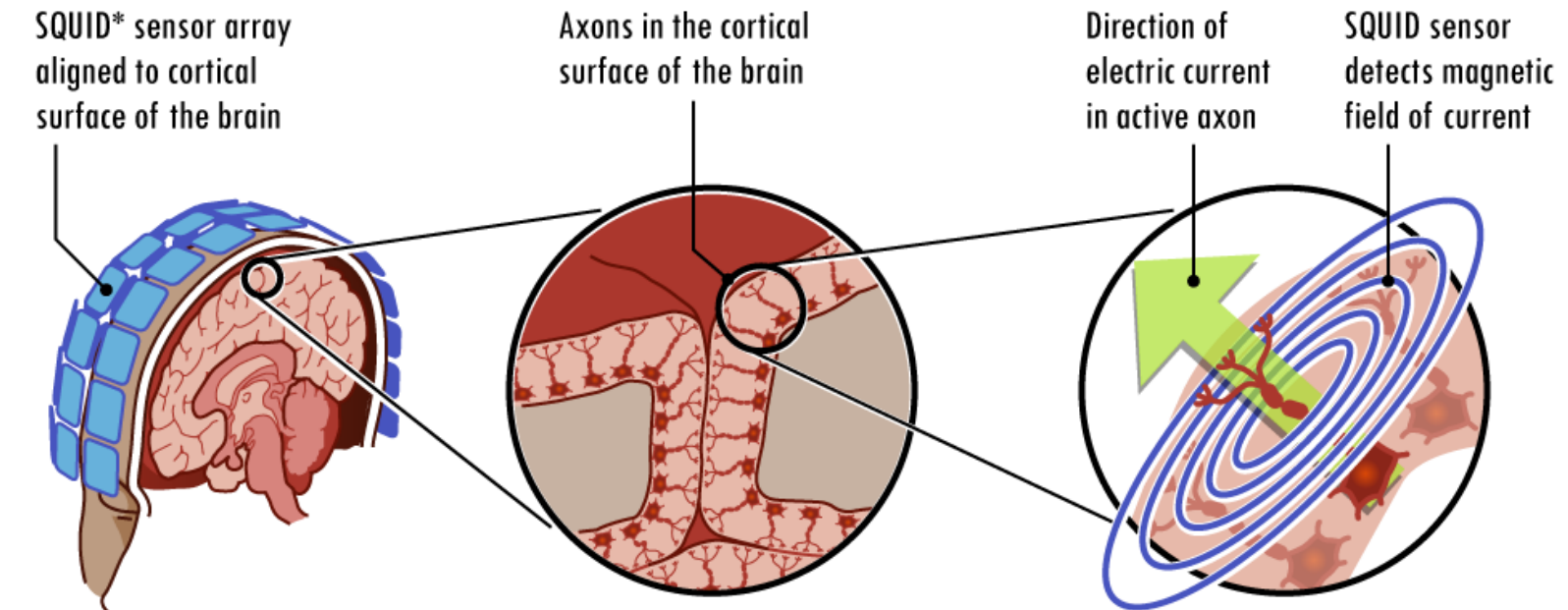
An electrical bridge occurs when two EEG electrodes are joined together by a low-impedance electrical connection (e.g., excessive use of gel). When electrodes are electrically bridged, the information in their respective channels will be nearly identical, distorting the EEG topography.

- But
 - Easier to identify and remove artifacts
 - More accurate source modelling reconstruction

EEG Vs MEG

Both the MEG and EEG signals derive from the net effect of ionic currents flowing in the dendrites of neurons during synaptic transmission.

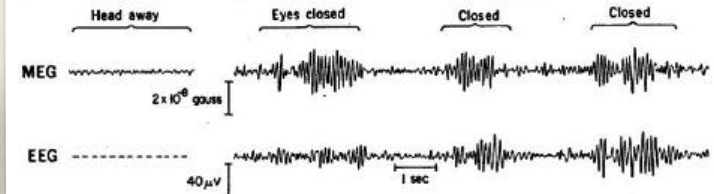
In accordance with **Maxwell's Equations** (*Electromagnetism*), any electrical current will produce a magnetic field.



* Superconducting Quantum Interface Device

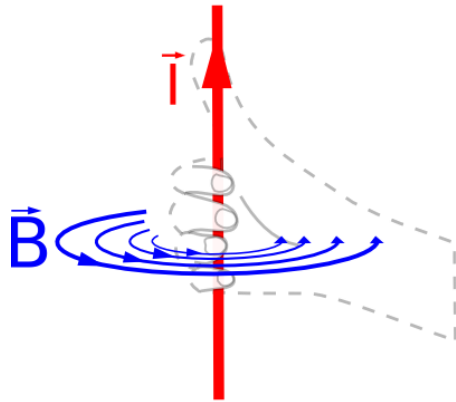
EEG vs MEG

measure the same underlying neural currents



- **EEG:** Measures differences in electric potential at the scalp
- **MEG:** Measures changes in magnetic flux density outside of the head
- Both MEG and EEG record synchronized neural activity at a very high temporal resolution, are non-invasive, cover the whole head.

MEG recording



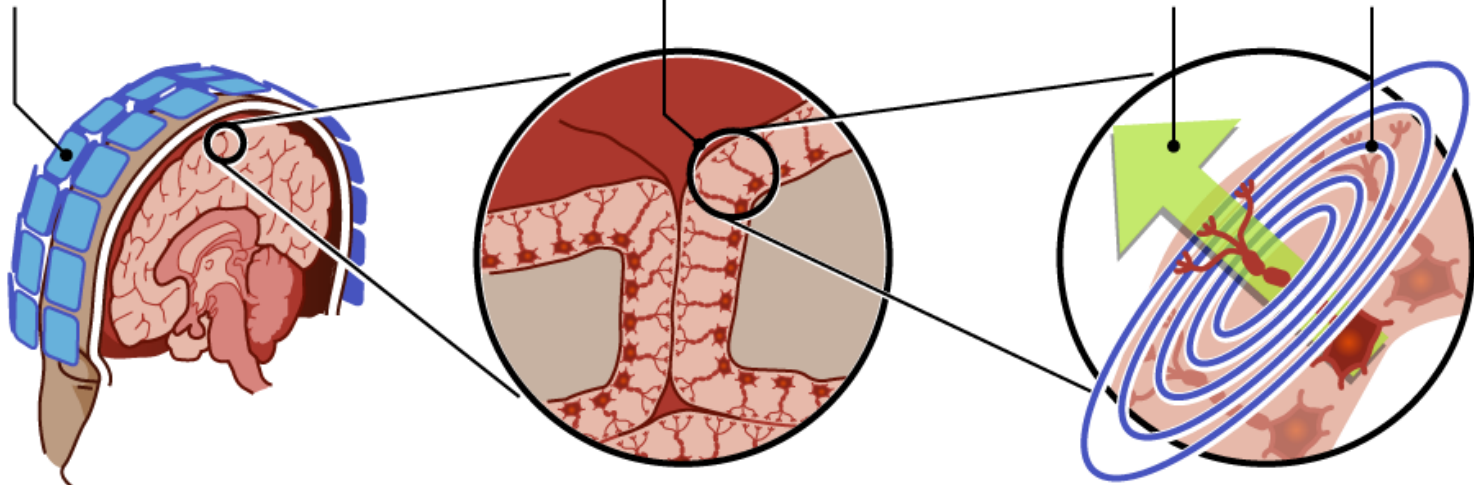
- **MEG** measures the changes in the magnetic fields induced mainly by primary currents based on excitatory activity (Okada et al. 1997)
- While **EEG** is more sensitive to **EXTRACELLULAR currents**, **MEG** is more sensitive to **INTRACELLULAR currents**

SQUID* sensor array aligned to cortical surface of the brain

Axons in the cortical surface of the brain

Direction of electric current in active axon

SQUID sensor detects magnetic field of current



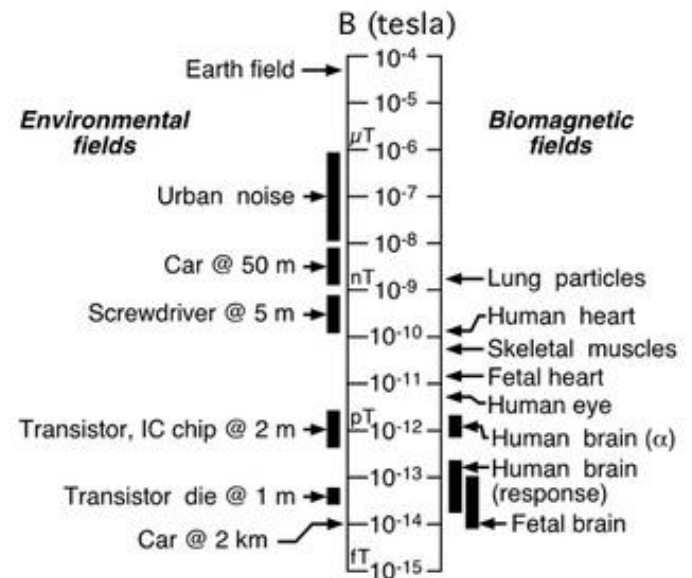
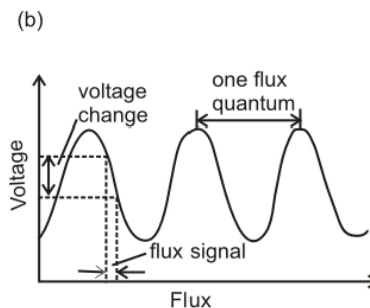
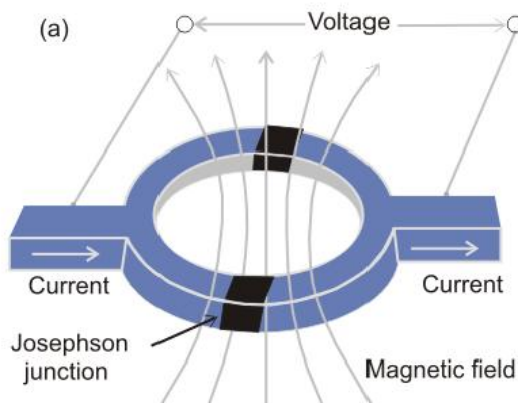
* Superconducting Quantum Interface Device

MEG recording

- **MEG** reveals fields of very small amplitude ($10^{-12} \div 10^{-14}$ T) compared to the Earth's naturally occurring magnetic field (10^{-4} T) and field generated by the equipment (10^{-7} T) or other devices.
- For this reason MEG requires very specialized test environment:
 - a purpose-built **magnetically-shielded room** (MSR), with all devices MEG-compatible
 - Ultrasensitive detectors of the magnetic flux:

Superconductive Quantum Interference Device (SQUID)

Cooling achieved by liquid Helium



MEG recording

- MEG sensors cover the whole head and are placed on a rigid helmet embedded in the **Gantry**, that contains also the cooling system and the helium
- Two main type of sensors:
 - **MAGNETOMETERS** = consist of a single superconductive coil measuring the magnetic flux at his surface (T)
 - **GRADIOMETERS** = consist of a “figure-of-eight” type coil configuration. The measured signal is the spatial gradient between the two loops of the eight (T/m)
- There are two orthogonal planar gradiometers at each locations.

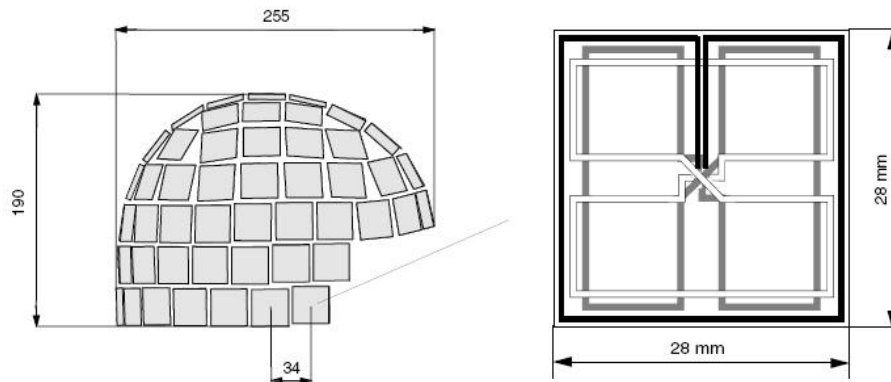
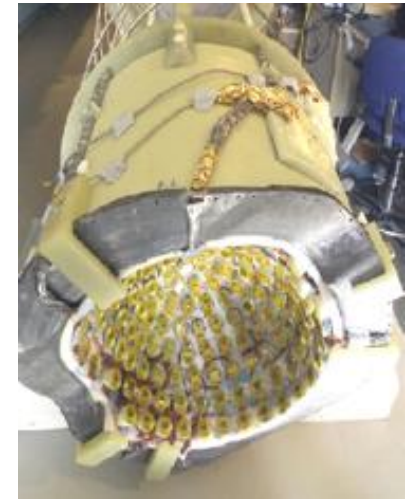


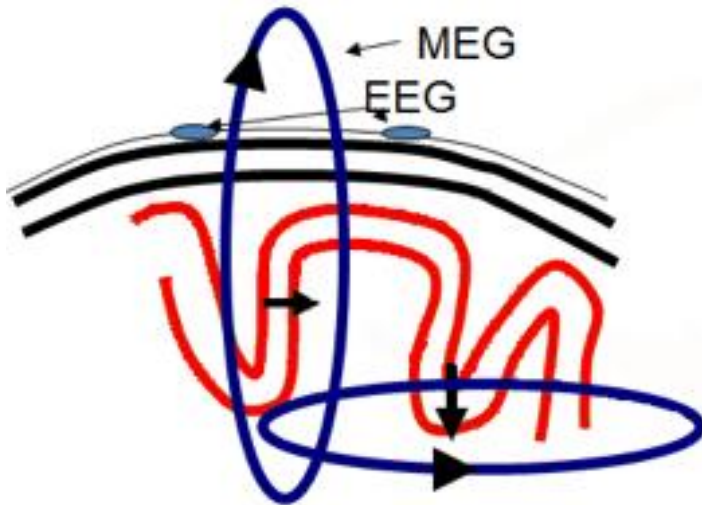
Figure 1.6. (left) Detector array, side view. Average distance between sensor elements : 34,6 mm. (right) Triple sensor detector unit.



Black =
magnetometer
Grey/white =
gradiometers

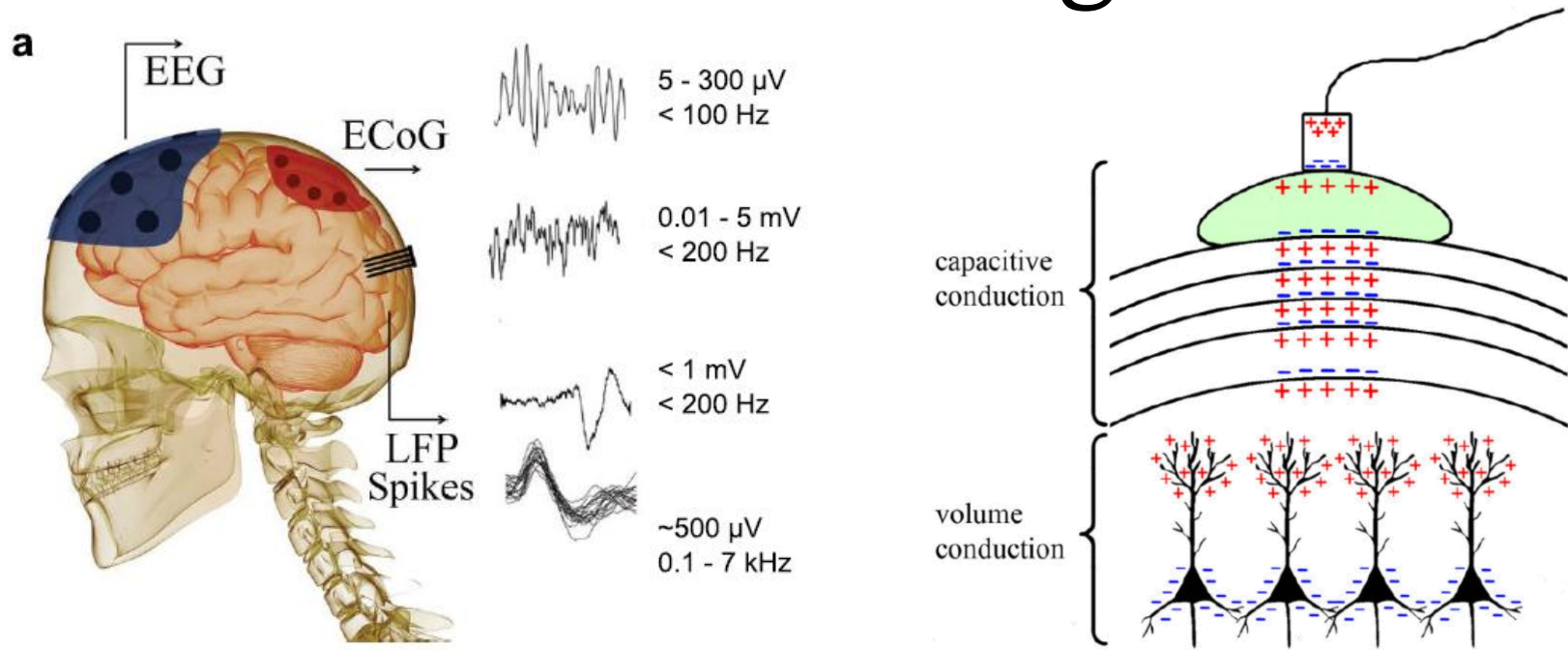


MEG Vs EEG recording



- **MEG** is more sensitive to currents that are **tangential** to the surface of the scalp. Since the net current direction on the cortex is directed to the cortical mantle, this means that MEG is **selectively sensitive to activity on the sulci walls**. MEG is less sensitive but not blind to radial sources.
 - **EEG** in contrast tends to be **more sensitive to activity in the gyri**, i.e. **radial sources**, even if it is sensitive to both radial and tangential sources. It is a more complex signal
-
- **MEG is reference-free**. This make interpretation easier.
 - Magnetic Fields are less distorted by tissues such as the skull and the scalp, which results in a better spatial resolution (lower collinearity between sensors)
 - Larger distance of the MEG sensor array and the neural sources themselves and more pronounced decay of magnetic field with distance -> MEG Is more sensitive to superficial cortical activity

EEG recordings



- EEG is a non-invasive field potential recording techniques
- In order to measured with an electrode outside the head, the electrical signal must travel from the brain, through the dura layers, through the skull layers, through the scalp and finally to the electrode
- The amplitude of EEG of a normal subject in the awake state is **10 – 100 μV (Compared to the brain signals, the amplitude of noise sources (internal and externa) are gigantic!)**
- High frequencies are heavily attenuated by the skull and scalp layers and may be overshadowed by muscular and ocular artifacts

EEG recordings

Amplifier



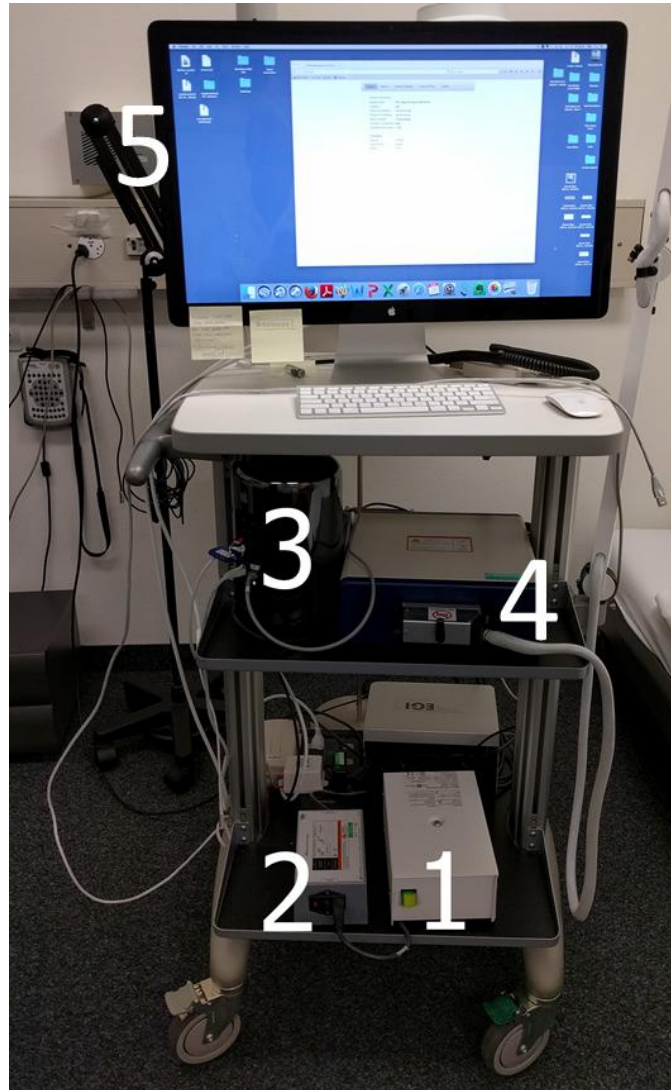
EEG recordings

**Data
Recording
and
Visualization
System**



EEG recordings

- 1 Transformer
(amplifier and
acquisition computer)
- 2 Power Supply
(amplifier)
- 3 Acquisition computer
(Mac)
- 4 Amplifier
- 5 Screen
(acquisition computer)
- 6 E-prime computer



EEG recordings



Number of sensors in EEG systems

19ch: The standard 10-20 system used in clinical research and practice.

32-64ch: Neuroimage research studies that rely on source localization or EEG imaging (with heavy EEG artifact filtering).

>64ch: High-density neuroimage research or EEG studies that rely heavily on source localization and signal processing filtering, or very localized brain processes.

EEG sensor Layout

We measure the EEG activity as the difference in voltage between two electrodes. As a general rule, one fixed electrode is chosen as the reference for all the other electrodes. Therefore, an EEG headset contains three types of electrodes:

Recording electrodes: placed over the specific scalp locations that we want to measure.

Reference electrode: one whose signal is subtracted from each of the recording electrodes.

Ground electrode: used to place both the amplifier and the body to the same potential and to reduce common-mode interference.

Reference Electrode

Amplifier inputs are differential. The voltage measured at every channel is the difference in voltage between the reference sensor and that channel's sensor.

The reference lead is the lead that connects the reference electrode; in EEG recordings, this electrode is usually placed at the ear or, in the case of “summed ears,” to a pair of electrodes, one at each ear. Choice of physical reference is arbitrary.

The measured electrical potential differences are ideally the voltage drops from the active electrode to the reference electrode.

Ground electrode

The **ground** is used for common mode rejection. The primary purpose of the ground is to prevent power line noise from interfering with the small biopotential signals of interest. A ground electrode for EEG recordings is often placed on the scalp (but could be placed anywhere else on the body; the location of the ground on the subject is generally irrelevant).

In EEG recordings with signal, ref, and ground you have 2 subtractions:

REF – GND : removes common mode signal. The human body is an antenna, and can pick up ambient interference. This is subtracted away from the REF signal, to provide a "cleaner" baseline.

SIGNAL – REF : provides your biopotential signal against this clean baseline REF signal.

EEG sensor Layout

When talking about the number of electrodes, it always refers to the EEG recording electrodes only, since the **reference** and the **ground** are always needed.

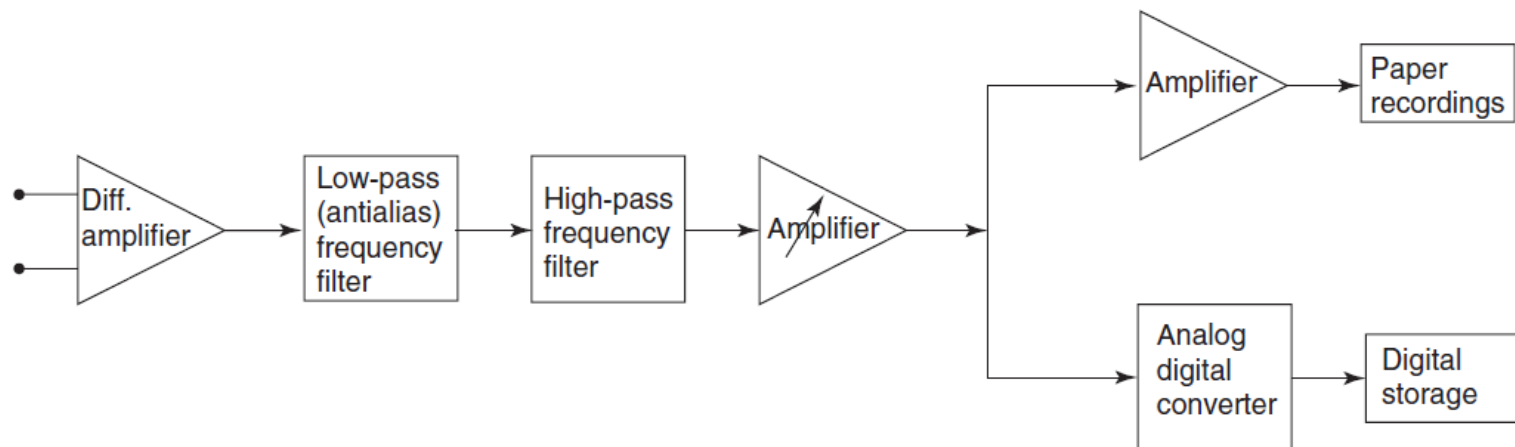
For example, a 32-channel EEG device has 32 recording channels, plus ground and reference.

Reference Electrode

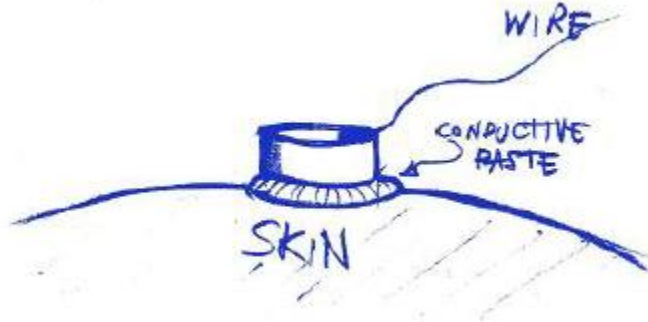
Ground
Electrode

EEG recordings

- The signal is picked up with a differential amplifier, that amplifies the difference between the input electrodes
- The signal recorded by both electrodes (common mode potential) is typically due mostly to capacitive coupling with power-line fields.
- The EEG system is designed to reject this common mode potential and amplify the potential difference between pairs of scalp locations.
- An EEG reference electrode placed on the scalp, nose or neck provide a reference voltage to the amplifier
- The purpose of the amplifier is to maximize the Signal to Noise Ratio (SNR) (CMRR)



EEG recordings

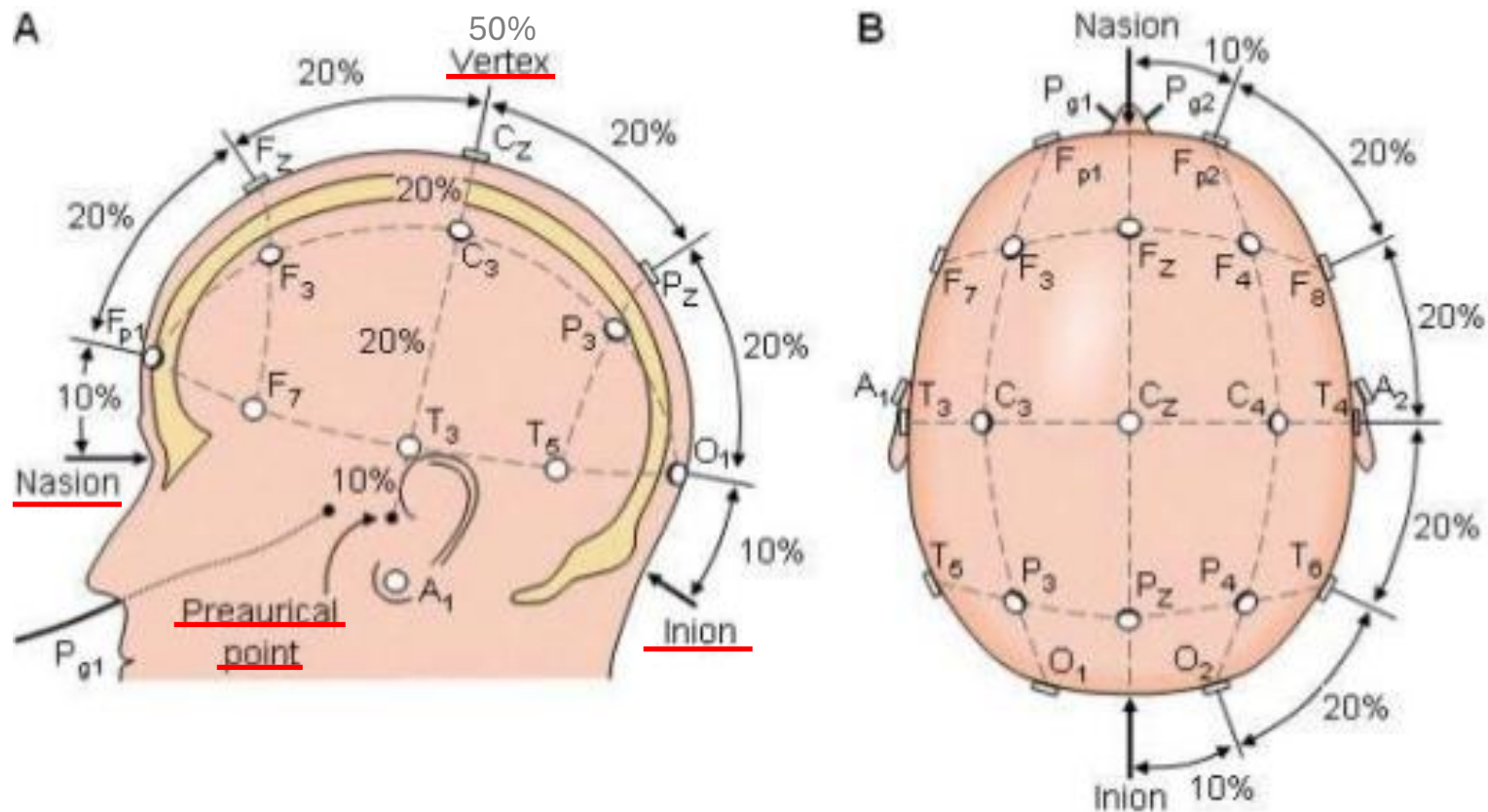


- The resistance of the electrode-scalp connection must be reduced as much as possible
- Wipe the area in order to remove dead skin cells or skin oils (some systems also abrading cream)
- Put the electrode on the skin not on hairs when is possible
- Fill the electrode cup with electrode past and press it firmly
- Highly conductive electrode gel will saturate the space beneath an electrode, filling the air pockets between hairs and providing a conductive path (contains electrolytes that provide good electrical contact) from scalp to the electrode
- Check electrode impedance ($< 5k\Omega$)

Standard Electrodes Placement

The **10–20 system** is an internationally recognized method used to define the location of scalp EEG electrodes. The "10" and "20" refer to the fact that the actual distances between adjacent electrodes are either 10% or 20% of the total front–back or right–left distance of the skull.

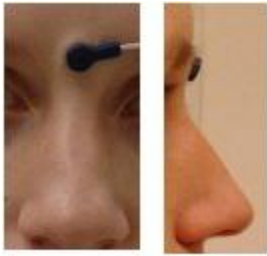
(common nomenclature: + make comparable results / facilitate communication)





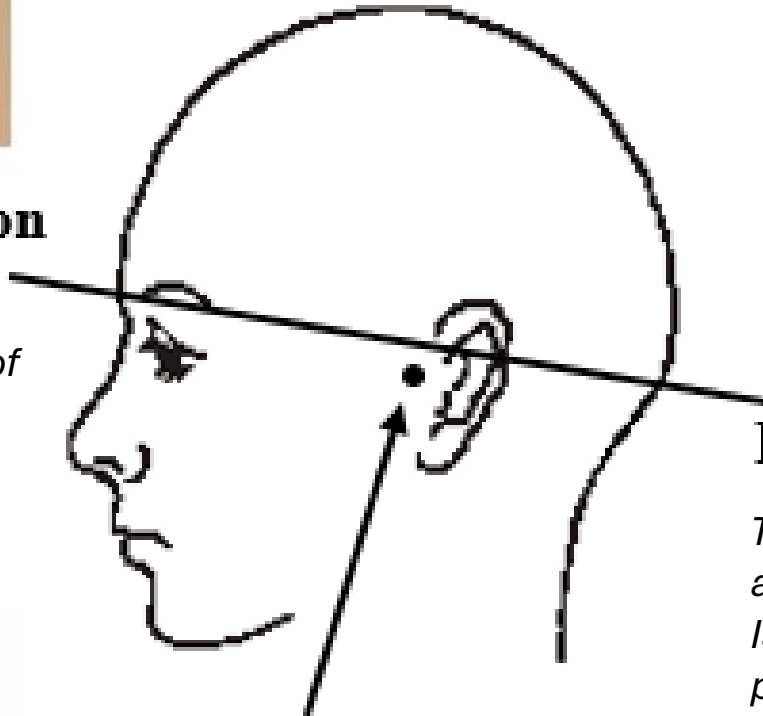
- The system is based on the relationship between the location of the electrode and the underlying area of the cerebral cortex. the starting letter of the label of each site identify the lobe (**F = Frontal** / **T = Temporal** / **C = central** / **P = Parietal** / **O = Occipital**)
- **The z = zero** refers to an electrode placed on the **midline sagittal plane of the skull**
- **EVEN numbers** refer to electrode positions on the **RIGHT HEMISPHERE**
- **ODD numbers** refer to electrode positions on the **LEFT HEMISPHERE**
- The letter codes A and Fp identify the **earlobes** and **frontal polar** sites respectively

Anatomical Landmarks



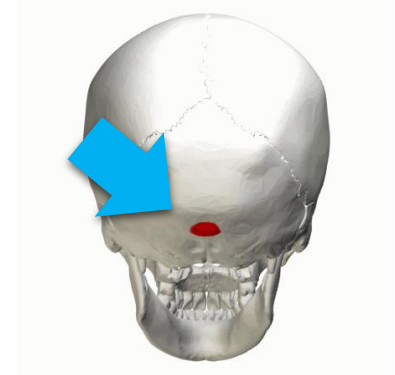
Nasion
Nasion

The intersection of the frontal bone and two nasal bones of the human skull



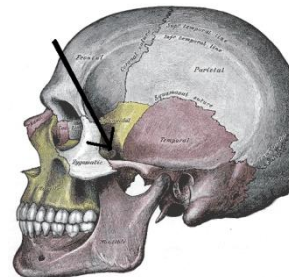
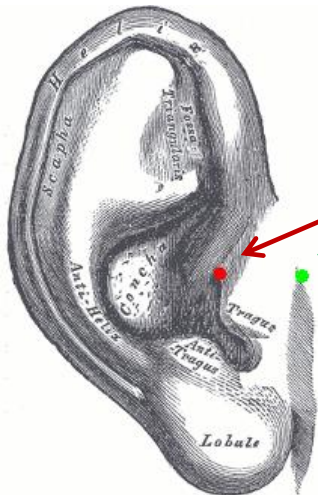
Inion

*The projecting part of the occipital bone at the base of the skull.
Is the highest point/the most prominent projection of the **external occipital protuberance**, which is located at the posteroinferior (lower rear) part of the human skull.*



Preauricular Point

The proper definition of the preauricular point (green) is: "a point of the posterior root of the zygomatic arch lying immediately in front of the upper end of the tragus" (LPA / RPA)



Left



Right

Preparation for an EEG recording session

① Session Setup	② Patient Preparation	③ Acquisition	④ Cleanup
☐ Gather the needed materials. ☐ Ensure that all GES equipment is operating correctly and prepared for this recording session. ☐ Confirm you have enough disk space for the acquisition data. ☐ Set up this session for this patient. ☐ Make sure that the patient's head is clean and dry.	☐ Measure the patient's head circumference. ☐ Select the correct Net size and Spandage size (if using). ☐ Locate and mark the vertex. ☐ Preheat the Elefix EEG paste. ☐ Apply the Net. ☐ Adjust the Net. ☐ Ensure the patient's comfort. ☐ Prefill the applicator(s) with Elefix EEG paste. ☐ Fill the sensor pedestals with Elefix paste. ☐ Plug in the Net. ☐ Check data quality and sensor impedances. ☐ If using Spandage, wrap the Net and head. ☐ Position the patient.	☐ Establish the maximum recording time and maintenance schedule for this patient. ☐ Start recording. ☐ Monitor the patient and data regularly. ☐ Protect the patient's skin, if needed. ☐ Add events, as necessary. ☐ Stop recording. ☐ Close session.	☐ Remove the Net. ☐ Rinse the patient's skin and wash the patient's hair. ☐ Debrief and release the patient. ☐ Rinse, disinfect, and dry the Net.

Getting started: head circumference

- Ask the subject **to arrive with freshly washed, but not wet, hair. Request that the subject not use any hair products (gels or hair spray, for example) or any facial foundation on the day of the session**
- Ask the subject to **remove earrings, glasses, barrettes, and hair ties**. Earrings can be especially uncomfortable with an high-density net applied. Glasses can be after Net application, as needed.
- Before applying the net to the subject, **measure the circumference of the subject's head** to determine the appropriate Net size
- With a measuring tape, measure head circumference at the glabella (the brow ridge) and 2.5 cm. (1 in.) above the inion (the bump on the back of the skull);
- Adult size ranges for the net are: • small (53.5–55 cm.) • medium (55.5–58 cm.) • large (more than 58.5 cm.)



Figure 6-1. Measuring head circumference

Net size

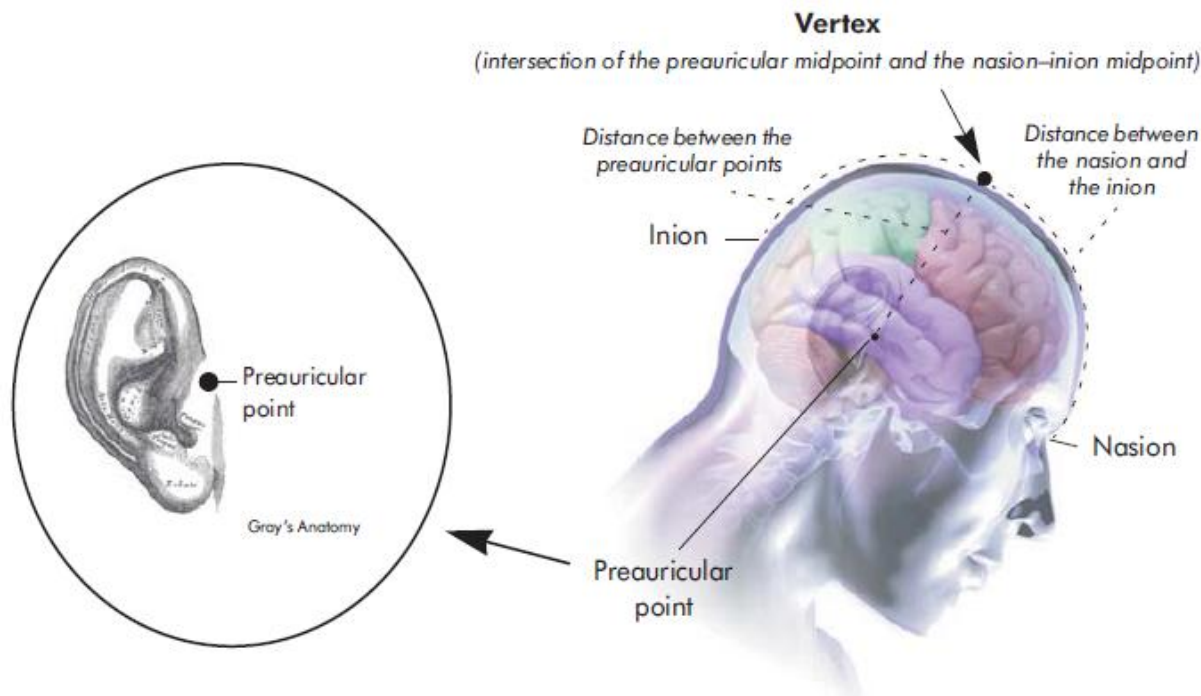
How to Use the Size Chart

Always base your Net size selection on the circumference of your patient's head, not on your patient's age.

Stage	Head Circum. and Net Size	General Age or Head Size		Channel Counts				Monitoring Type		
		Males	Females	32	64	128	256	Routine	LTM	MR
Preterm infant	28 – 30 cm	Preterm Small		✓	✓	✓		✓		
	30 – 32 cm	Preterm Medium		✓	✓	✓		✓		
	32 – 34 cm	Preterm Large		✓	✓	✓		✓		
Infant	34 – 36 cm	Neonate	Neonate	✓	✓	✓		✓		✓
	36 – 37 cm	Birth – 2 weeks	Birth – 1 month	✓	✓	✓		✓		✓
	37 – 38 cm	2 weeks – 1 month	1 – 2 months	✓	✓	✓		✓		✓
	38 – 40 cm	1 – 2 months	2 – 3 months	✓	✓	✓		✓		✓
	40 – 42 cm	2 – 3 months	3 – 6 months	✓	✓	✓		✓		✓
	42 – 43 cm	3 – 5 months	6 – 7 months	✓	✓	✓		✓		✓
	43 – 44 cm	5 – 7 months	7 – 9 months	✓	✓	✓		✓		✓
	44 – 47 cm	7 – 15 months	9 – 21 months	✓	✓	✓		✓		✓
Pediatric	47 – 51 cm	Pediatric Small		✓	✓	✓		✓	✓	✓
	51 – 54 cm	Pediatric Large		✓	✓	✓	✓	✓	✓	✓
Adult	54 – 56 cm	Adult Small		✓	✓	✓	✓	✓	✓	✓
	56 – 58 cm	Adult Medium		✓	✓	✓	✓	✓	✓	✓
	58 – 61 cm	Adult Large		✓	✓	✓	✓	✓	✓	✓
	61 – 64 cm	Adult Extra Large		✓	✓	✓	✓	✓	✓	✓

Applying the net: measuring and marking the vertex

- Determine the vertex point as the midpoint between the nasion (the indented area where the bridge of the nose meets the skull) and inion, centered between the preauricular points (the indented area in front of each ear flap, where the jaw meets the skull)



Applying the net

- *Ask the subject to remove earrings, glasses, barrettes, and hair ties. Earrings can be especially uncomfortable with an high-density net applied. Glasses can be after Net application, as needed.*
- *Ask the subject to notify you of any discomfort or irritation from the Net.*
- *Ask the subject or an assistant to hold the Hypertronics connector during the initial application*
- *Ask the subject to close his or her eyes and to keep them closed until told otherwise.*
- *Stretch the perimeter band to a size just larger than the subject's head. Do this by spreading out your thumbs and little fingers so that the perimeter band forms a circle. Do not overstretch the Net, as this could damage the geodesic tension structure.*



- Keeping moderate tension on the perimeter band with your thumbs and littlefingers, set the GSN on the subject's head so that the colored vertex sensor lies on the vertex mark on the head.
- Slowly pull straight downward keeping the *vertex* electrode over the vertex and the *nasion* electrode aligned with the nasion. Standing directly in front of the subject makes this easier.

Net adjustment

VI. Adjusting the Net

Check and adjust (if needed) the placement of the Net's sensors.

Note: Given the variation in head and face shapes, exact placement of some sensors (e.g., nasion and infraorbital) is not always possible, but get as close as instructed.

While adjusting the Net's sensors, gently shift sensors and make sure that:

- ☐ You grasp groups of sensors, not individual sensors.
- ☐ You never pinch or damage the wires.
- ☐ You often check and, if needed, adjust the chinstrap.
- ☐ The Net is positioned symmetrically on the head and that the midline is straight.
- ☐ The reference (REF/Cz) sensor is positioned on the vertex.
- ☐ The ears are sitting correctly in the ear holes.
- ☐ The mastoid sensors are positioned on the mastoids.
- ☐ The nasion sensor is positioned on the nasion.
- ☐ The infraorbital sensors are positioned directly below the pupils.
- ☐ The facial sensors are evenly spaced.



Net adjustment

VII. Ensuring Patient Comfort

- 1 Ask the patient whether the Net is comfortable. Adjust it, if necessary.
- 2 Make sure that the chinstrap adjustment beads are not positioned on the jawbone.
- 3 Place a pad of gauze between the chin and the chinstrap.
- 4 Place a pad of gauze (or a latex-free bandage) between the nasion and the nasion tube.



VIII. Prefilling the Applicators with Preheated Elefix EEG Paste

At this stage, you will be filling the applicators with the Elefix EEG paste that was preheated in Section IV, "Preheating the Elefix EEG Paste."

- 1 Trim 0.25 inch (0.635 cm) off the tip of the curved-tip syringe applicator(s).
Note: Remember that each applicator (if using more than one) fills 45 to 50 sensors.
- 2 Fill the applicator(s) with the preheated paste.

Net preparation

IX. Filling the Sensor Pedestals with Elefix Paste

- 1 Start with the reference (REF/Cz) and common (COM) sensors, then proceed in numerical order.
- 2 Move each sensor side to side in a scrubbing motion to part the hair so that the sensor makes direct contact with the scalp.
Note: For long hair, it may be necessary to push away the hair gently with the tip of the applicator.
- 3 While keeping a sensor anchored on the scalp, tilt it, insert the tip of the applicator into the pedestal, and fill it with paste to just above the rim.
- 4 Turn the sensor upright and press gently to make sure that it is directly seated on the scalp.



Starting the recording session

X. Plugging in the Net

- 1 Plug the Net's Hypertronics connector into the articulated arm or the Net Amps 300 amplifier.
- 2 Twist the Net's connector lever 180° clockwise to lock the connection.

Note: The connector is detached by twisting the lever counterclockwise.



XI. Checking Data Quality and Sensor Impedances

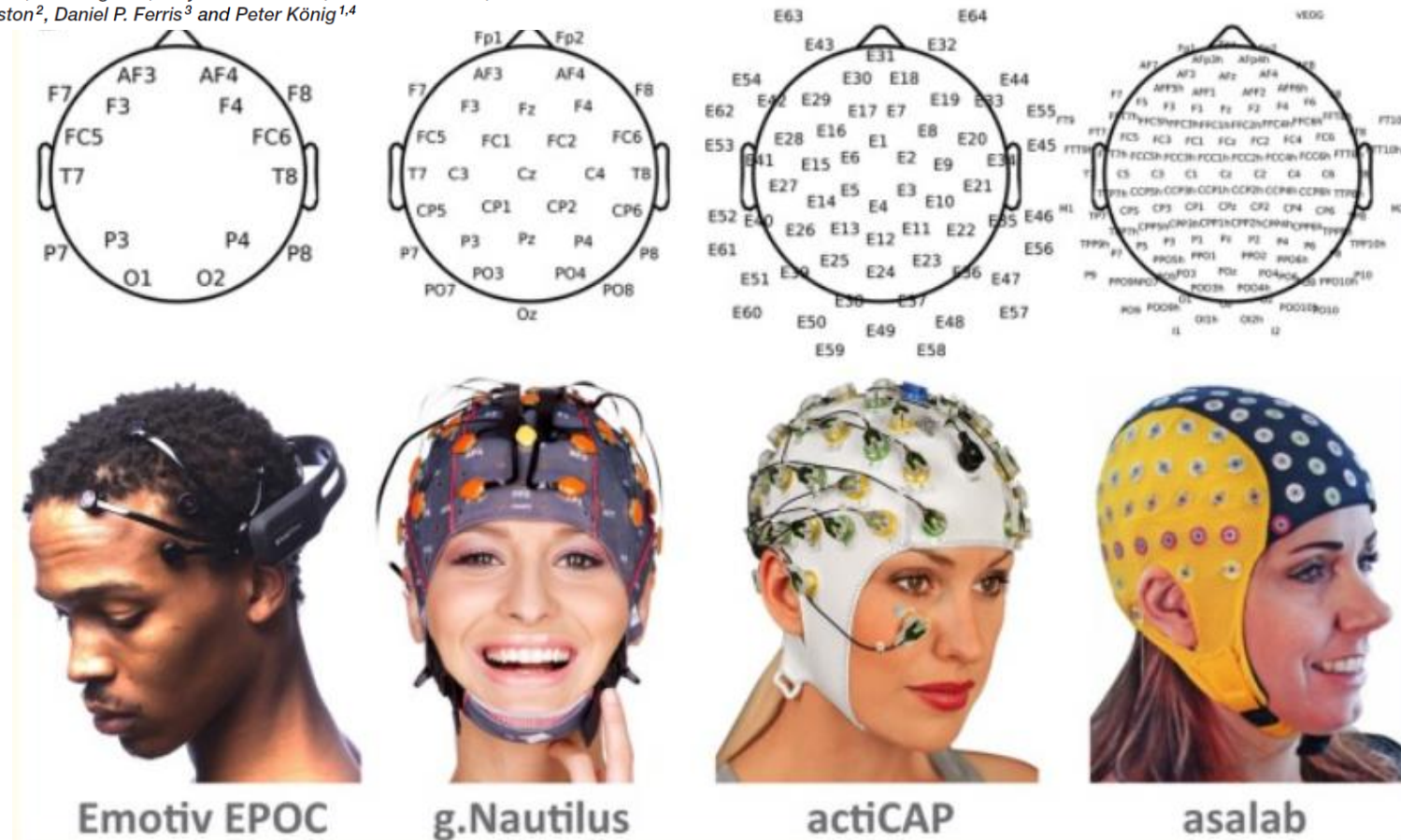
- 1 Within Net Station, check the data quality and impedance values of all the Net's sensors by:
 - Check data quality by scrolling through the EEG waveforms.
 - Check the impedance values by clicking **Panels > Impedance > Measure Impedance** and setting the impedance threshold to 100 k Ω (unless advised otherwise).
 - Make sure that most impedances are below the threshold and that they do not vary greatly between sensors.

Note: Remember that hair length, thickness, and cleanliness, as well as skin texture and whiskers, can affect impedance.

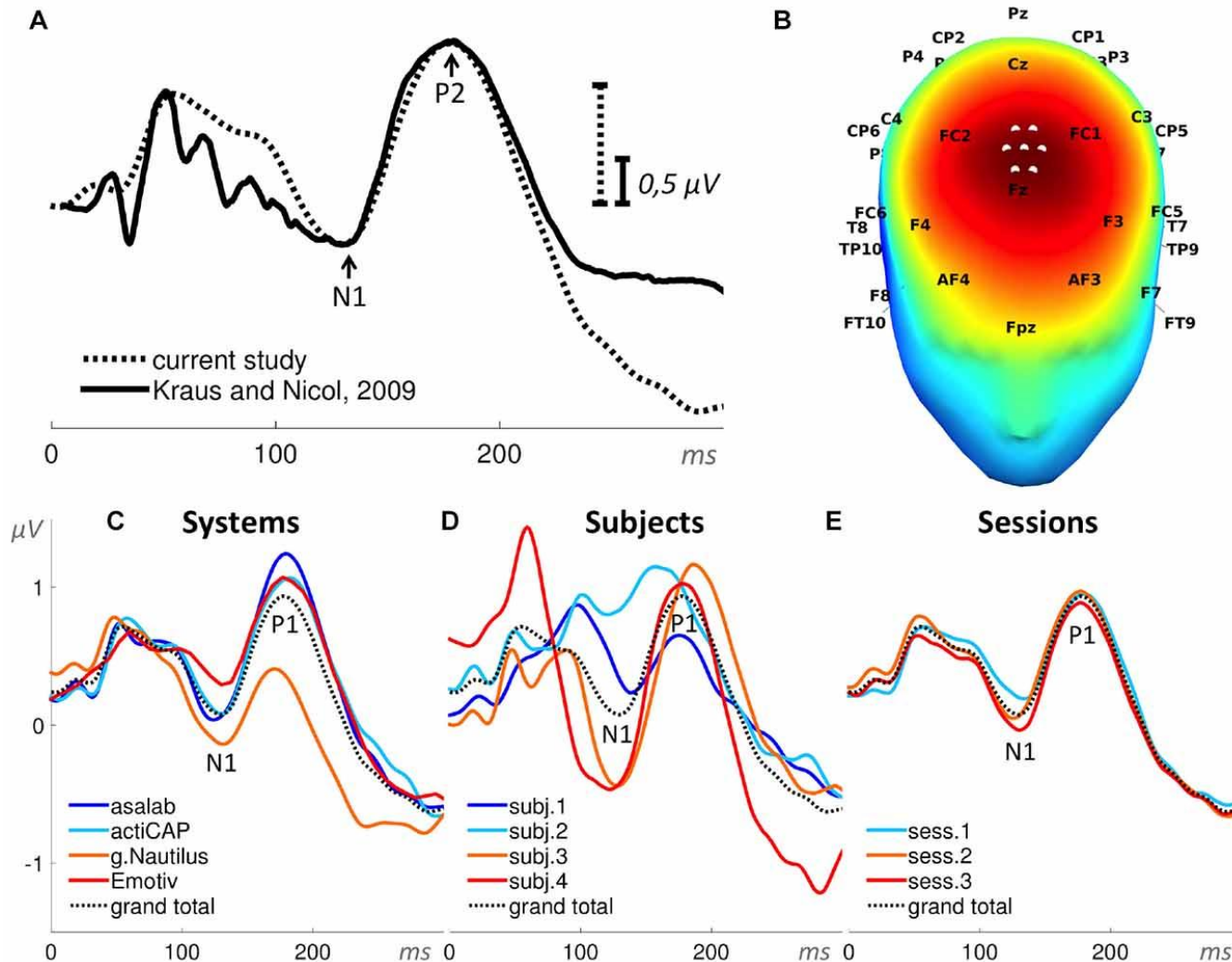
Systems, subjects, sessions

Systems, Subjects, Sessions: To What Extent Do These Factors Influence EEG Data?

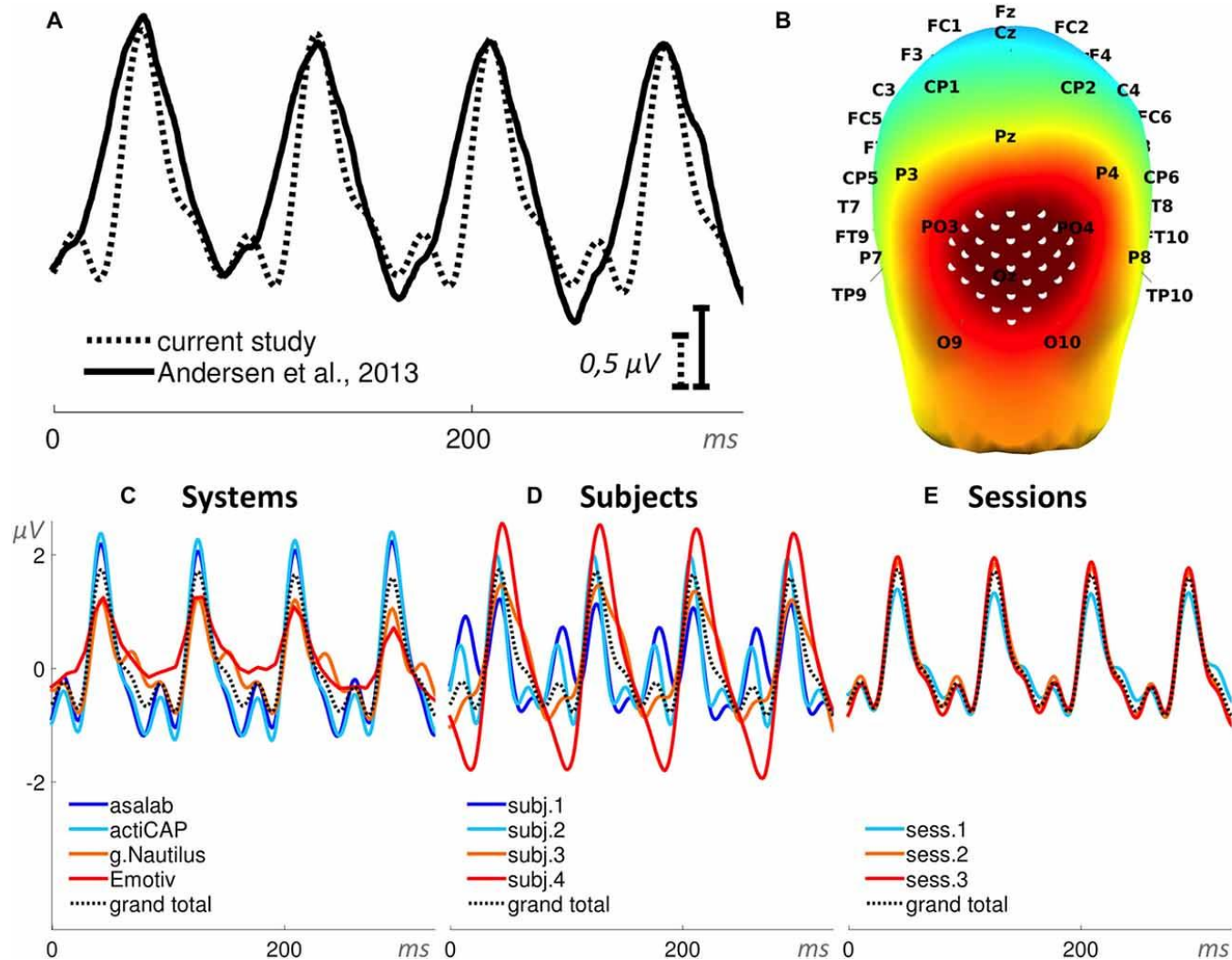
Andrew Melnik^{1*}, Petr Legkov¹, Krzysztof Izdebski¹, Silke M. Kärcher¹,
W. David Hairston², Daniel P. Ferris³ and Peter König^{1,4}



Systems, subjects, sessions



Systems, subjects, sessions



EEG sensor Layout

Commercial EEG-based headsets feature either **fixed** or **interchangeable** sensor positions.

The sensors of systems with **fixed positions** cannot be moved from one location to another, while the systems with **interchangeable positions** can be moved to accommodate different experimental setups.

EEG sensor Layout

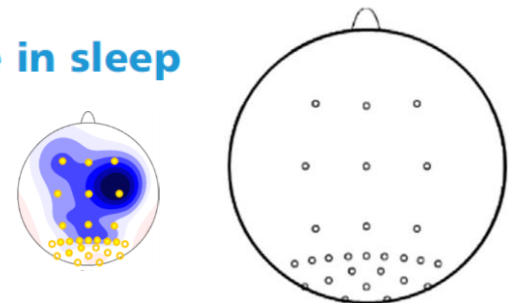
EEG headset with variable layout

These systems allow for moving their interchangeable positions to accommodate different experiments.



Blindfolding during wakefulness causes decrease in sleep slow wave activity

Eva Magdalena Korf¹, Matthias Mölle², Jan Born^{1,3} & Hong-Viet V. Ngo^{3,4}

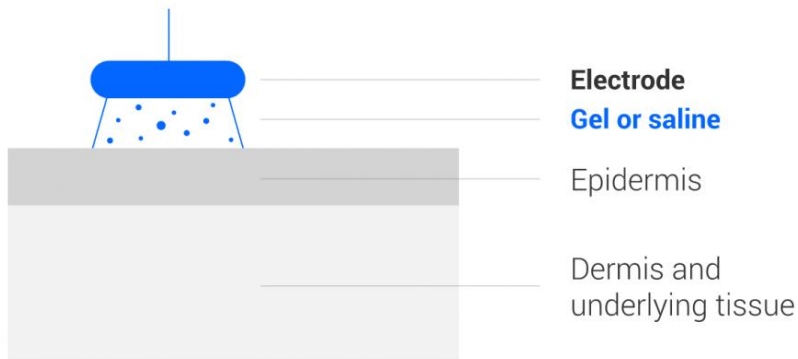


EEG sensor types

Dry EEG, semi-dry EEG, saline EEG, gel EEG electrodes: The difference between these electrodes is the electrolytic substance applied between the electrode and the scalp, used to improve conductivity.

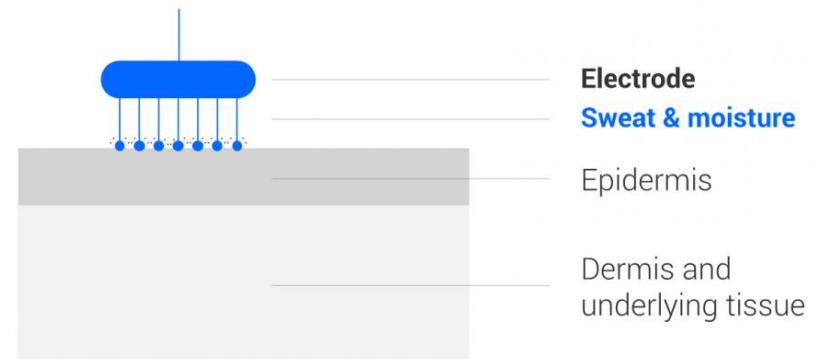
Wet or semi-dry electrode

(A)



Dry electrode

(B)



EEG sensor types

Dry Electrodes

Dry electrodes do not require the use of any electrolytic substance, making the contact directly with the scalp. The main advantage is that they are **fast to place, do not require any additional instruments** like syringes, **do not require cleanup** of the head, and **do not require heavy hygienic procedures** on the equipment afterward.

The main disadvantage is the **high contact impedance** between the sensor and the skin, which requires the amplifier to deal with **more noise and artifacts**.

To achieve the same signal quality than those systems that use electrolytic substances, the **amplifier** must have better performance:

- It must **have higher input impedance** to avoid signal attenuation.
- It must provide **active shielding** techniques to minimize coupled artifacts, such as power line interface or movement artifacts as examples.

EEG sensor types



EEG sensor types

Wet Electrodes

Semi-dry, saline or gel-based EEGs require the **application of an electrolytic substance** between the scalp and the electrode to obtain good contact and reduce the impedance of the skin-electrode interface. It allows the system to improve features and performance by **reducing noise and artifact sensitivity**.

Beyond the typical electrolytic gel systems, there are also **semi-dry electrodes** that use only tap water (also denoted water-based), and others that use **saline solutions** to increase water conductivity properties.

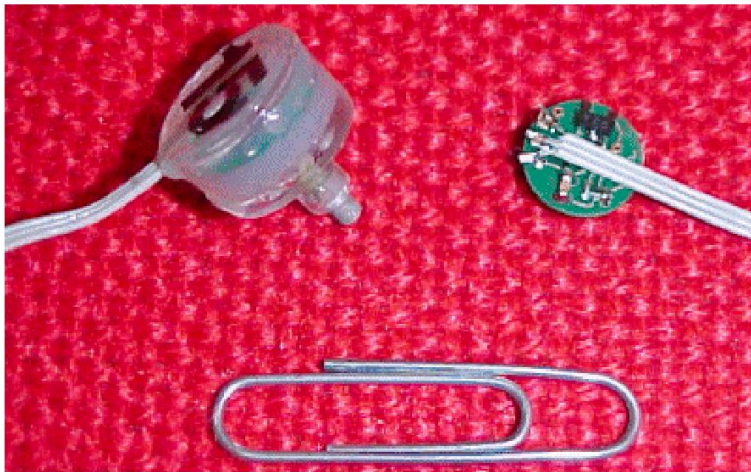
EEG sensor types

	Dry	Semi-dry (Tap water)	Semi-dry (Saline solution)	Wet (Gel)
Contact impedance	High	Medium/Low	Medium/Low	Low
Movement artifacts	High	Medium/Low	Medium/Low	Medium/Low
Electrode preparation (time consuming)	No	Medium/Low	Medium/High	High
Skin preparation (abrasion)	No	No	No	Yes
Application of electrolyte solution or gel	No	Yes	Yes	Yes
User reluctance to electrolyte solution	No	Low	Medium/Low	High
Dehydration of the contact electrolyte (impact in signal quality in long term recordings)	No	Yes	Yes	Yes
Short circuits due to solution smearing	No	Yes	Yes	Yes
Cleaning and hygienic procedures	Low	Medium/Low	Medium/High	High

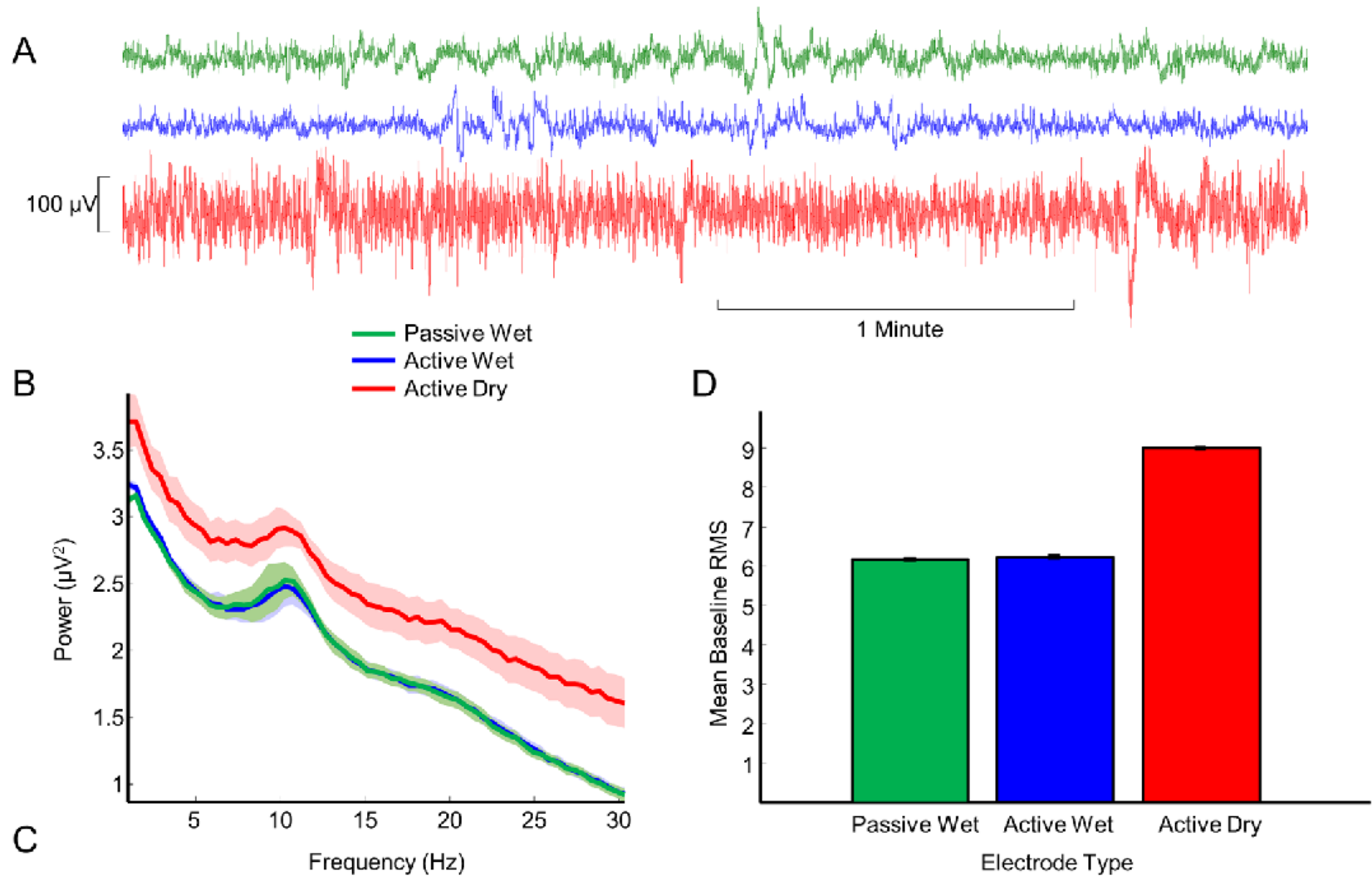
EEG sensor shielding

Active vs. Passive electrodes

The main difference between these two technologies is that active electrodes have an **electronic circuit** embedded **between the sensory part and the wire**, while in passive electrodes these two sections are welded directly together.



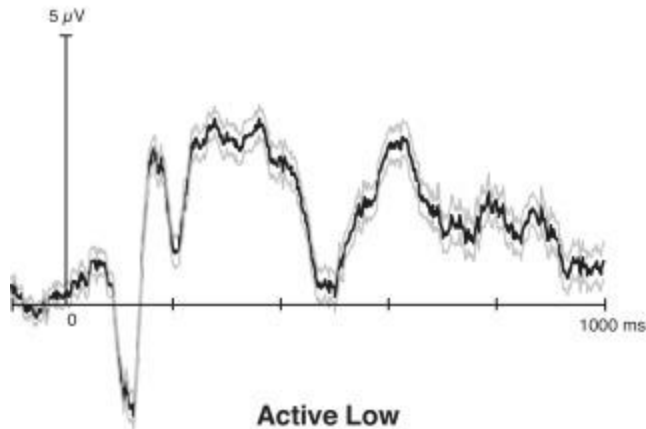
EEG sensor shielding



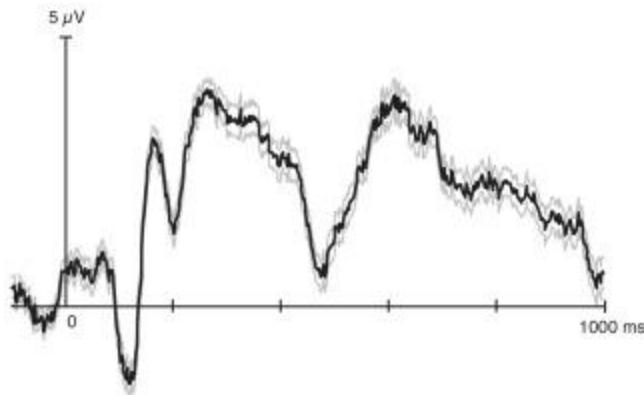
EEG sensor shielding

Low impedance ($\sim 2\text{k}\Omega$)

Passive Low

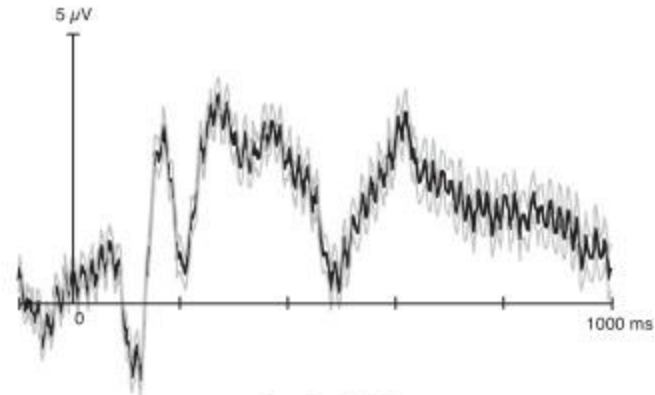


Active Low

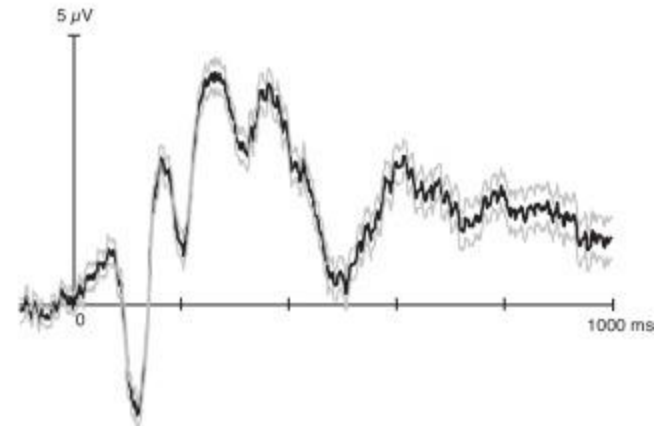


High impedance ($\sim 50\text{k}\Omega$)

Passive High



Active High



— Commons (OOO)
— Standard Deviation

EEG sensor shielding

Active vs. Passive electrodes

	Active electrodes	Passive electrodes
Possible breaks in electronic elements	Yes	No
Electrode cost	High	Low
Coupled artifacts in wires (need to shield)	Low (can work with basic shielding or no shielding)	Yes (require active shielding)
Effects on Common Mode Rejection Ratio (CMRR)	Yes (could affect CMRR if not perfectly calibrated)	No

EEG connectivity

Cable powered: The amplifier is connected directly to the power line through a power supply. The main advantage is that the experiment can last as long as necessary, with no limit on functioning time imposed by a battery charge. The main drawback is that the amplifier must be plugged into an outlet, so the participant will be tethered with a power cable that will constrain both local movement and reach.

Battery powered: The amplifier is powered by batteries, so it can be attached to the participant, allowing free movement and more natural behavior. The main drawback is that the maximum recording duration will be limited by battery capacity. The battery can be internal or swappable.

EEG connectivity

Wired: Most wired EEG devices, if not all, use **USB technology**. The main advantages of this kind of data transfer is that there is almost no limit in the amount of EEG data that can be transmitted, in real-time, to the computer - with very **small latency** and **high robustness against packet loss or corruption**. The main drawback is the same as with any wired interface, there is always a cable that constrains mobility.

Wireless: Modern wireless communication technologies convert the amplifier into a mobile device that can be attached to the participant. The two main standards are **Bluetooth** and **WIFI**, plus there are manufacturers who use their **proprietary technology**.

EEG connectivity

Disadvantages of wireless systems (vs. wired systems):

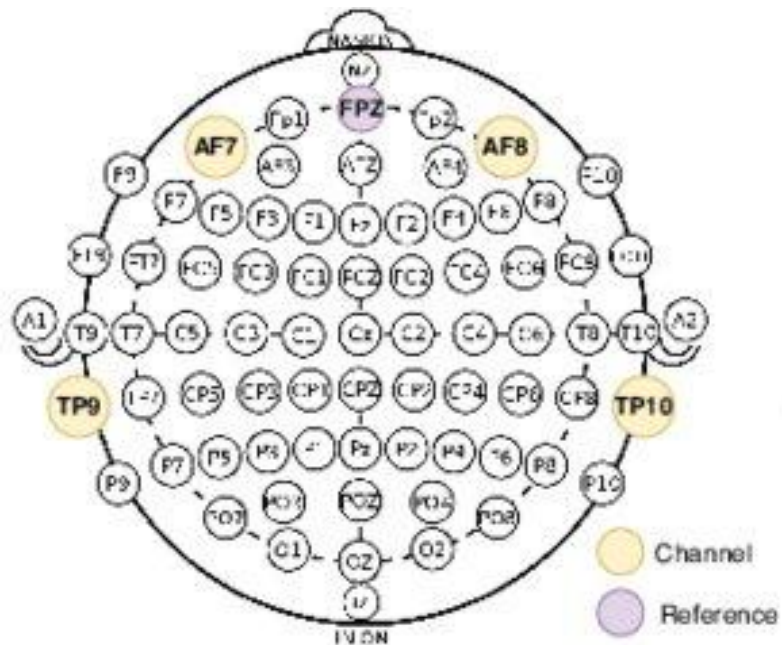
- Higher power consumption
- Higher data transfer/visualization latency
- Possible disconnections

All wireless EEG headsets are prone to **disconnections between the emitter and receiver** when there is coverage loss or if the electromagnetic space is very “**crowded**” (i.e. many devices like cell phones or computers, using the same type of connections at the same time in the same place). Thus it is very important that EEG systems implement specific protocols that guarantee immediate reconnection and preservation of data integrity in the event of a connection loss.

Portable EEG devices

- **Portability & Comfort:** Unlike bulky, stationary clinical EEG machines, portable devices are often designed as **lightweight headsets or headbands**, enhancing user comfort and allowing for use outside of a lab or clinic setting.
- **Electrode Types:** Many modern portable devices use **dry electrodes** (such as conductive polymers or brush-type sensors) that do not require messy conductive gel, minimizing setup time and improving user experience.
- **Wireless Connectivity:** Most portable systems use wireless technologies like Bluetooth Low Energy or WiFi to transmit data to a computer or mobile device for analysis.
- **Fewer Channels:** To improve wearability, these devices typically have fewer channels (e.g., 4, 8, or 12 channels) compared to the 21+ electrodes used in a full clinical 10-20 system.
- **Applications:** portable EEGs are used in research for emotion recognition, attention assessment in educational settings, and BCI development, neurofeedback and others

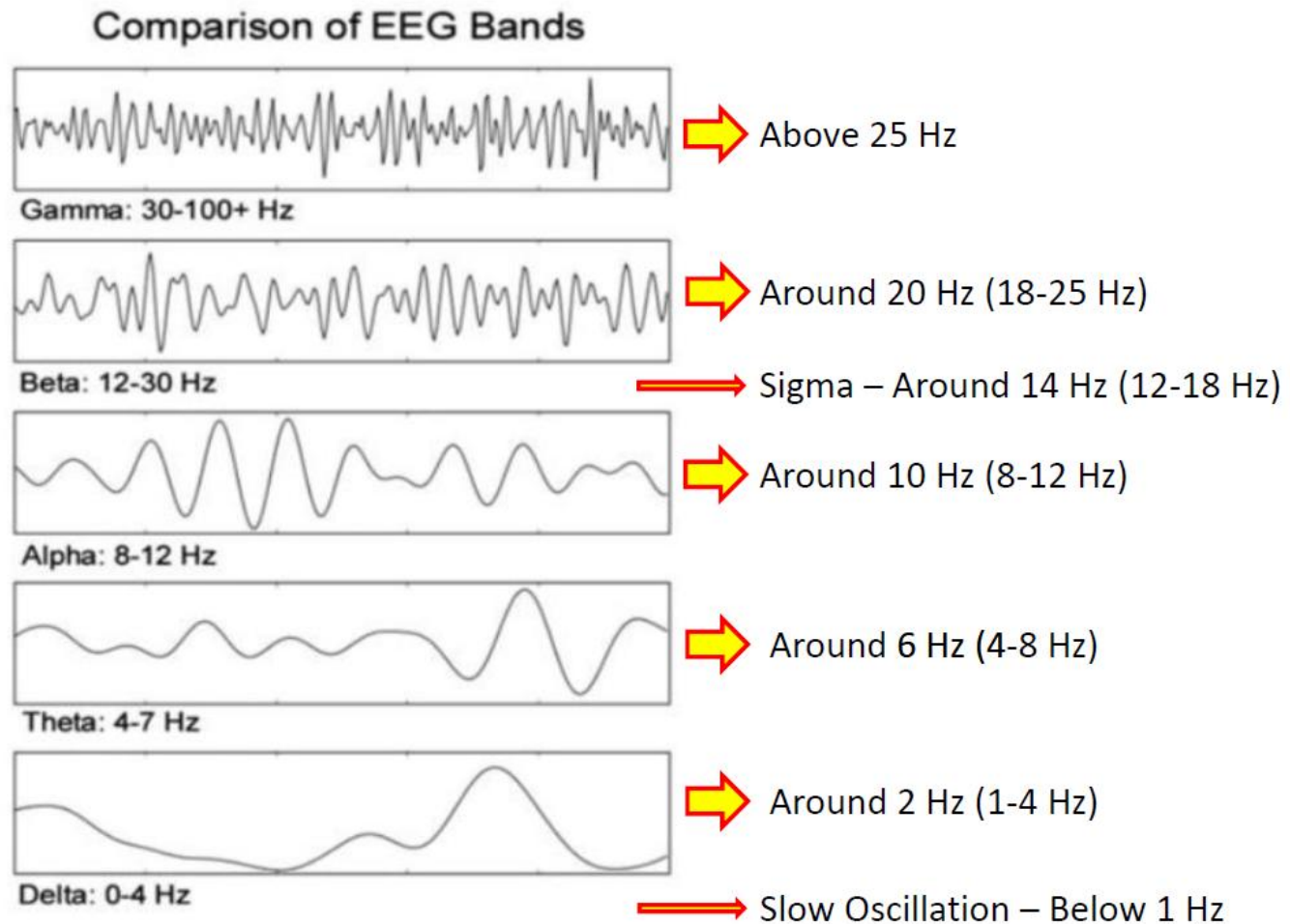
Portable EEG devices



<https://muse-eeg-app.web.app/>

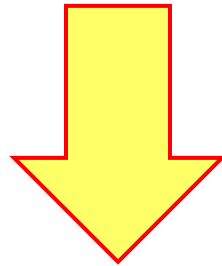
Typical EEG data Analysis

One of the most widely used method to analyze EEG data is to decompose the signal into functionally distinct frequency bands, such as delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–100 Hz).



EEG data analysis: a premise

Many powerful artifacts affect the scalp EEG
(Up to 99% of the EEG variance is not
related to brain activity)



Ongoing EEG is often heavily contaminated,
thus no information can be derived directly
The preprocessing is mandatory

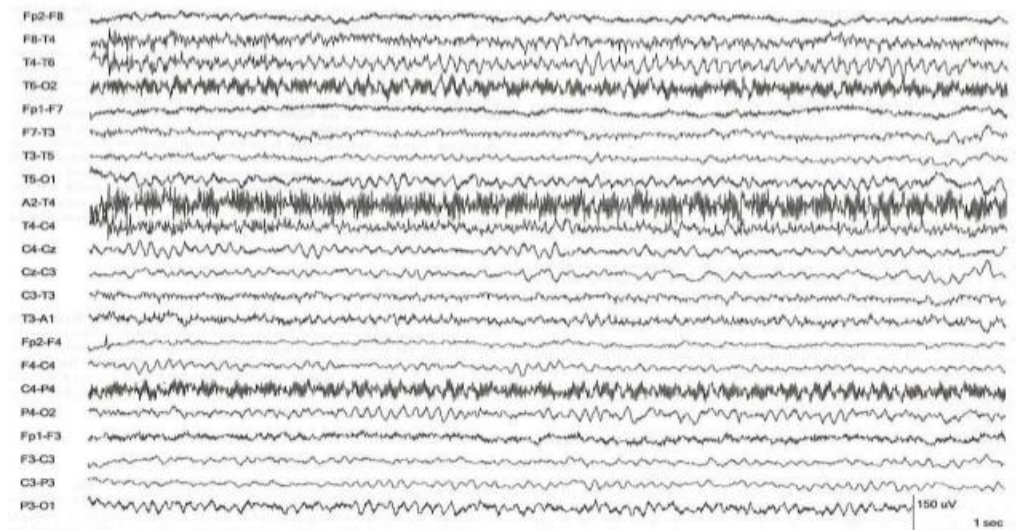
EEG data analysis: a premise

Always try to minimize artifacts **DURING data acquisition:**

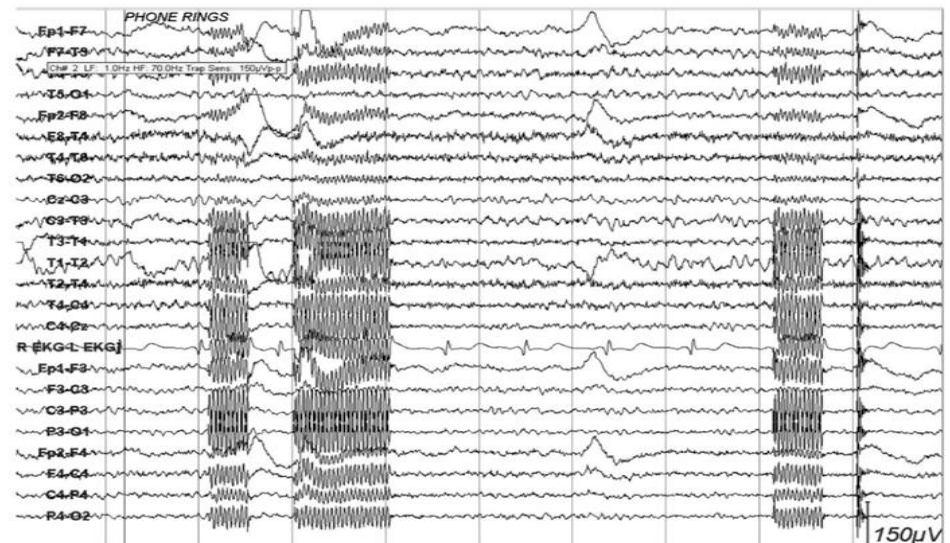
- Buy an electrically-shielded room (high cost!).
- Instruct the volunteer to remain still and not blink during recordings.
- Prepare the experimental setup so that the subject can sit/lay relaxed and only minimal movements are required.
- Be careful during the preparation of the EEG net (e.g., avoid putting too much or too few gel in the electrodes).

Main Artefacts: Non-physiological artefacts

**Electrical Line Artifact
(50/60 Hz, + armonics)**

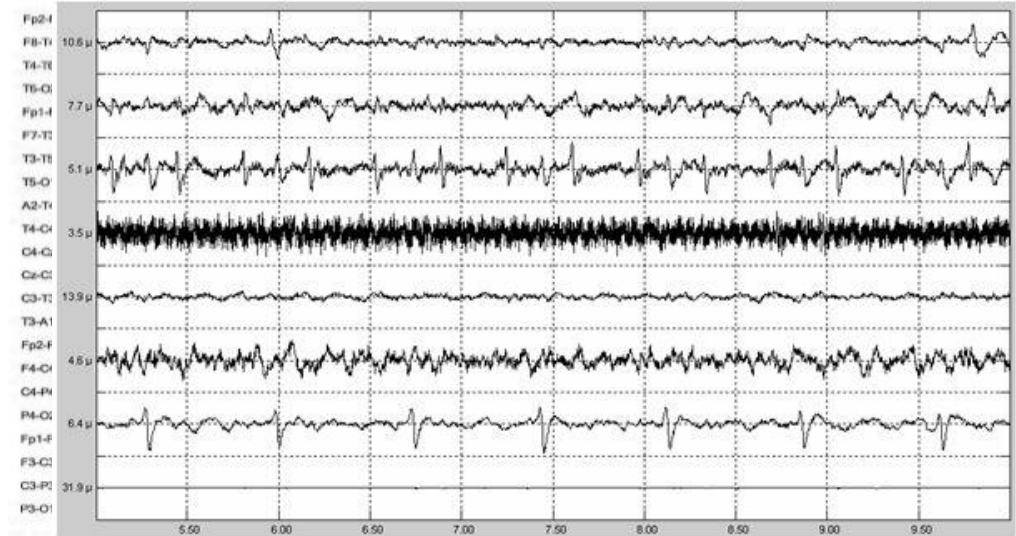


Electronic Components

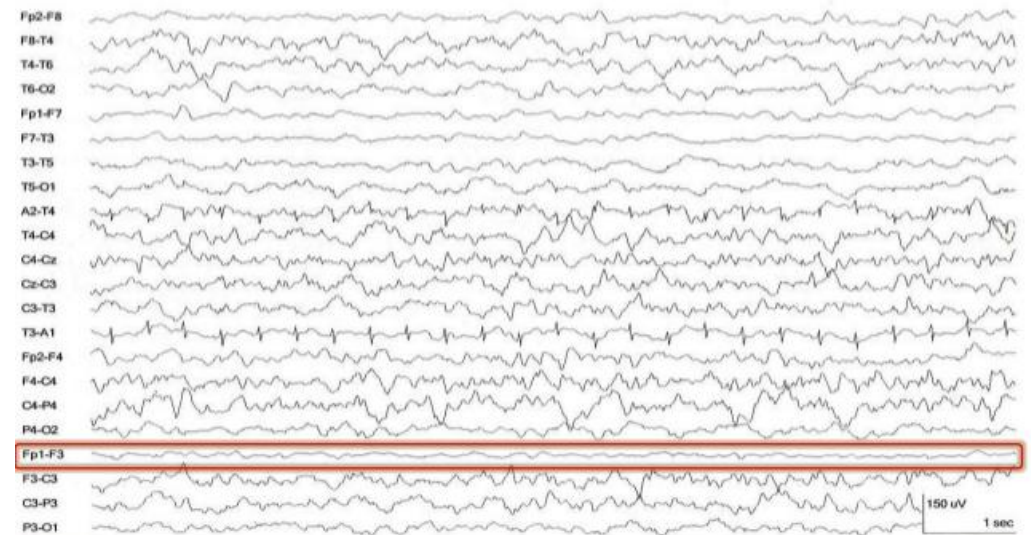


Main Artefacts: Non-physiological artefacts

Noisy EEG channel

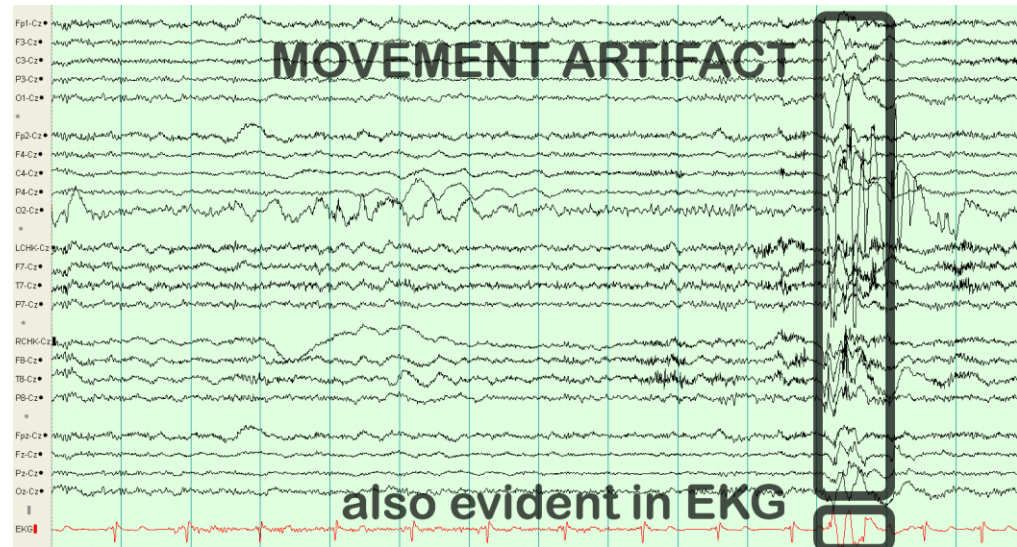


Flat EEG channel

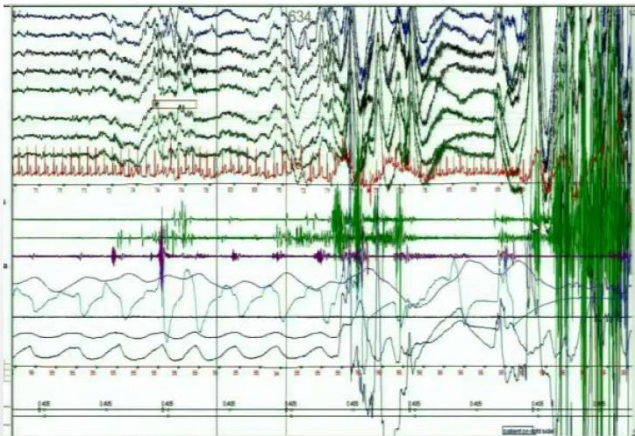


Main Artifacts: Physiological artefacts

**Movement
(broadband)**

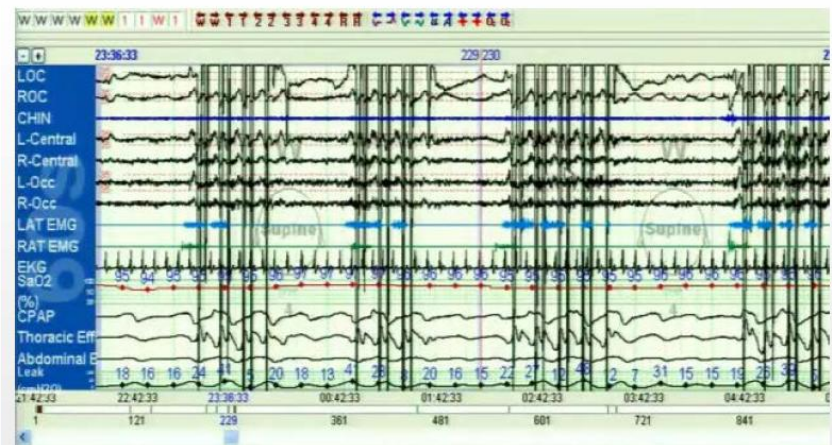


Movements during awakenings



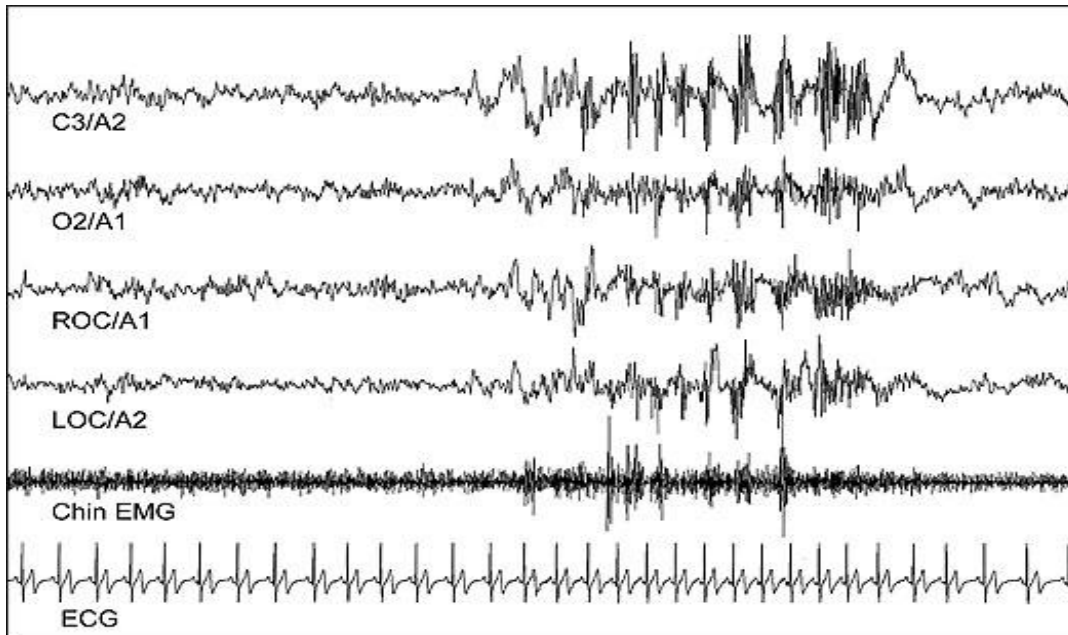
Kelly Carden, MD

Rhythmic Movement



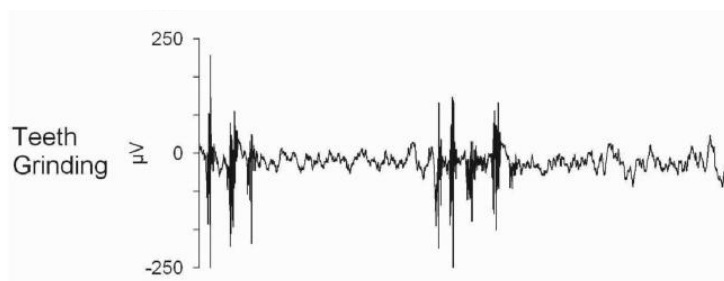
Main Artefacts: Physiological artefacts

EMG (10-200 Hz)



- Generated by the propagation of myogenic potentials from the mainly facial and neck muscle
- Generally associated with chewing, swallowing, smiling, talking ...
- Easily distinguish from the background
- They can be localized or widespread, and occur in a variety of forms

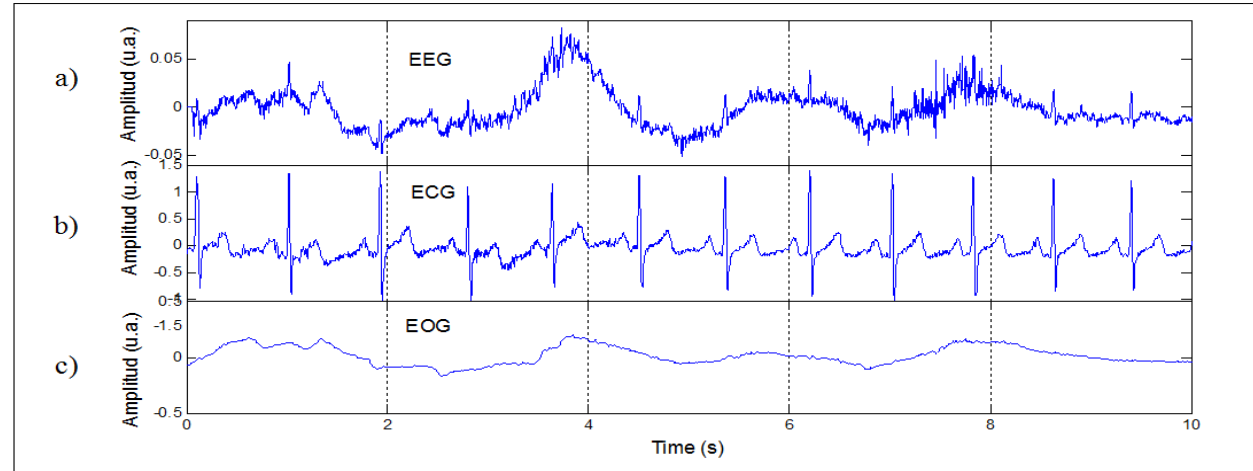
Teeth Grinding



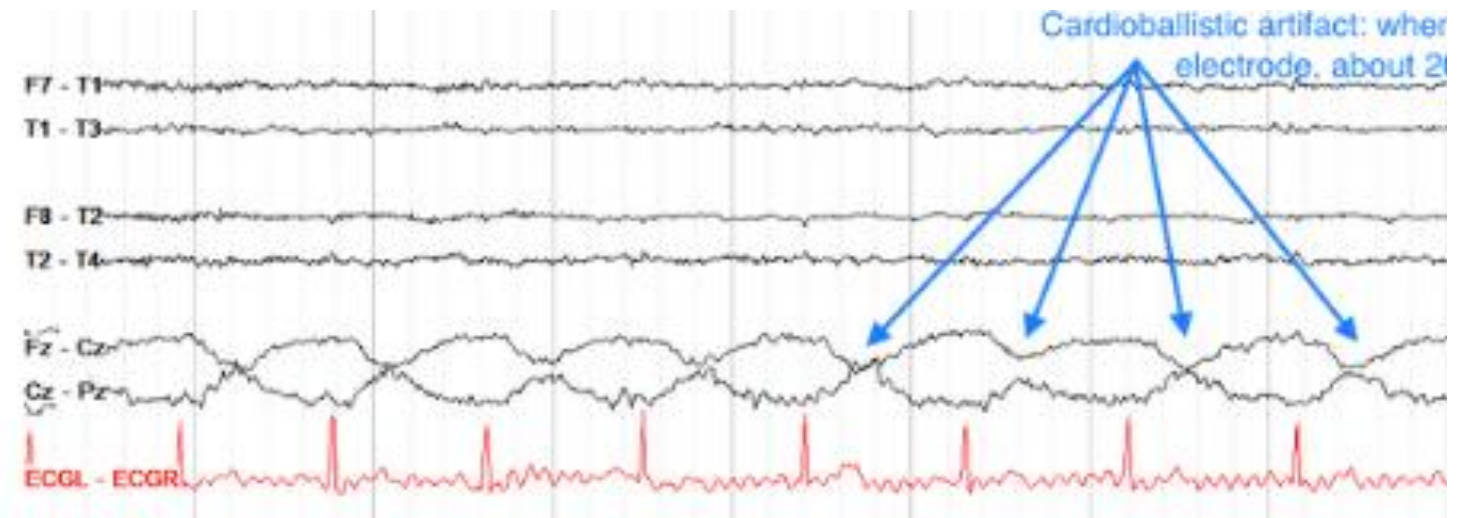
Regular periodic bursts of high-frequency EMG activity are seen in the EEG trace during teeth-grinding

Main Artefacts: Physiological artefacts

ECG/pulse
(1-2 Hz)

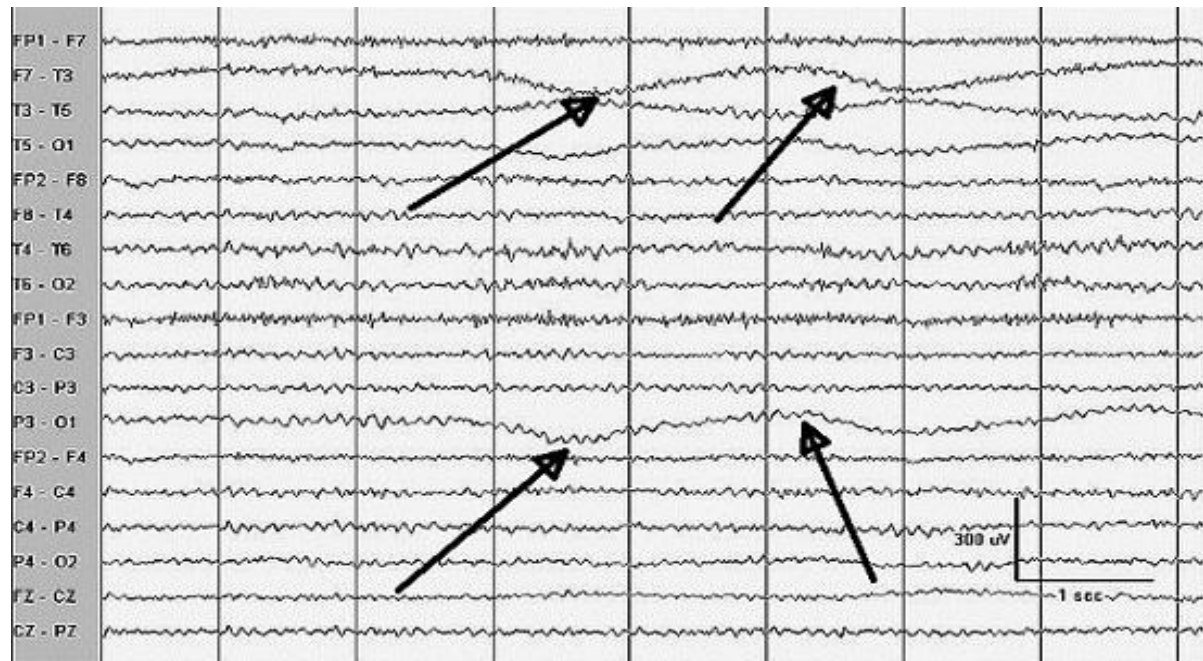


Cardioballistic artifact



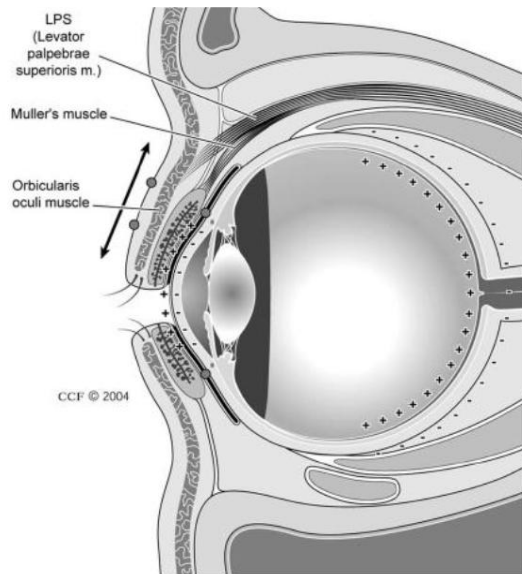
Main Artefacts: Physiological artefacts

Sweating
(0.1-1 Hz)

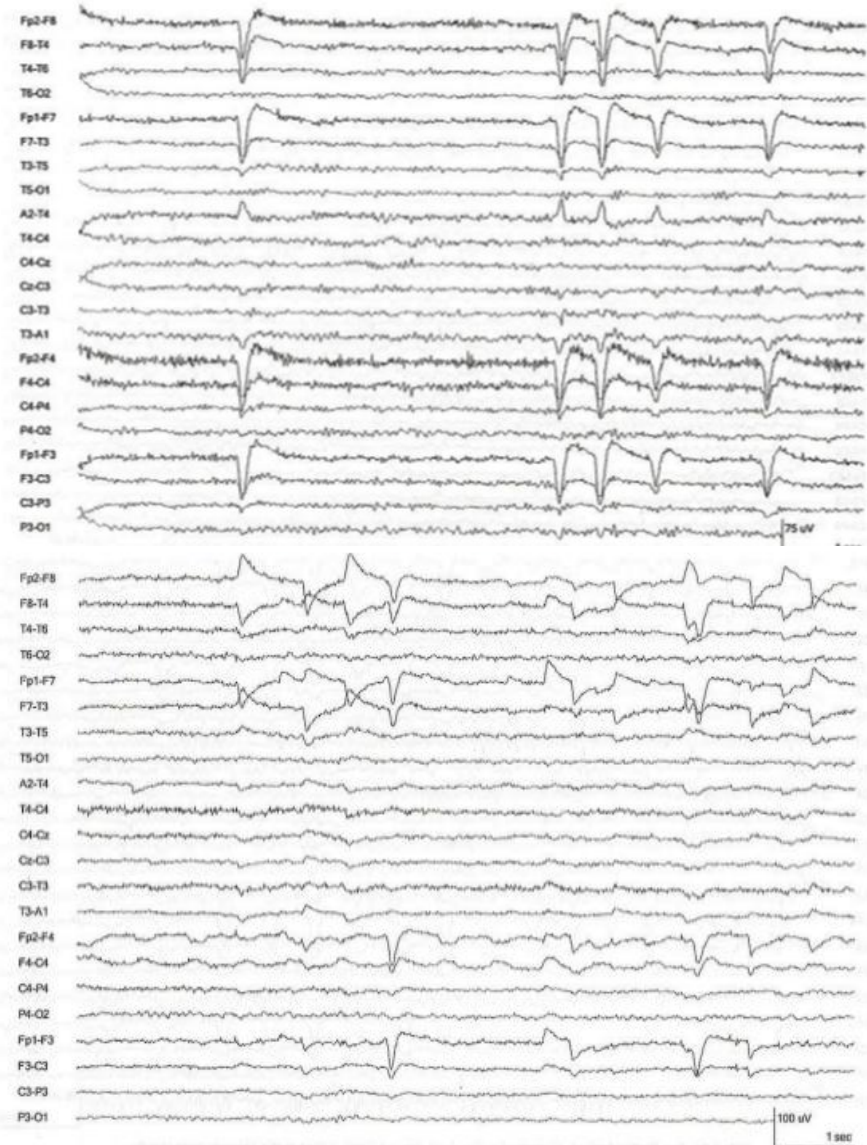


Main Artefacts: Physiological artefacts

Blink during wakefulness
(0.5-12 Hz)



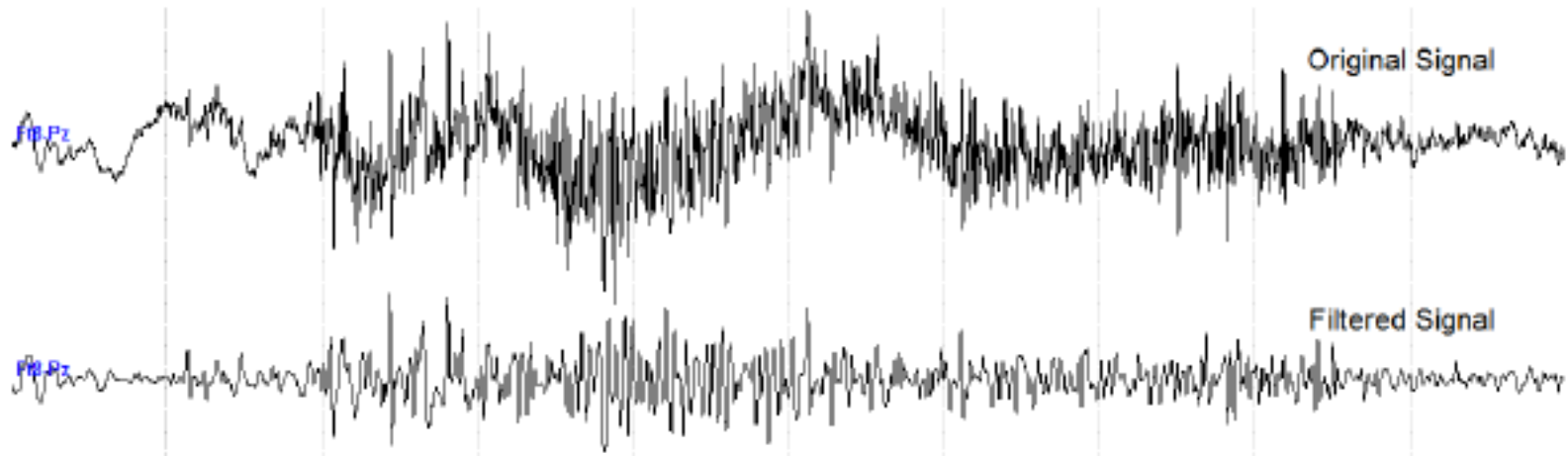
REMs during REM sleep
(0.5-12 Hz)



Preprocessing: Main steps

1. Filtering (band-pass + Notch)
2. Bad epoch and channel rejection
3. Artifact removal (blinks, heartbeat, etc.)
4. Bad Channel Interpolation
5. Power Spectral Density Computation
6. Waveform Detection
7. Source-modeling/Connectivity analysis ?

Filtering

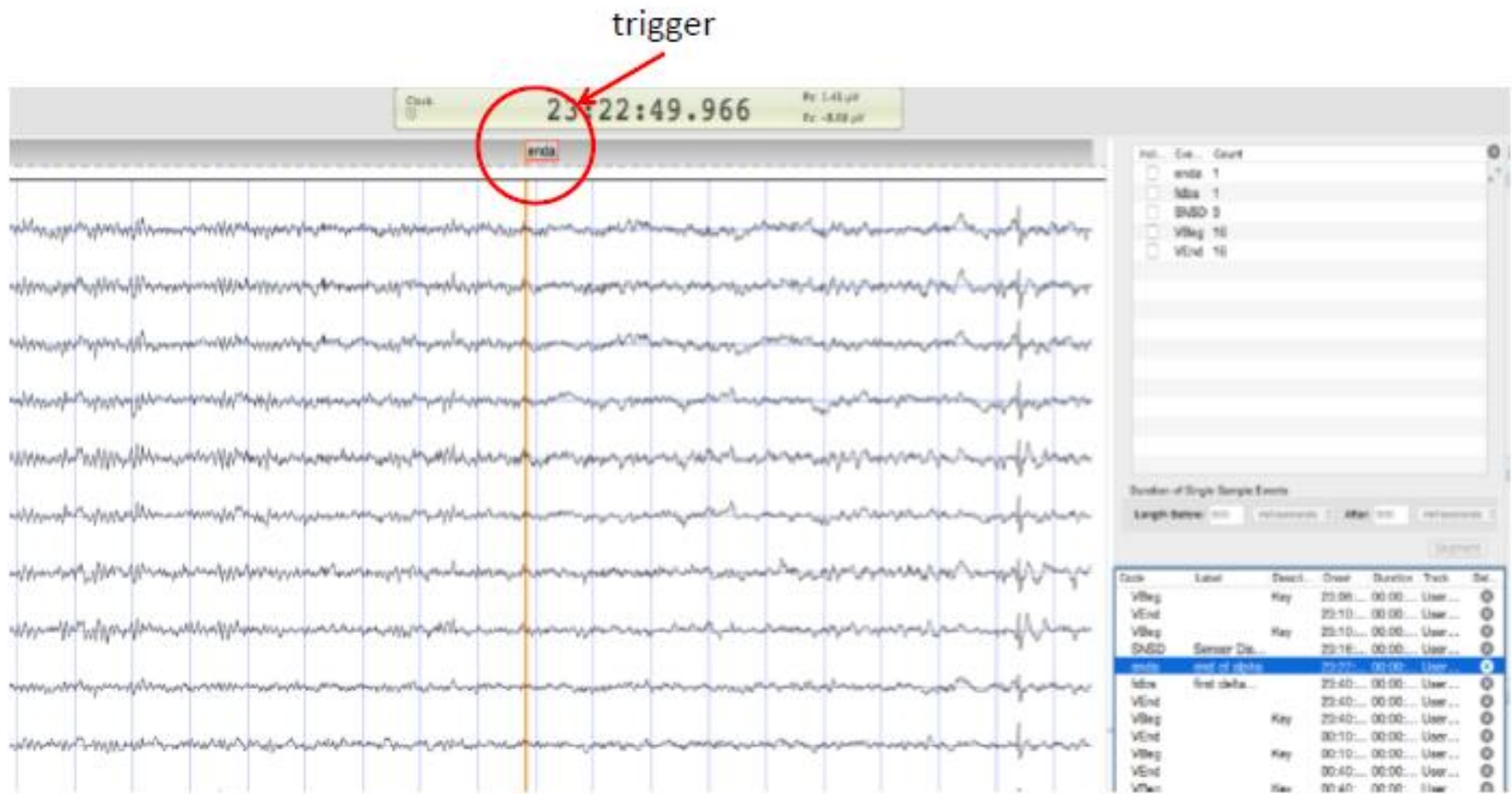


Improved data stationarity, reduction of artifacts and linear trends

Possibility to select specific frequency ranges of interest

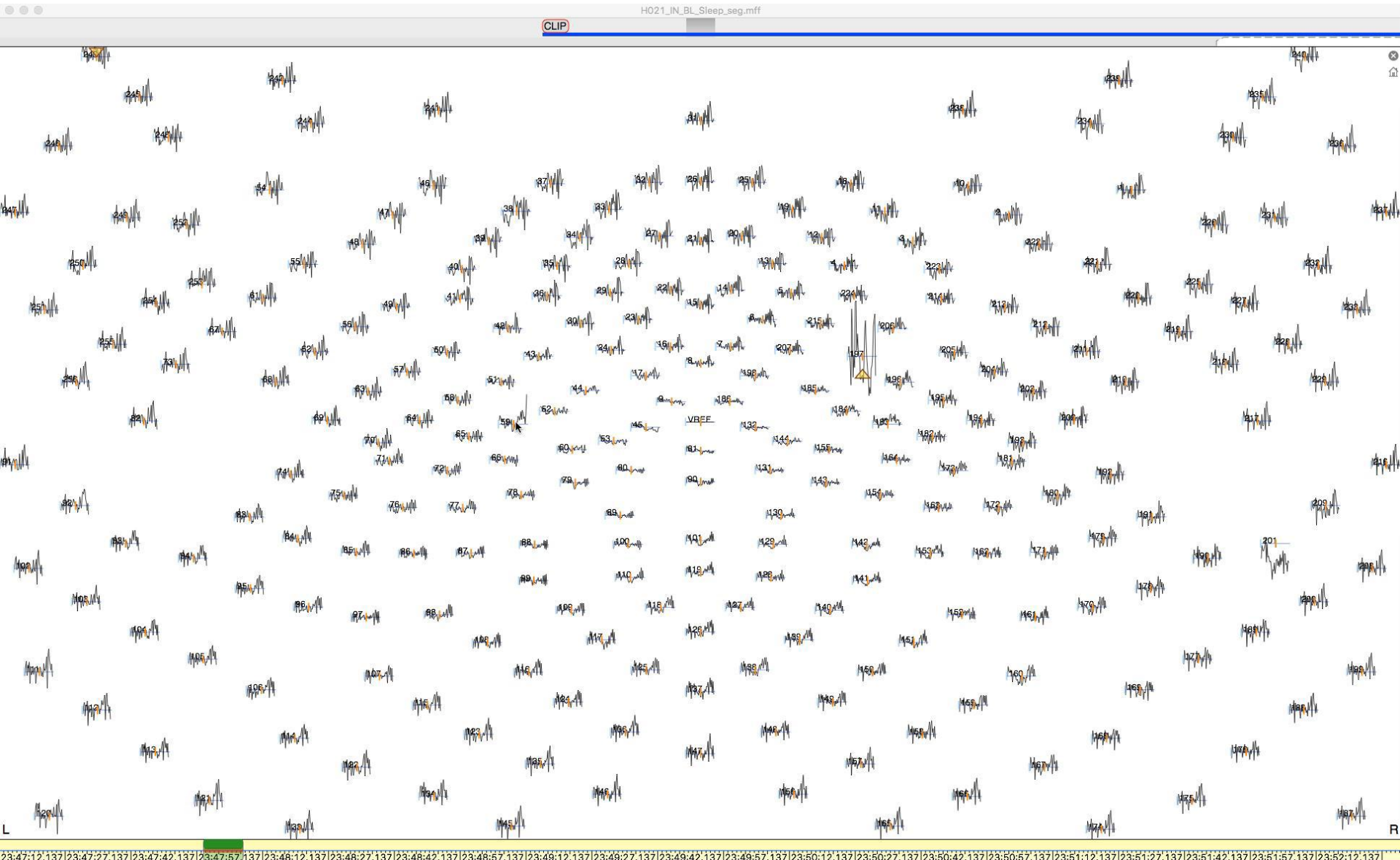
Exclusion of frequency ranges commonly containing non-physiological signals
(e.g., < 0.1 Hz and > 50 Hz for scalp EEG)

Segmentation and bad epoch detection

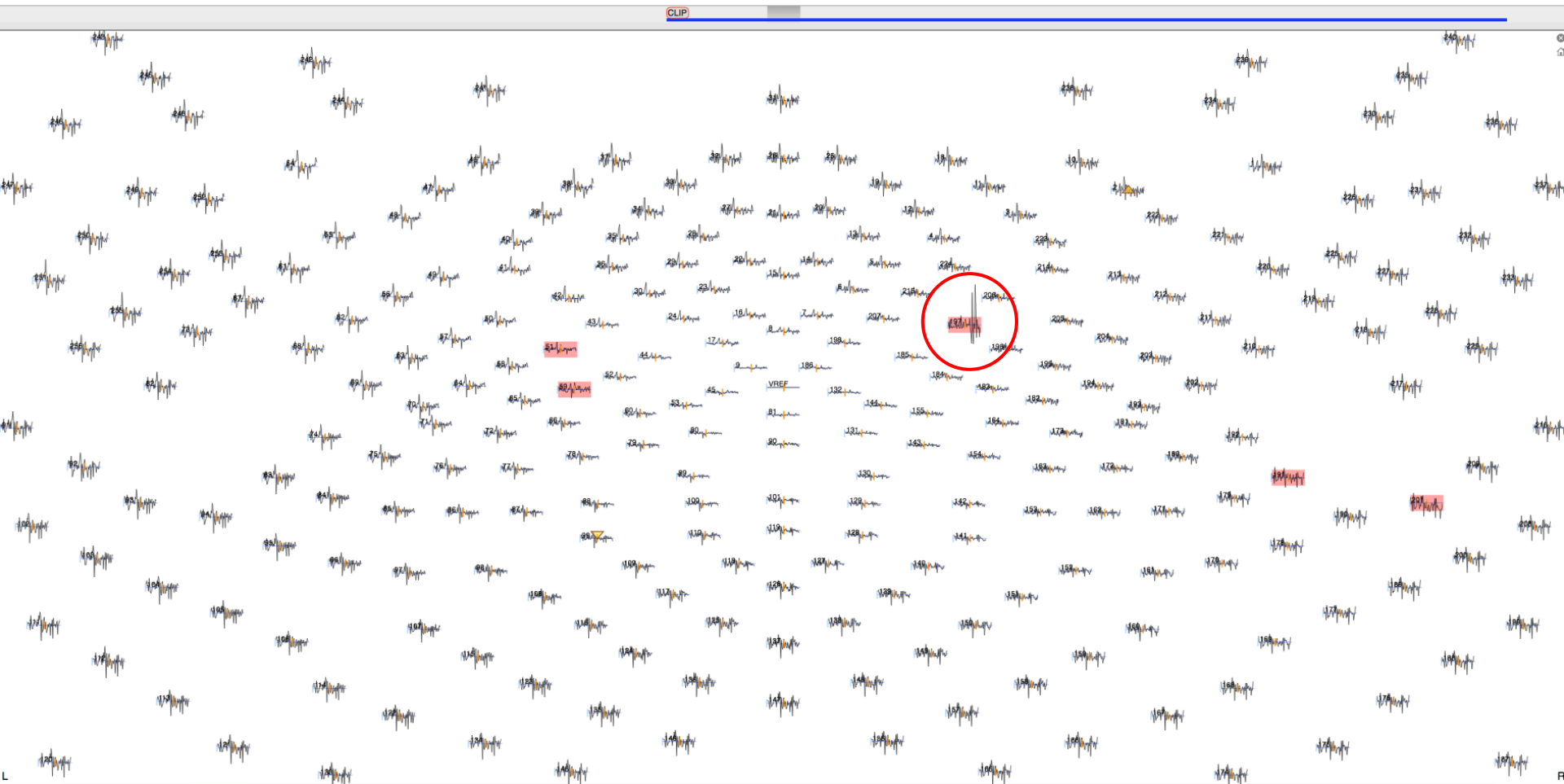


Manual or automatic methods based on statistical metrics across epochs (standard deviation, kurtosis, probability) or artifact subspace reconstruction
Softwares: EEGLAB (matlab), MNE (python)

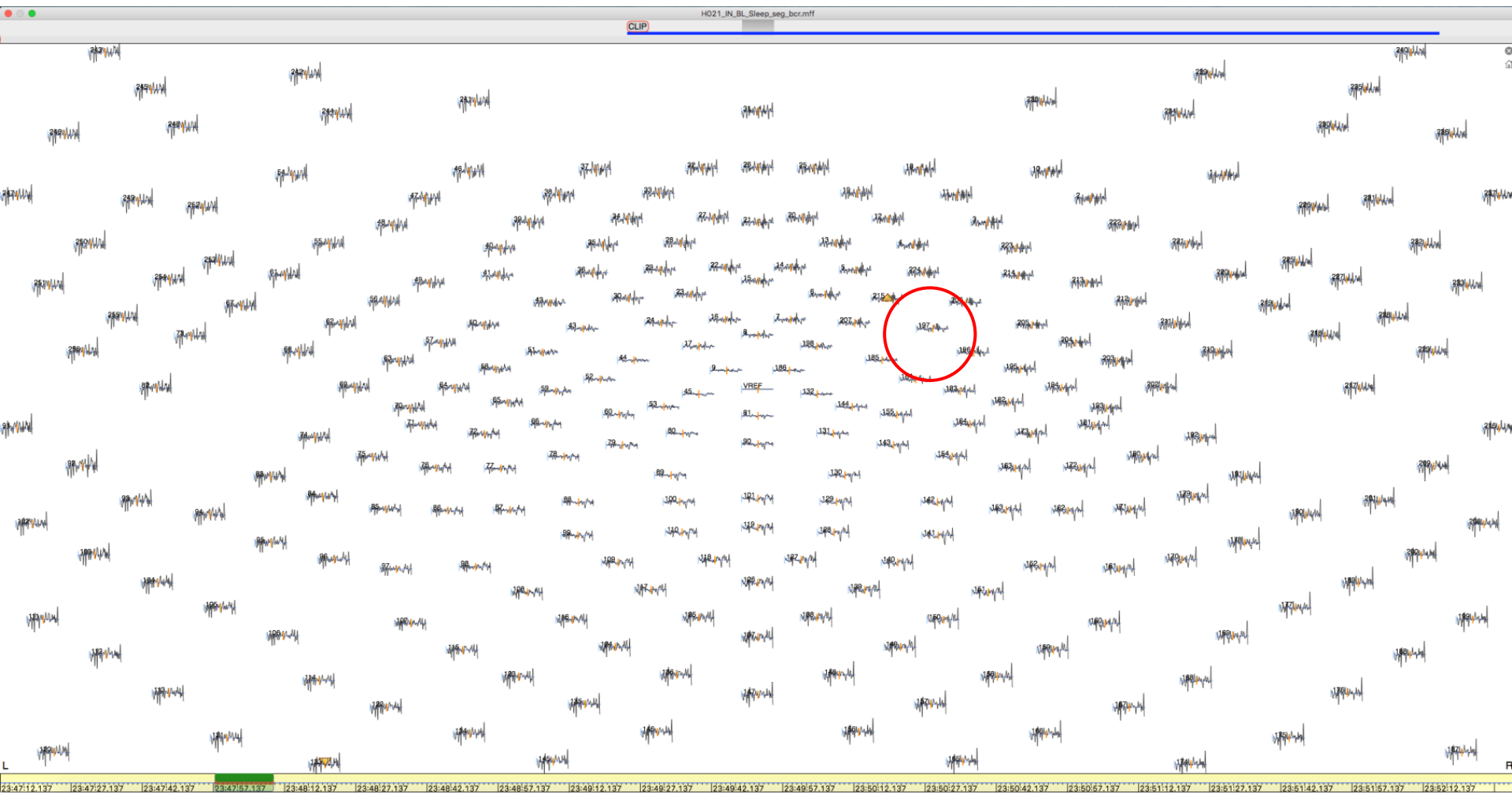
Bad channel marking



Bad channel interpolation (pre)

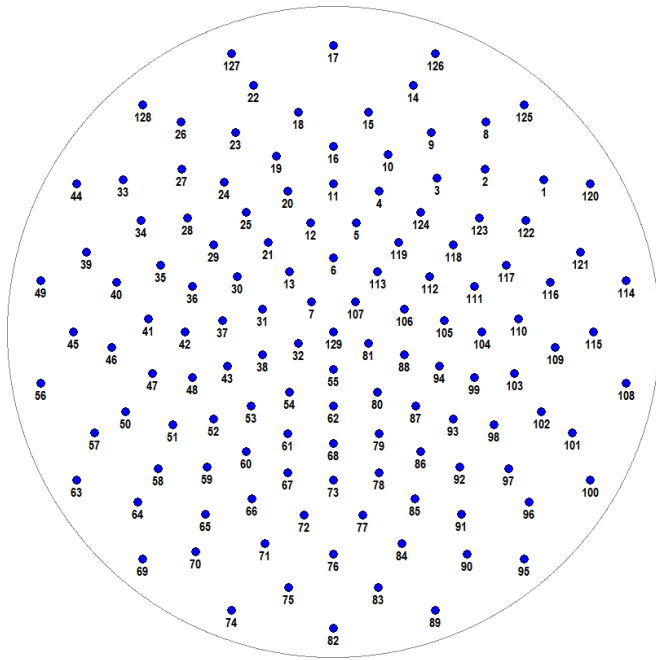


Bad channel interpolation (post)



Bad Channel Interpolation

Spherical splines [Perrin et al. (1989)]



The functions $P_n(x)$ are the Legendre polynomials of order n , which form a complete set of basis functions on a spherical surface.

$$V(\vec{r}) = c_0 + \sum_{i=1}^{128} c_i g_m(\cos(\hat{r} \cdot \hat{r}_i))$$

$$g_m(x) = \frac{1}{4\pi} \sum_{n=1}^{\infty} \frac{2n+1}{(n(n+1))^m} P_n(x)$$

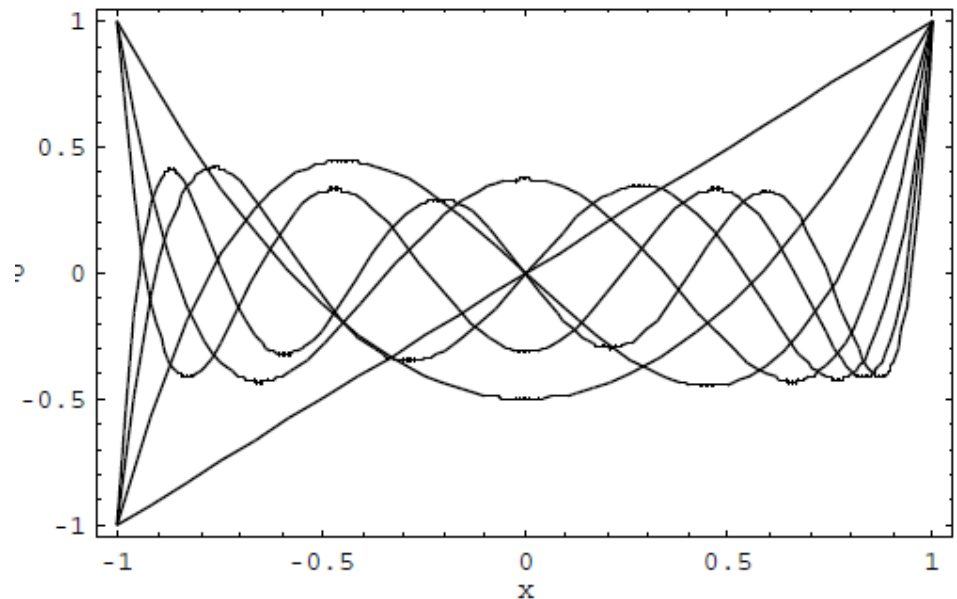
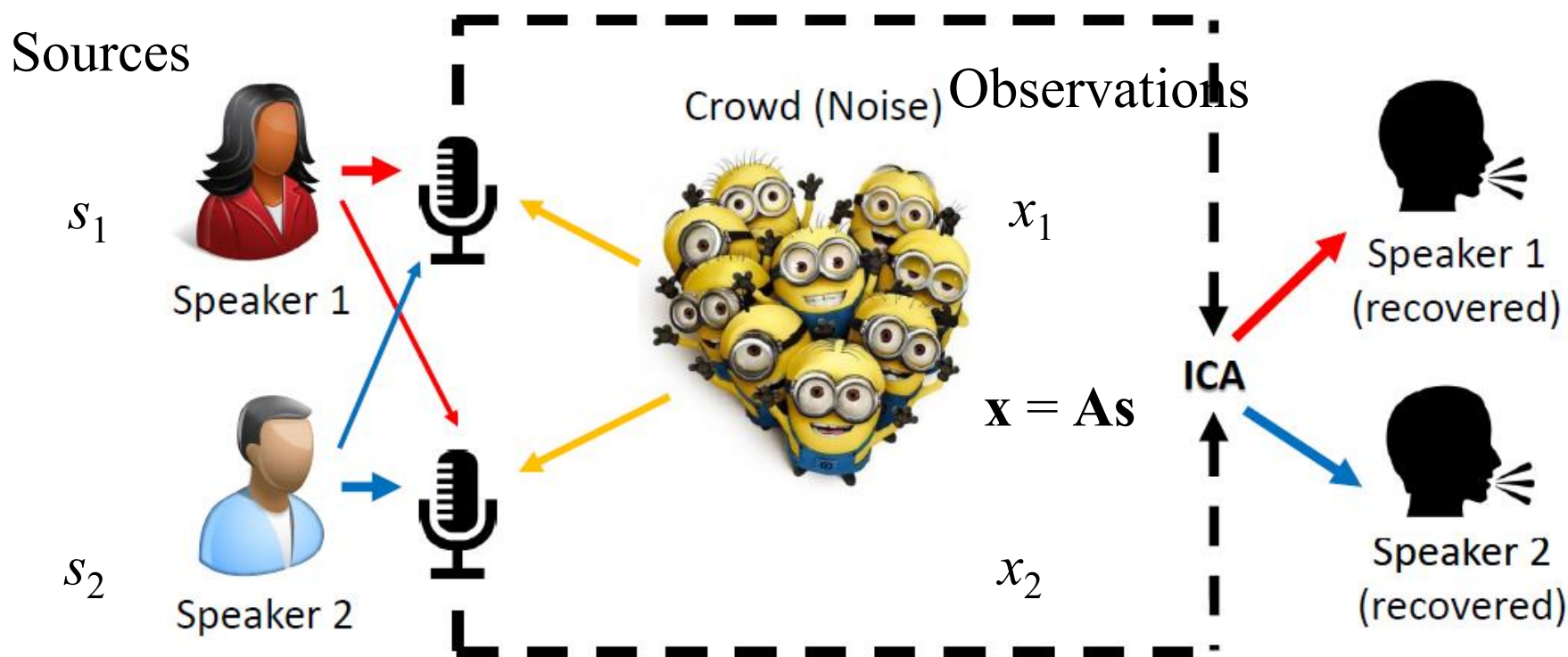


Figure 1. The first seven Legendre polynomials.

Artifact Removal/Reduction

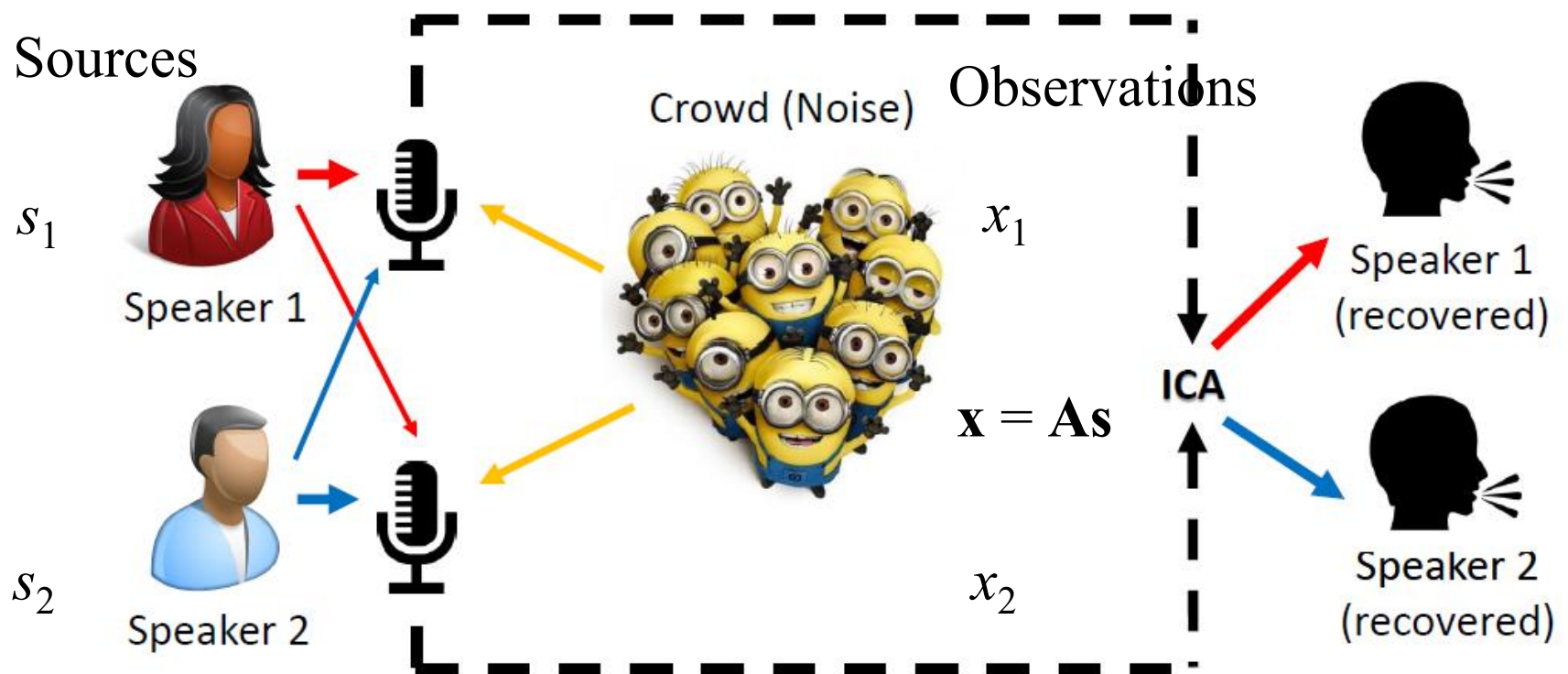
Independent component analysis (ICA) attempts to decompose a multivariate signal into **independent non-Gaussian signals**. It is a special case of blind source separation.

The Cocktail party problem:



ICA model

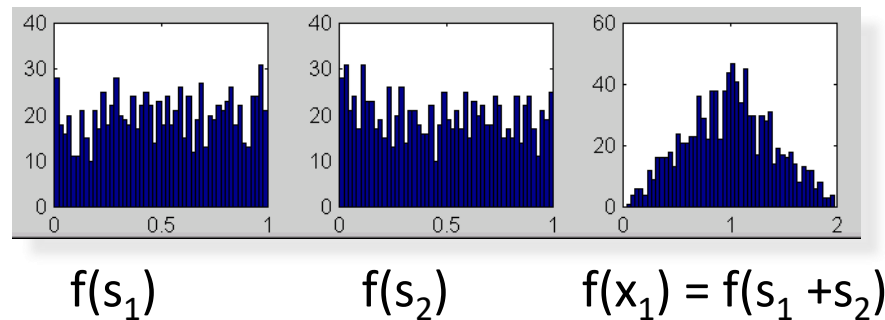
- Sources s and Mixing matrix A are unknown
- Task: estimate A and s using only the observable random vector x
- The ***independent components (IC's)*** are the estimates of s
- **Ill posed problem**, unless assumptions are made. This means **multiple (often infinite) solutions are possible**



ICA assumptions

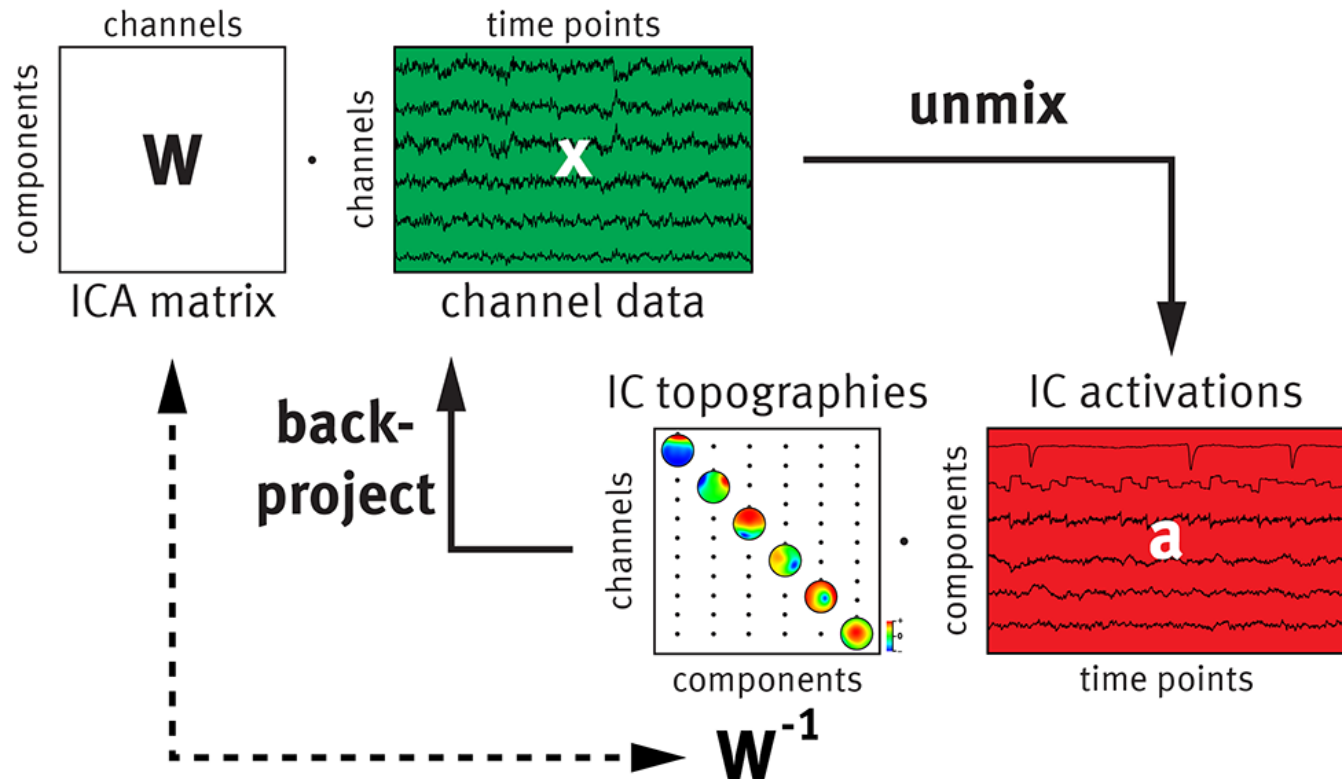
INDEPENDENCY + NON-GAUSSIANITY

- **Source signals are statistically independent:** time series of one does not give any information about the others.
- Hence the name *Independent Component Analysis*.
- **Signals are linear combination of source signals**
- Within the brain, ICA assumptions usually holds, then the source signals can be recovered up to permutation and scaling.
- Key to estimating A is **non-gaussianity**
- The distribution of a sum of independent random variables tends toward a Gaussian distribution. (By Central Limit Theorem)



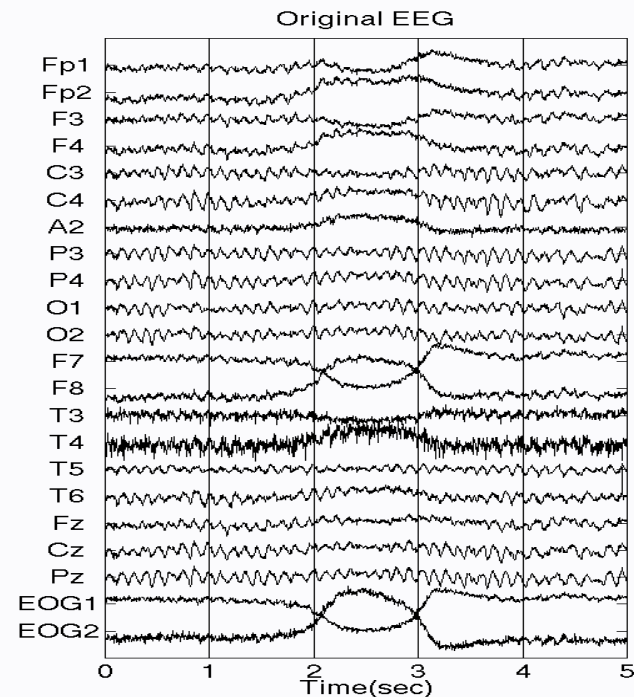
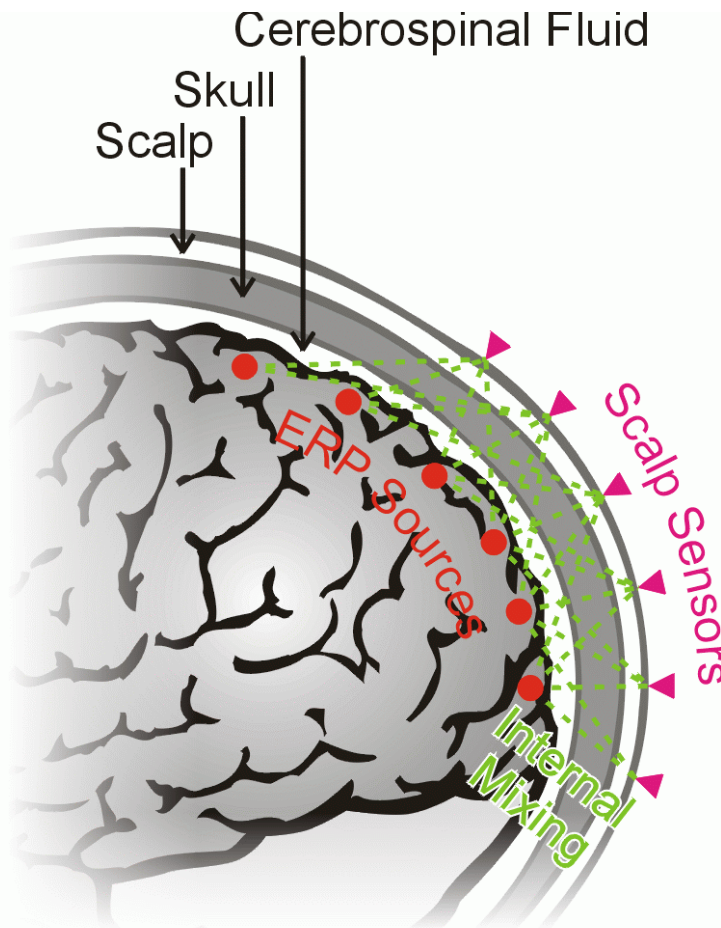
ICA model

- Under some mathematical conditions: $\mathbf{x} = \mathbf{A}\mathbf{s} \rightarrow \mathbf{s} = \mathbf{A}^{-1}\mathbf{x}$,
 $\mathbf{A}^{-1}$ often called $\mathbf{W}$, the unmixing matrix
- Since sum of two indep r.v. is more gaussian than individual r.v., so we could take $\mathbf{w}$ as a vector which maximizes the non-gaussianity of $\mathbf{w}^T\mathbf{x}$.

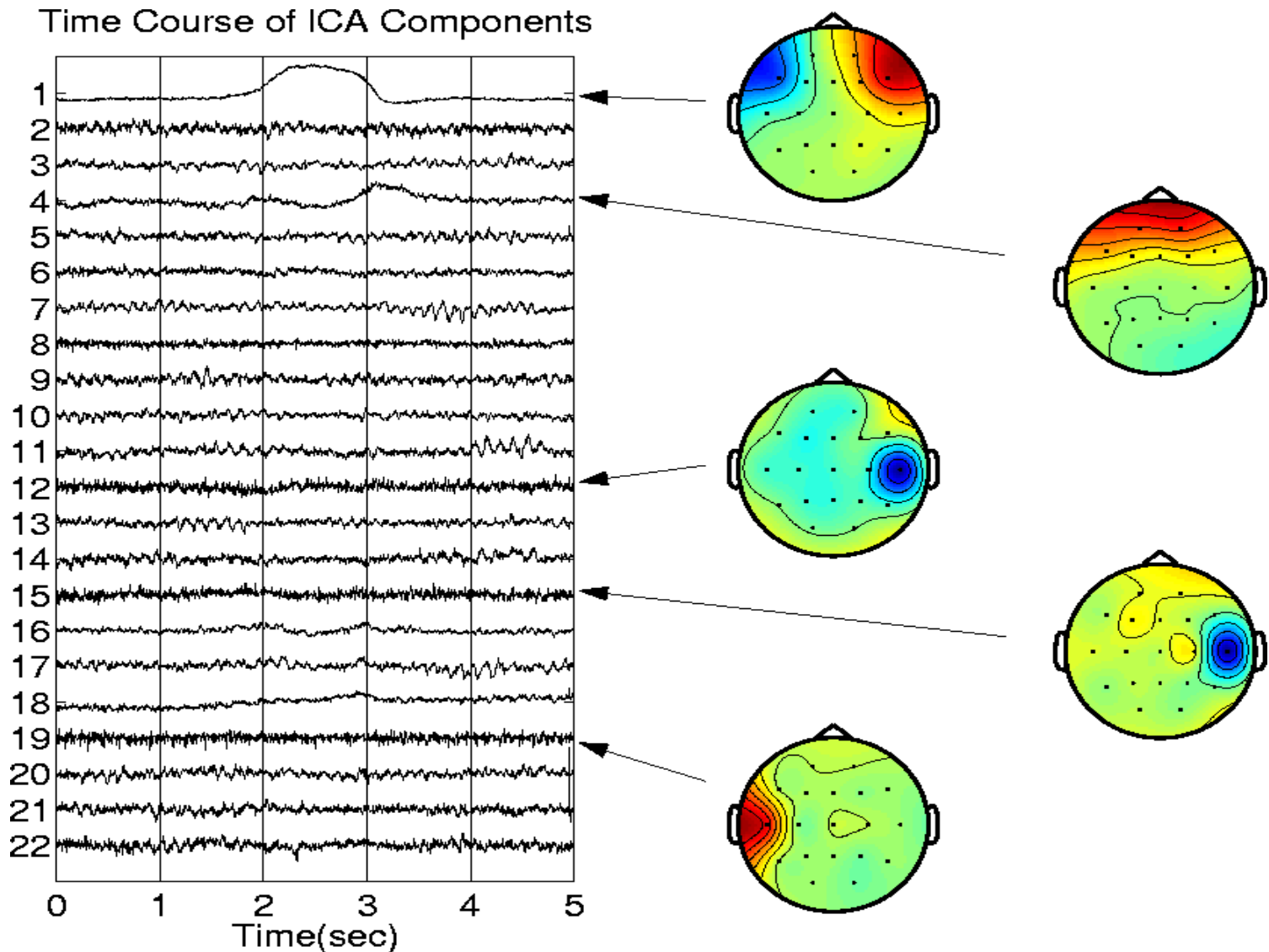


ICA applied to brainwaves

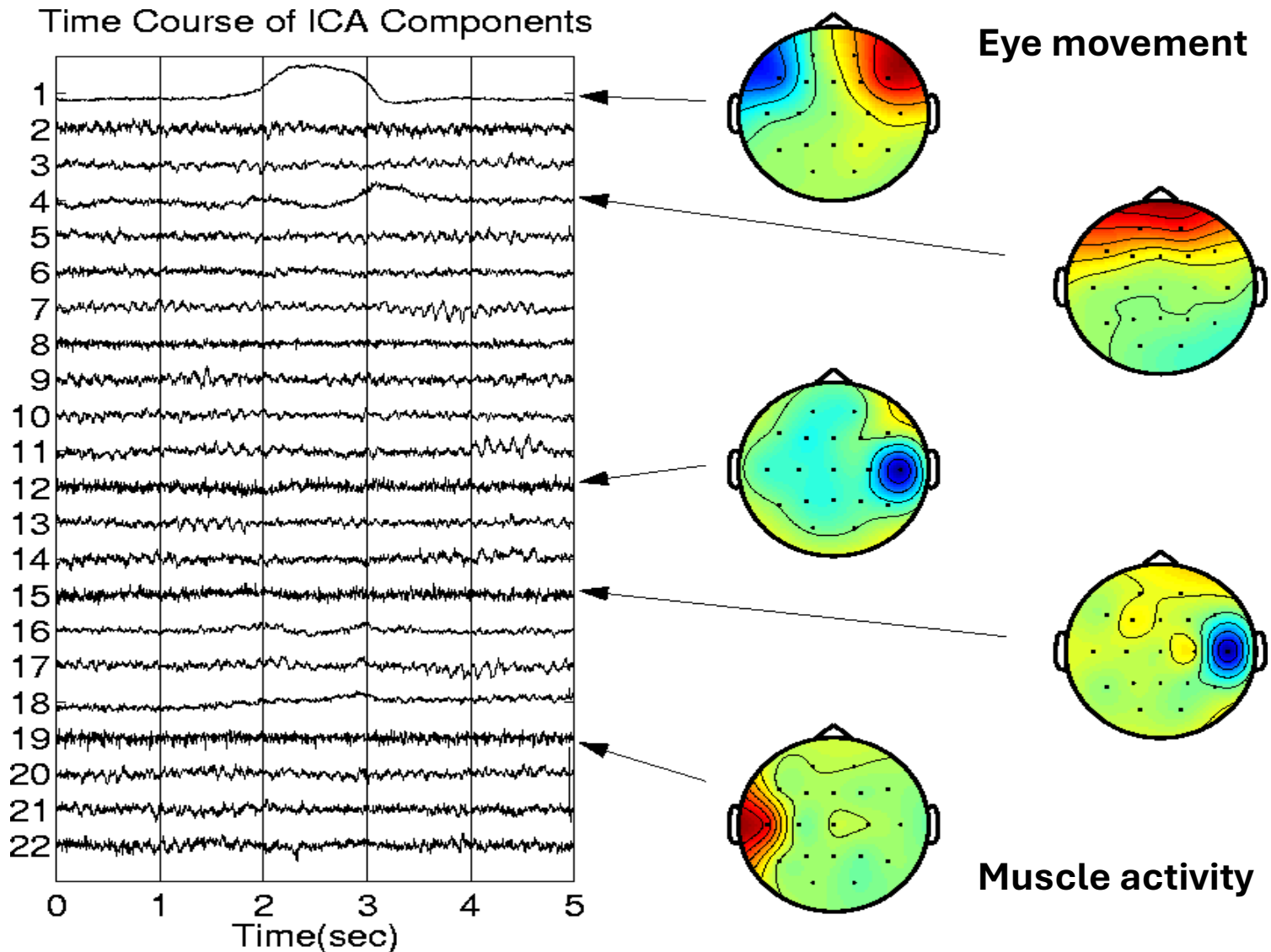
An EEG recording consists of activity arising from many brain and extra-brain processes. ICA can be used to exclude blink, eye movement artifact, cardiac artifacts, etc



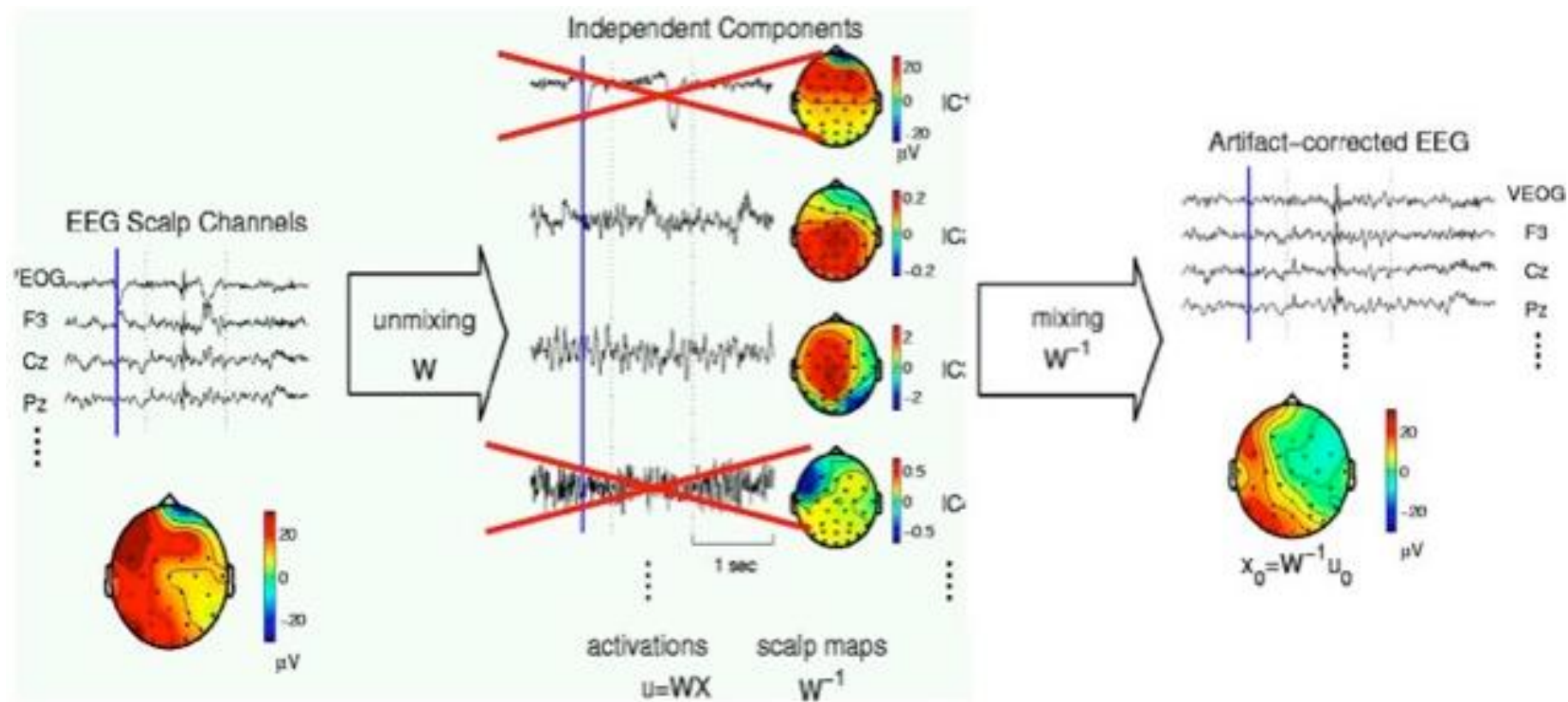
ICA applied to brainwaves



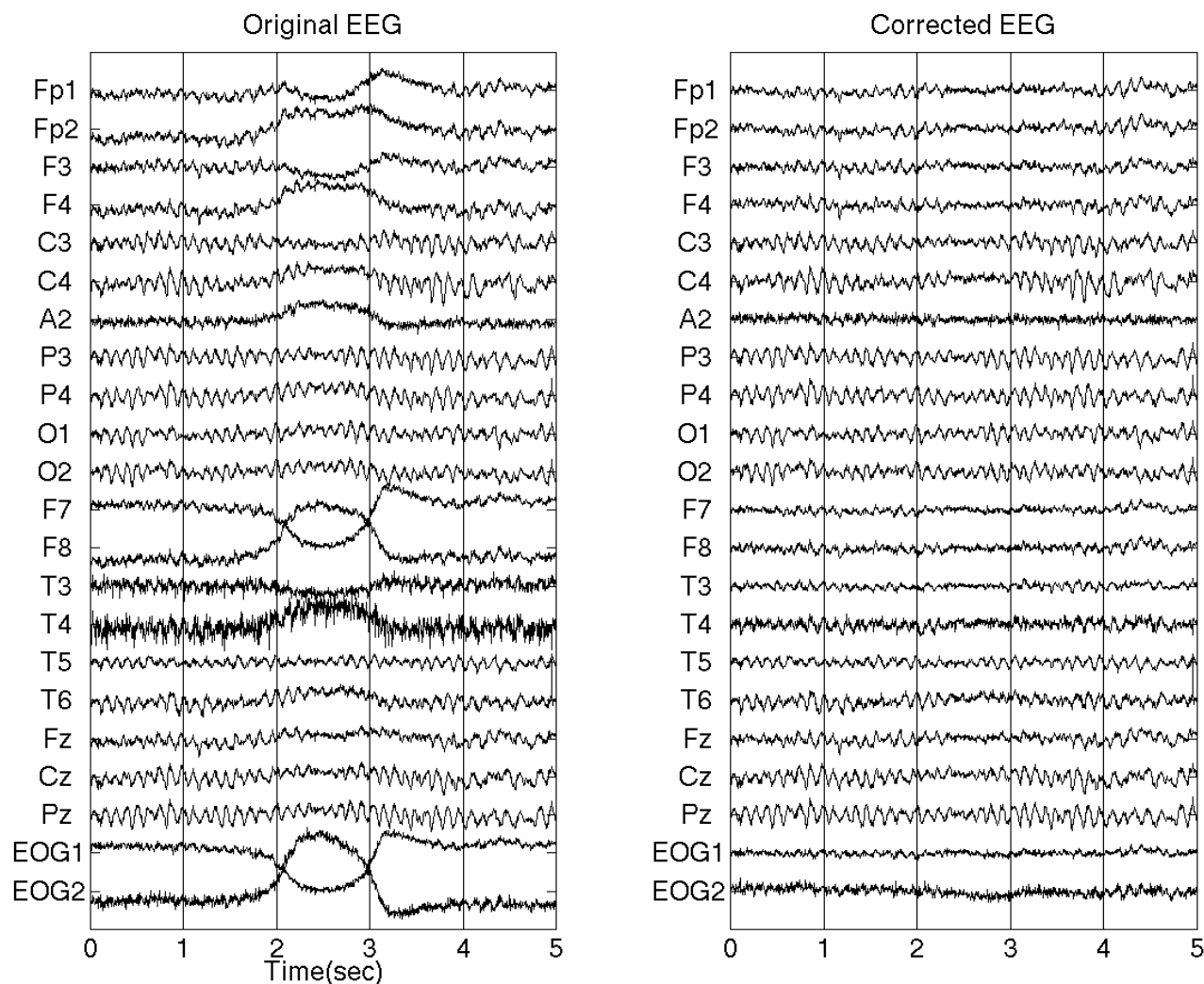
Identify artefactual components



ICA for artifact removing

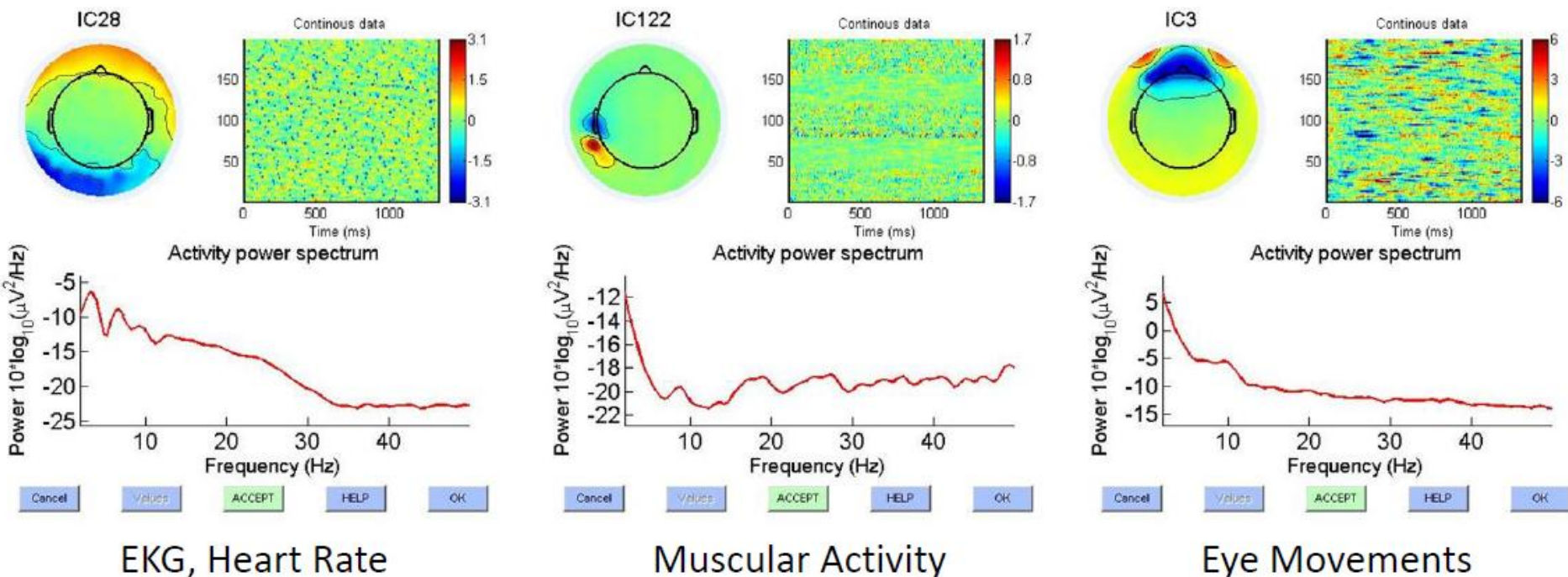


ICA for artifact removing



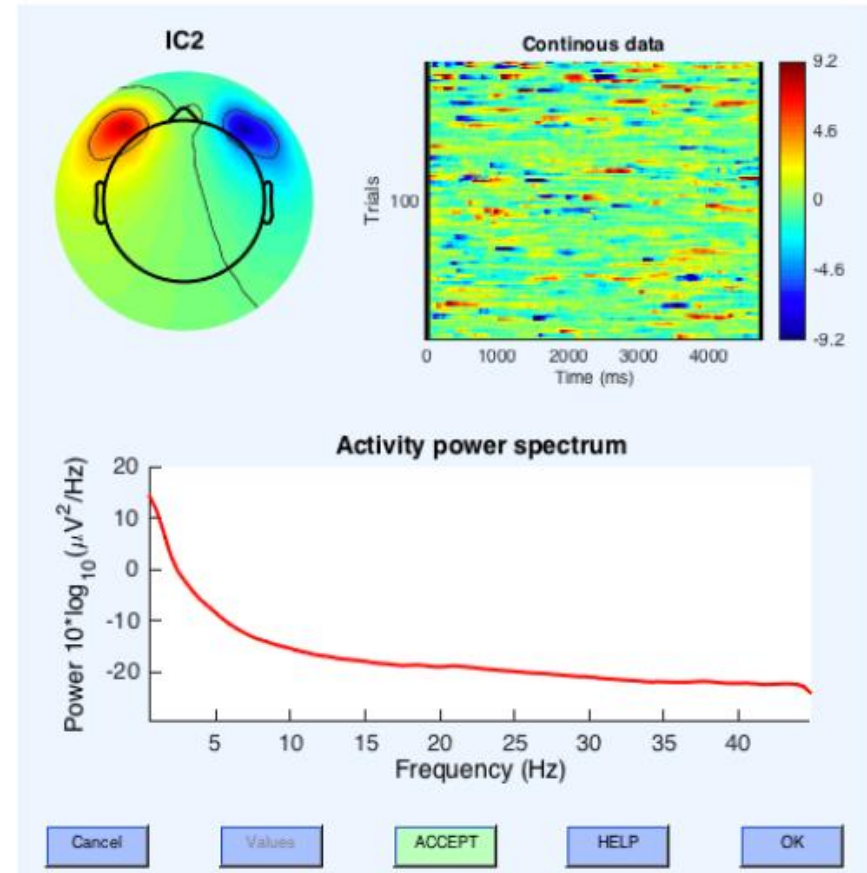
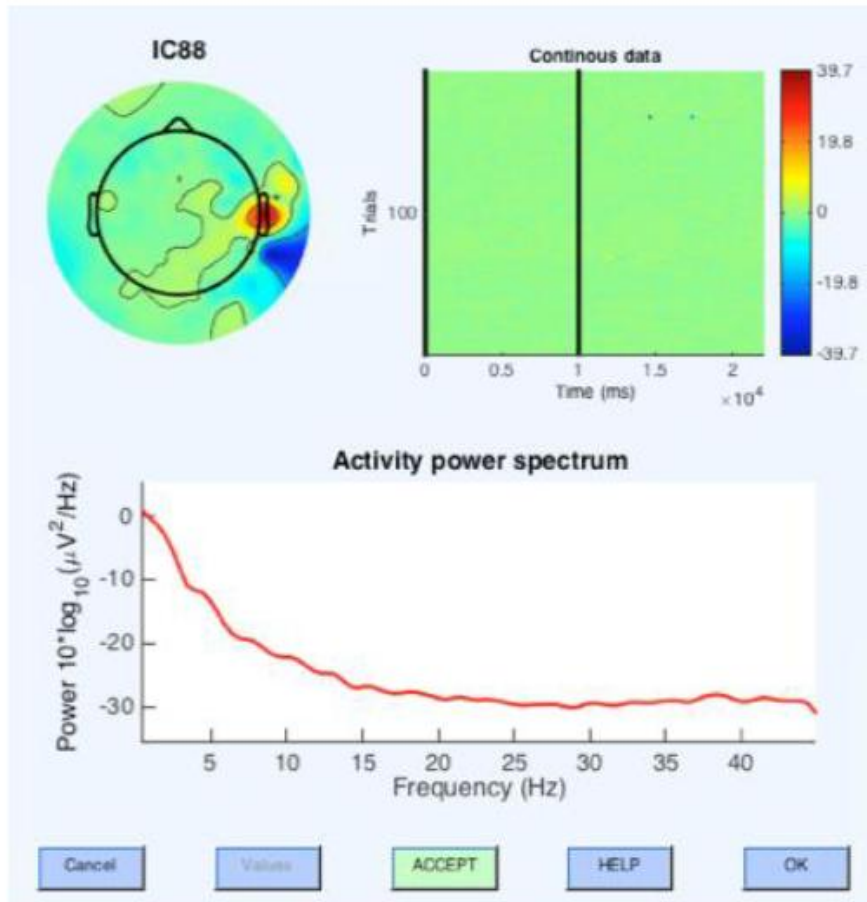
Identify artefactual components

May be used to remove several artifact sources [EEGLAB] on the logic that certain (stereotyped) artifacts will be temporally independent from neural sources.

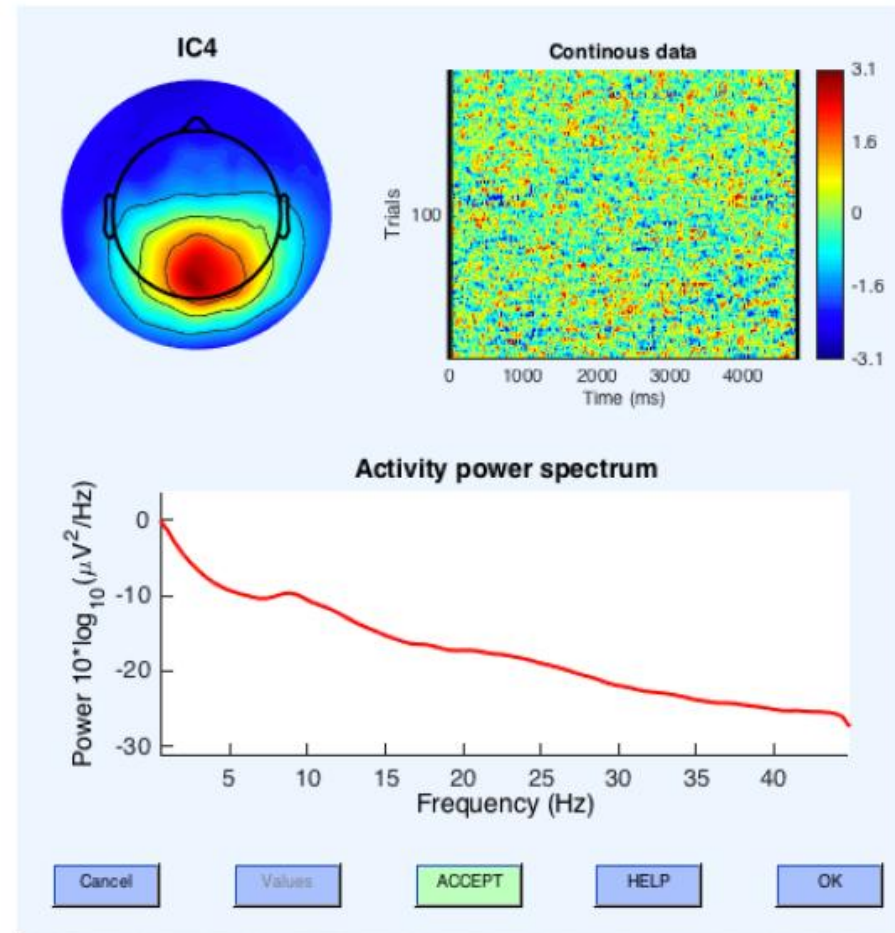
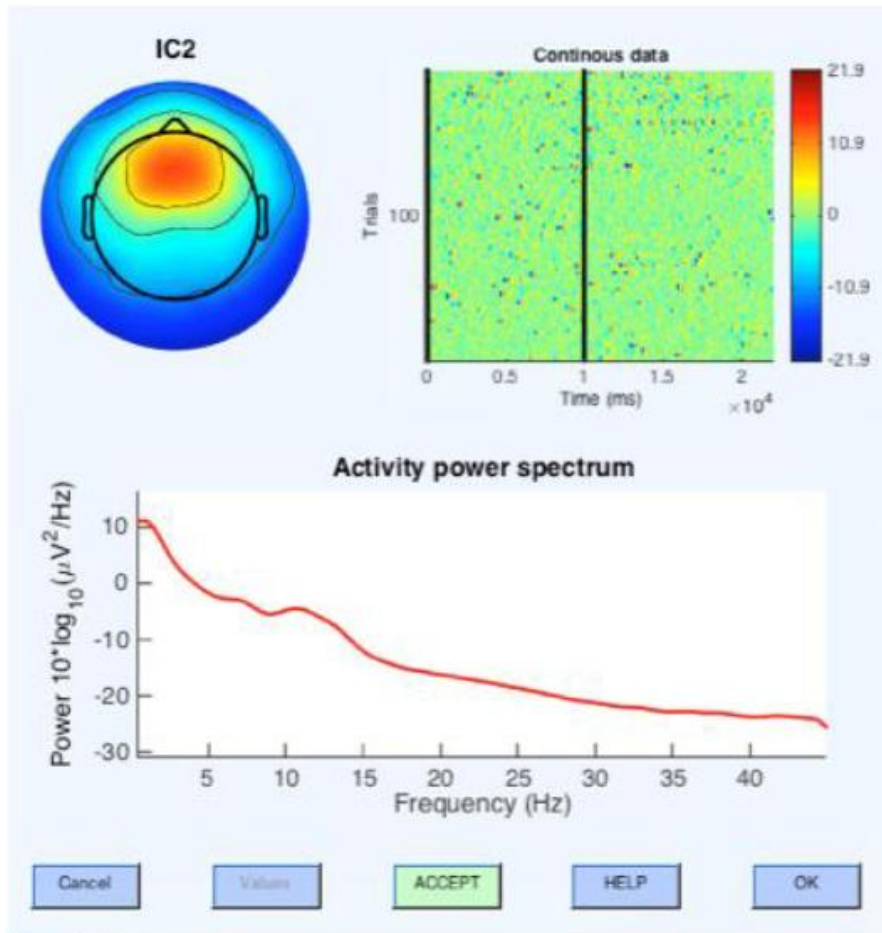


Note: This procedures should be applied AFTER removal of bad channels and/or bad epochs (potential bad channel interpolation should be performed AFTER the ICA)

Identify artefactual components



Identify artefactual components

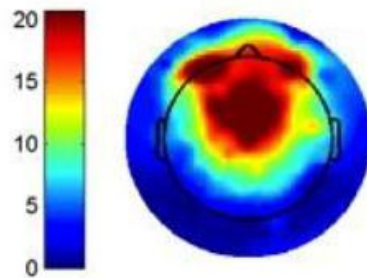


<https://labeling.ucsd.edu/tutorial/labels>

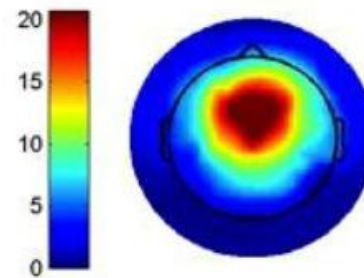
ICA for artifact removing

Topographical distribution of power in the delta range

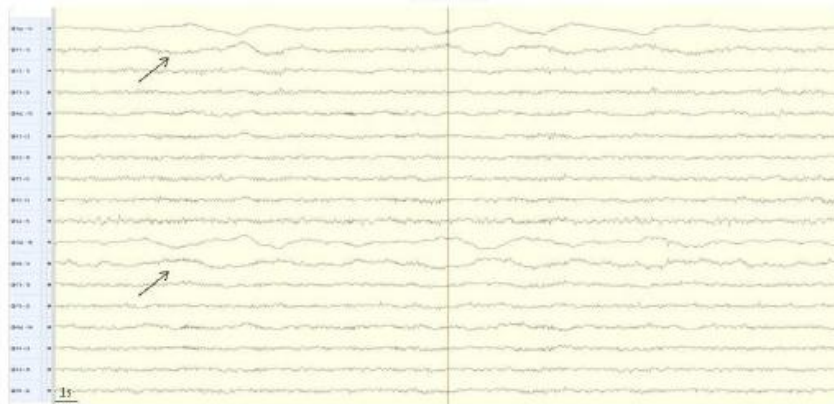
Pre-artifact removal (ICA)



Post-artifact removal (ICA)



PRE-ICA



POST-ICA



How many point for ICA running

As a general rule, **finding N stable components** (from **N -channel data**) typically **requires *more than* kN^2** data sample points (at each channel), where **N^2 is the number of weights in the unmixing matrix** that ICA is trying to learn and k is a multiplier. In our experience, **the value of k increases as the number of channels increases**. In our example using **32 channels**, we have 30800 data points, giving $30800/32^2 = 30$ **pts/weight points**. However, to find 256 components, it appears that even 30 points per weight is not enough data.

Can you use too much data? The short answer is no. However, suppose you have **two tasks in your experiment** and are planning to process them separately (for two papers, for example). In that case, it is probably **better to run ICA on them separately because they may contain different ICA brain components**. Because **the number of components is limited to the number of channels**, processing the **two tasks together might mean that you might have fewer components specific to each task**. Also, if you have different sessions for the same subject (data collected on different days), these should be processed separately because of the slightly different electrode placements.

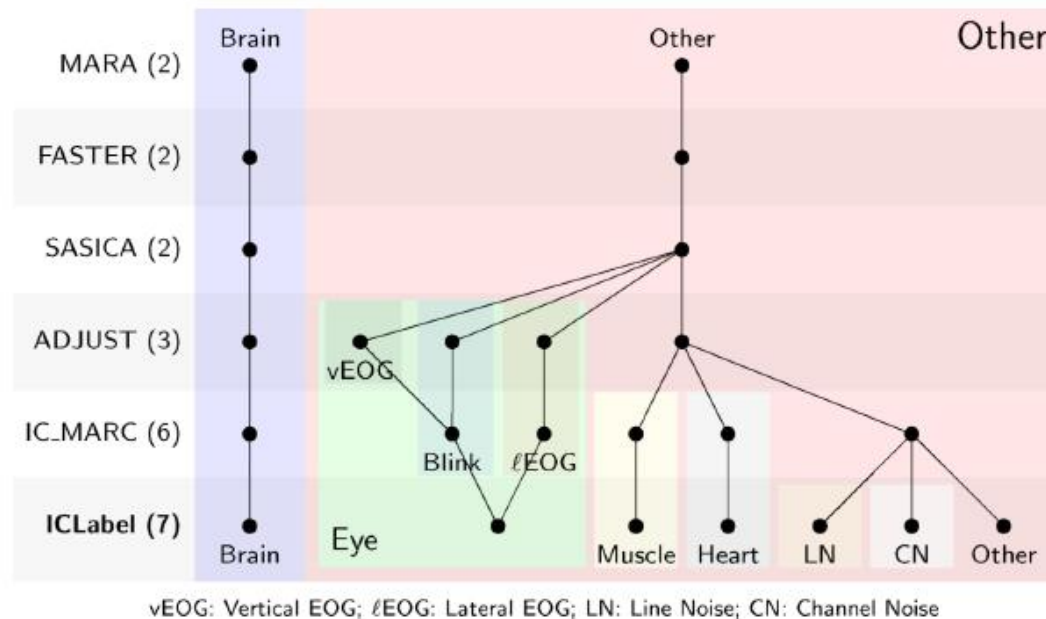
How to obtain a better ICA decomposition

- ICA takes all training data into consideration. **When too many types (i.e., scalp distributions) of noise - complex movement artifacts, electrode *pops*, etc – are left in the training data, these unique and irreducible data features will *draw the attention* of ICA**, producing a set of component maps including many single-channel or noisy-appearing components. **The number of components (degrees of freedom) devoted to the decomposition of brain EEG alone will be correspondingly reduced.**
- Therefore, **presenting ICA with as much *clean* EEG data as possible** is the best strategy. Note that blink and other stereotyped EEG artifacts do not necessarily have to be removed since they are likely to be isolated as single ICA components.
- Here ***clean* EEG data means data after removing noisy time segments** (does not apply to removed ICA components).

Automatic ICA classification

ICLabel: An automated electroencephalographic independent component classifier, dataset, and website

Luca Pion-Tonachini^{a,b,*}, Ken Kreutz-Delgado^{b,c}, and Scott Makeig^a



- The ICLabel classifier compute probabilities across seven classes: BRAIN, MUSCLE, EYE, HEART, LINE NOISE, CHANNEL NOISE, OTHER
- ICLabel dataset used to train and evaluate the ICLabel classifier encompasses a wide variety of EEG datasets from multiple paradigms (6352 recordings)
- Scalp topography, PDS and autocorrelation function, current dipole model fit, ...
- Crowd labeling collected through the ICLAB website + Expert-labeling